



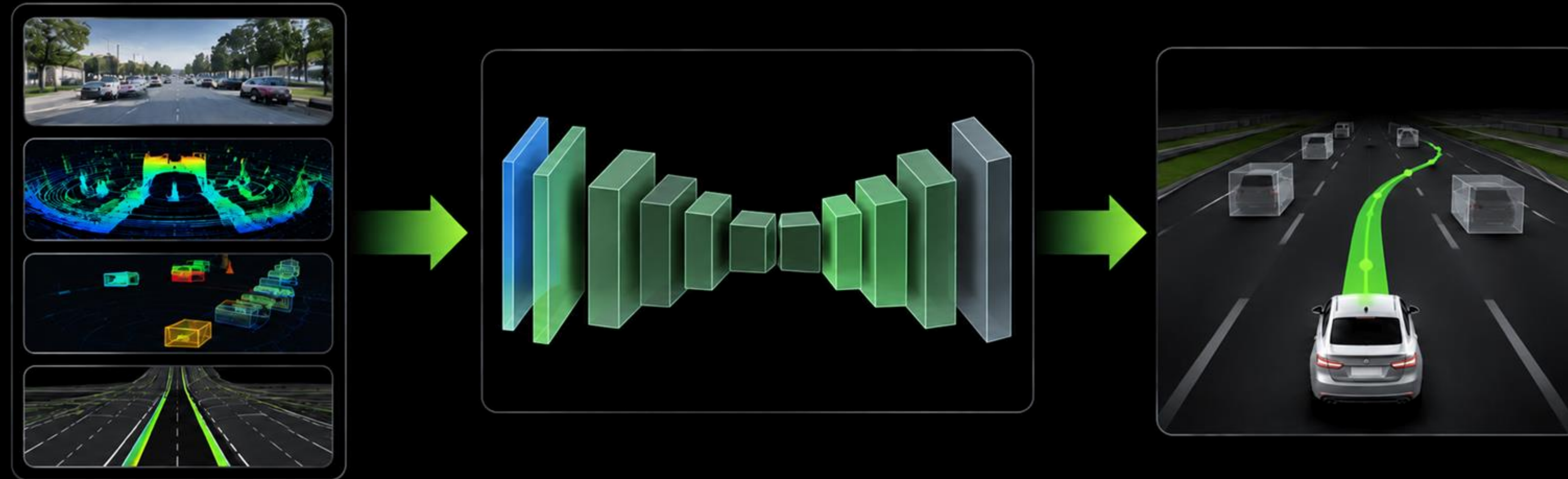
End-to-End Models for Autonomous Driving:

From Direct Prediction to
Counterfactual Deliberation on the Edge

Jose M. Alvarez,
Director AI Research, NVIDIA
ELLIS Summer School on Autonomous Driving, Sept. 2026

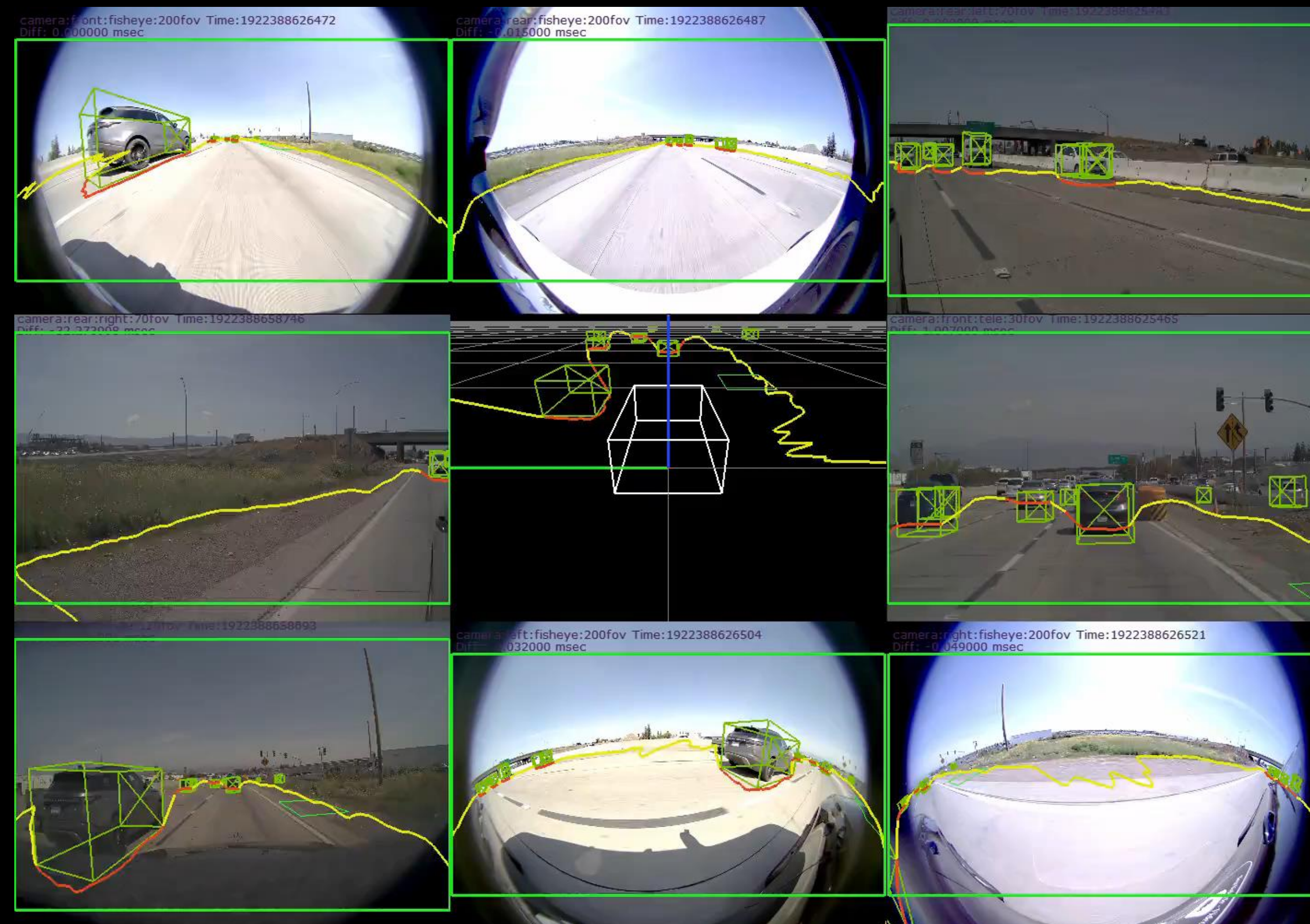
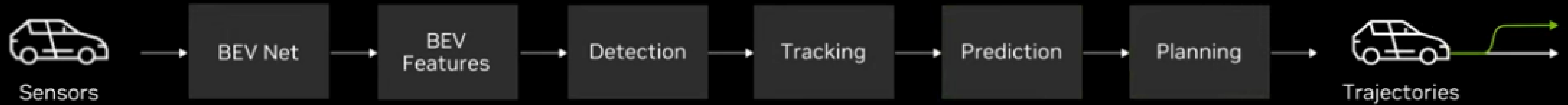


End-to-End Models in Autonomous Driving



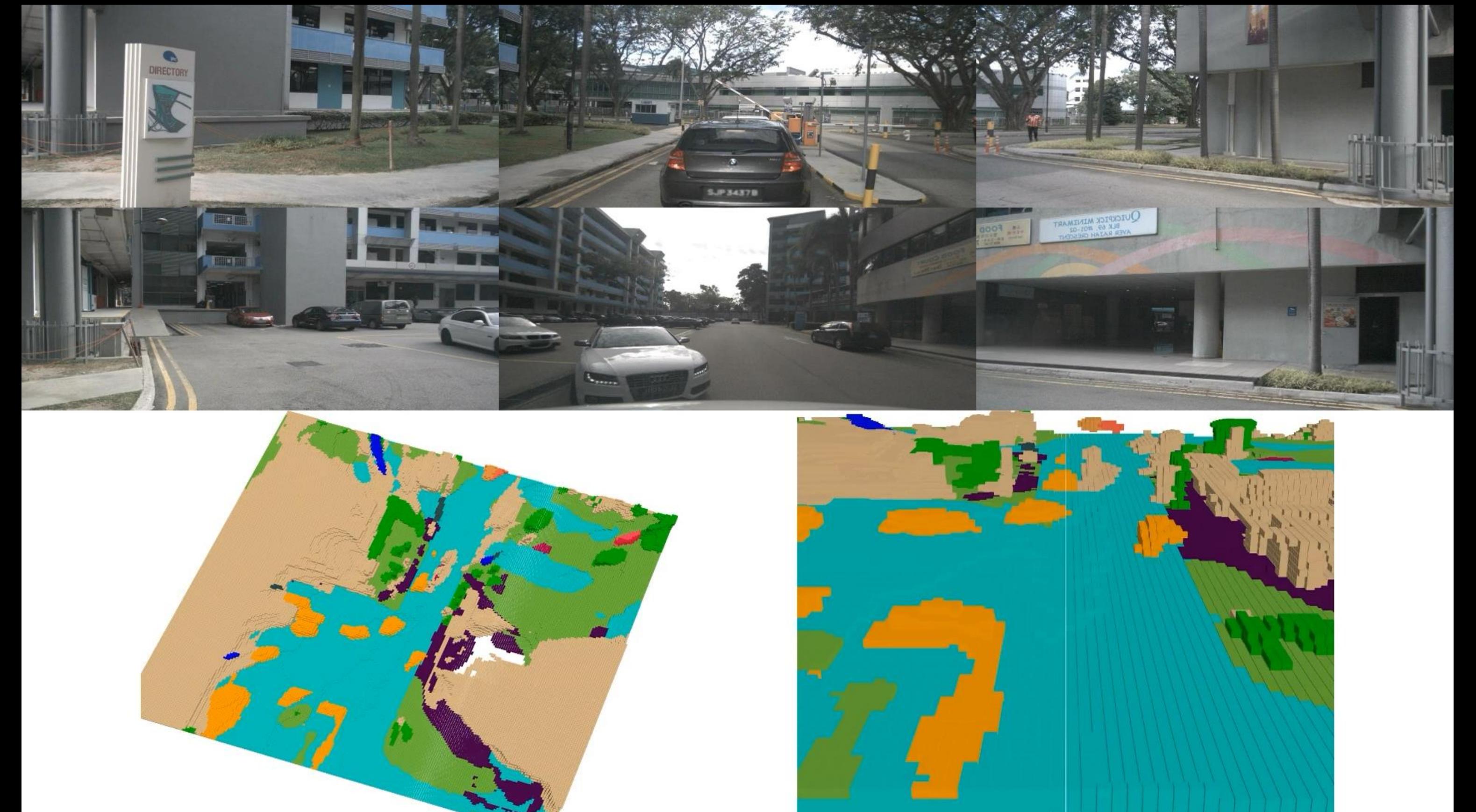
- **Direct mapping:**
 - Takes raw inputs like camera pixels, LiDAR point clouds, and radar data.
- **Direct outputs:**
 - Generates physical control actions like steering angles, throttle, and braking, or motion trajectories.
- **No manual staging:**
 - Replaces separate, hand-engineered stages with one learnable neural network.

Modular vs End-to-End for Autonomous Driving



3D Obstacle Detection

Shen et al. [Hardware-aware latency pruning for real-time 3d object detection](#) NeurIPS 2023

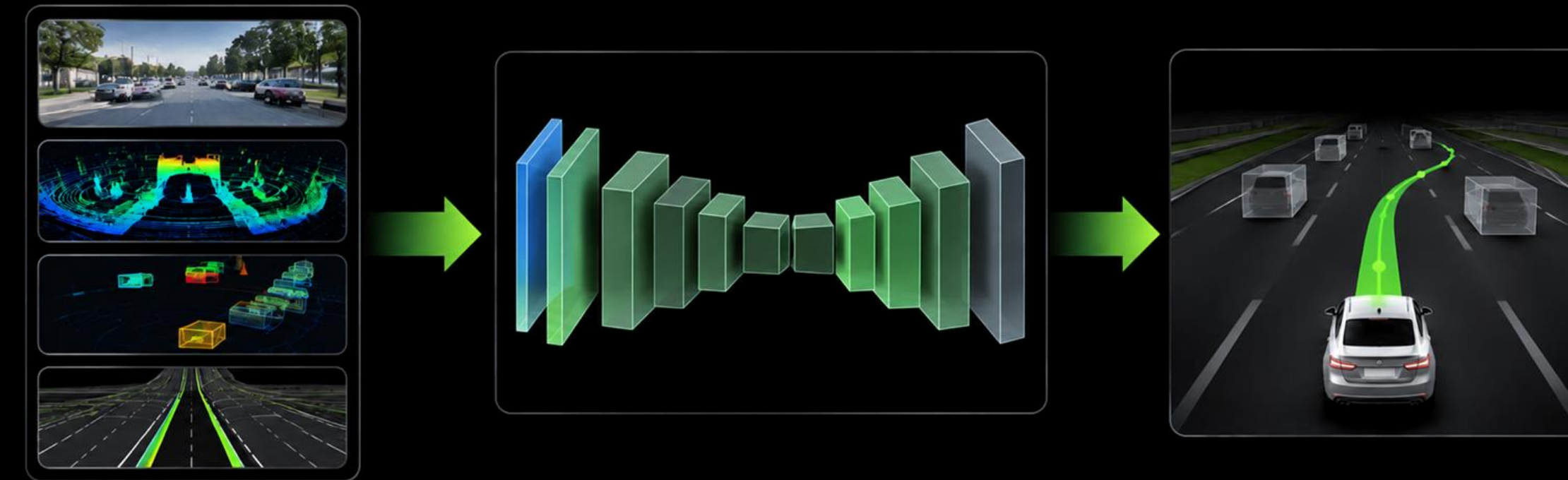


Occupancy Prediction

Li et al. [FB-OCC: Forward-backward occupancy prediction](#). CVPR 2023

Outstanding Champion and Innovation Award in [Autonomous Driving Challenge](#) at CVPR 2023 End-to-End Autonomous Driving Workshop and Vision-Centric Autonomous Driving Workshop.

End-to-End Models in Autonomous Driving

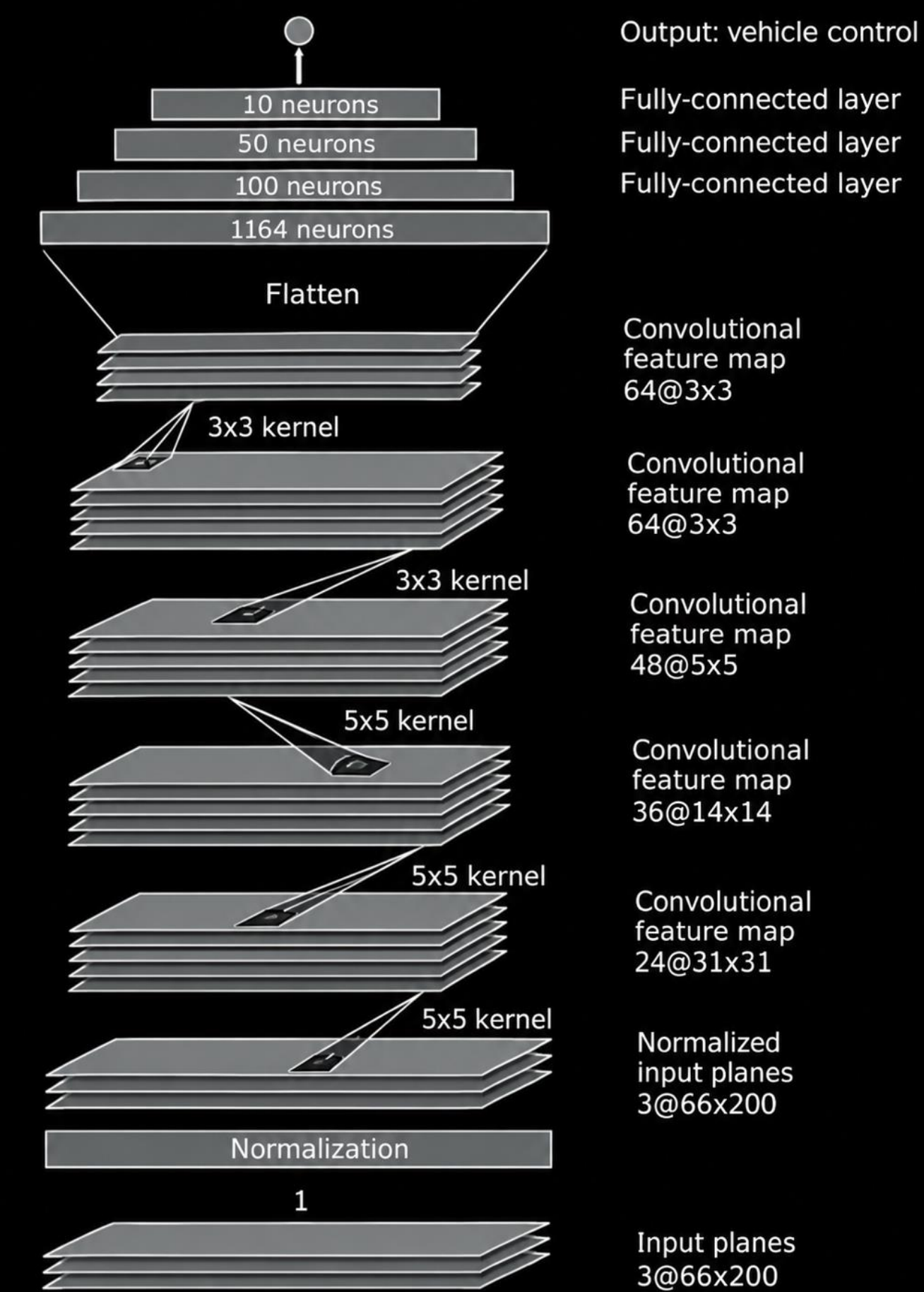


- Strengths:
 - Directly optimizes what matters: Safe-driving behavior
 - Global (End-to-end) optimization
 - Learns implicit representations
 - Data-driven and scalable
 - Simplicity, without intermediate modules
- Weaknesses:
 - Data hunger, requires large and diverse datasets
 - Interpretability & transparency
 - Requires careful evaluation for safety and long-tail
 - Mostly open-loop training based on human demonstrations
 - Lack of supervisory intermediate signals

From Monolithic to Structured End-to-End Architectures

Vanilla End-to-End

Direct mapping from sensor inputs to driving commands

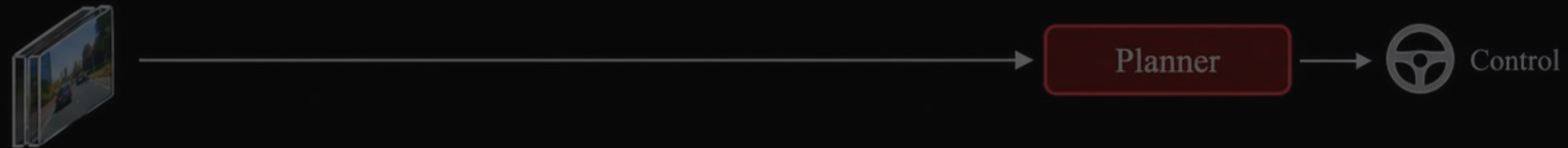


PilotNet 2016
250k parameters

From Monolithic to Structured End-to-End Architectures

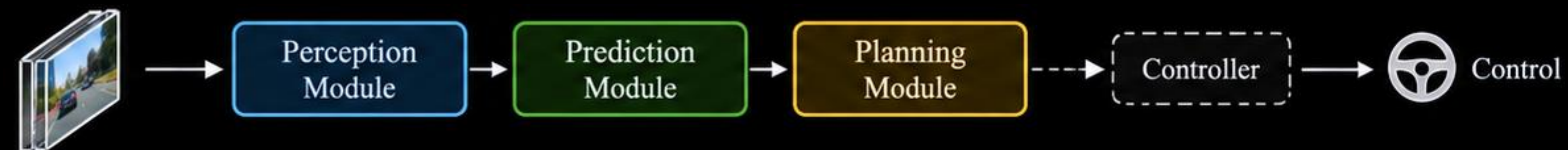
Vanilla End-to-End

Direct mapping from sensor inputs to driving commands



Modular End-to-End Driving with explicit design

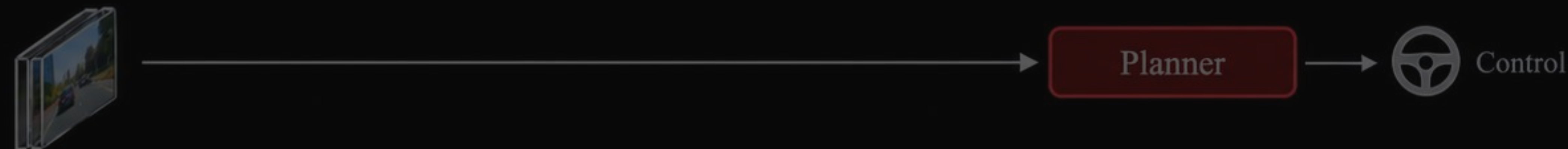
Differentiable pipeline with explicit modules



From Monolithic to Structured End-to-End Architectures

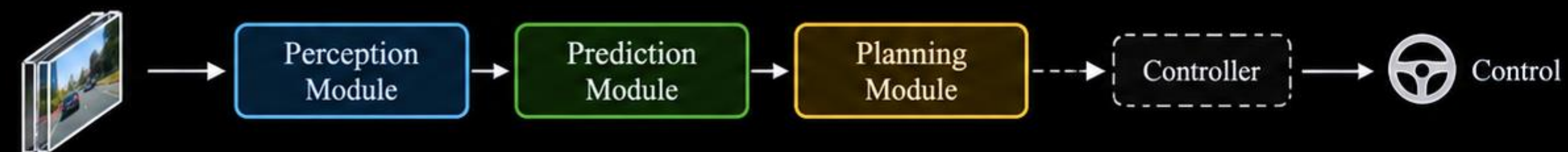
Vanilla End-to-End

Direct mapping from sensor inputs to driving commands



Modular End-to-End Driving with explicit design

Differentiable pipeline with explicit modules



Unprotected left turn



Right turn

From Monolithic to Structured End-to-End Architectures

Vanilla End-to-End

Direct mapping from sensor inputs to driving commands



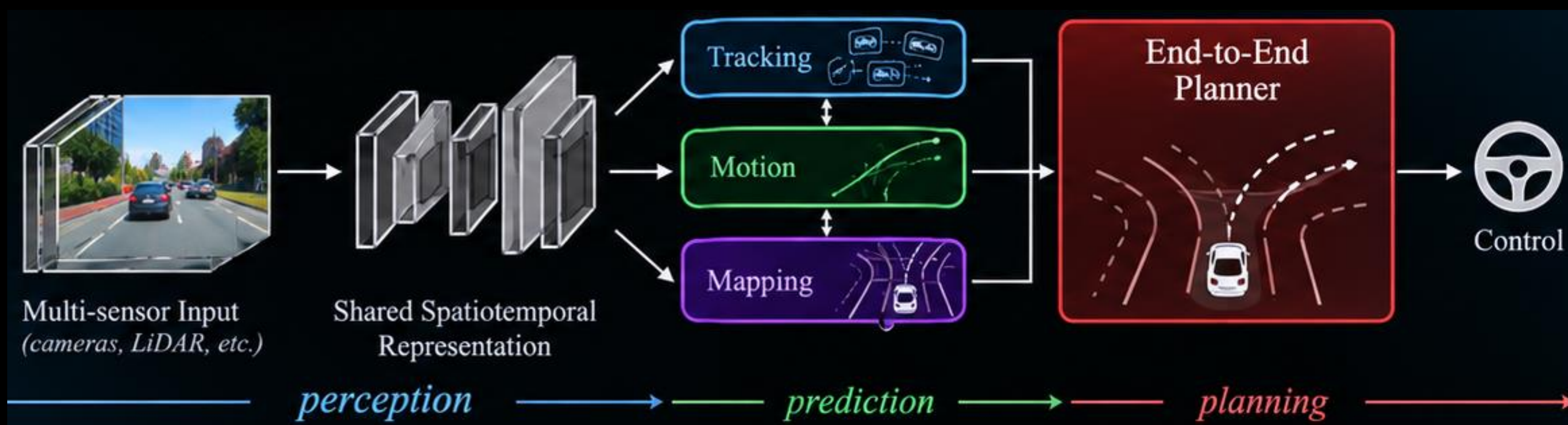
Modular End-to-End Driving with explicit design

Differentiable pipeline with explicit modules



Modern End-to-End Driving

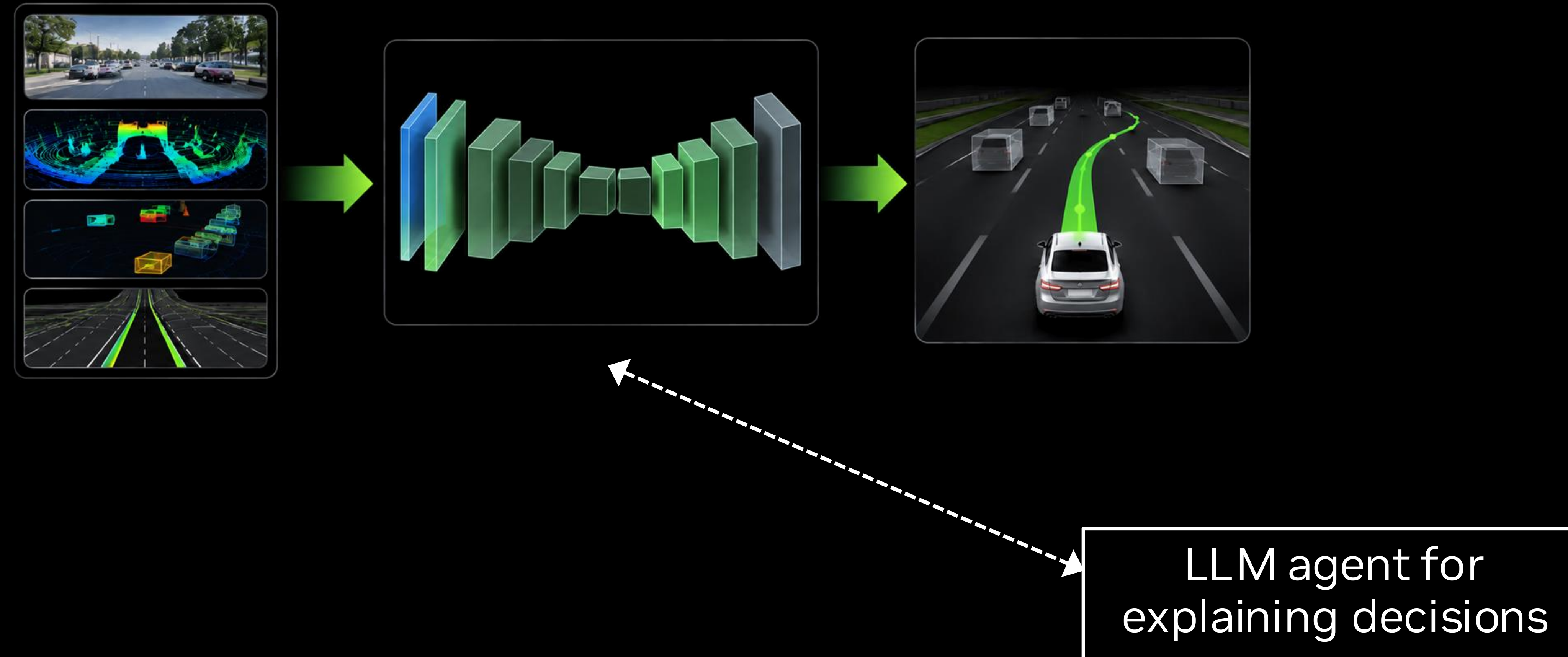
Unified model to jointly understand, predict, and plan



Interpretability Driving with Language

End-to-End Models for Autonomous Driving

Driving with Language



Driving with Language

Explainability of model decisions



I'm proceeding slowly in the current lane; the road is clear, and there are no obstructions or traffic.

Driving with Language

Human-Computer Interaction



Interactive Conversation with Ego Vehicle

Counterfactuals

A hypothetical future that answers: “What would happen if I took a different action?”

Counterfactual Reasoning of
Planning Behaviors



OmniDrive-Agent

If you follow the trajectory [PT, (+3.92, -1.15), (+9.18, -1.59), (+14.12, -1.61), (+20.35, -1.62), (+28.59, -2.22), (+37.89, -2.75)], what would happen?

Following this trajectory would result in **running the red light**, which is against traffic laws and poses a significant safety risk. Additionally, this path would lead to a **collision with a stopped car** located at (+12.2, -0.2) in the same lane. This action is unsafe and violates traffic regulations, highlighting the importance of adhering to traffic signals and maintaining a safe distance from other vehicles.

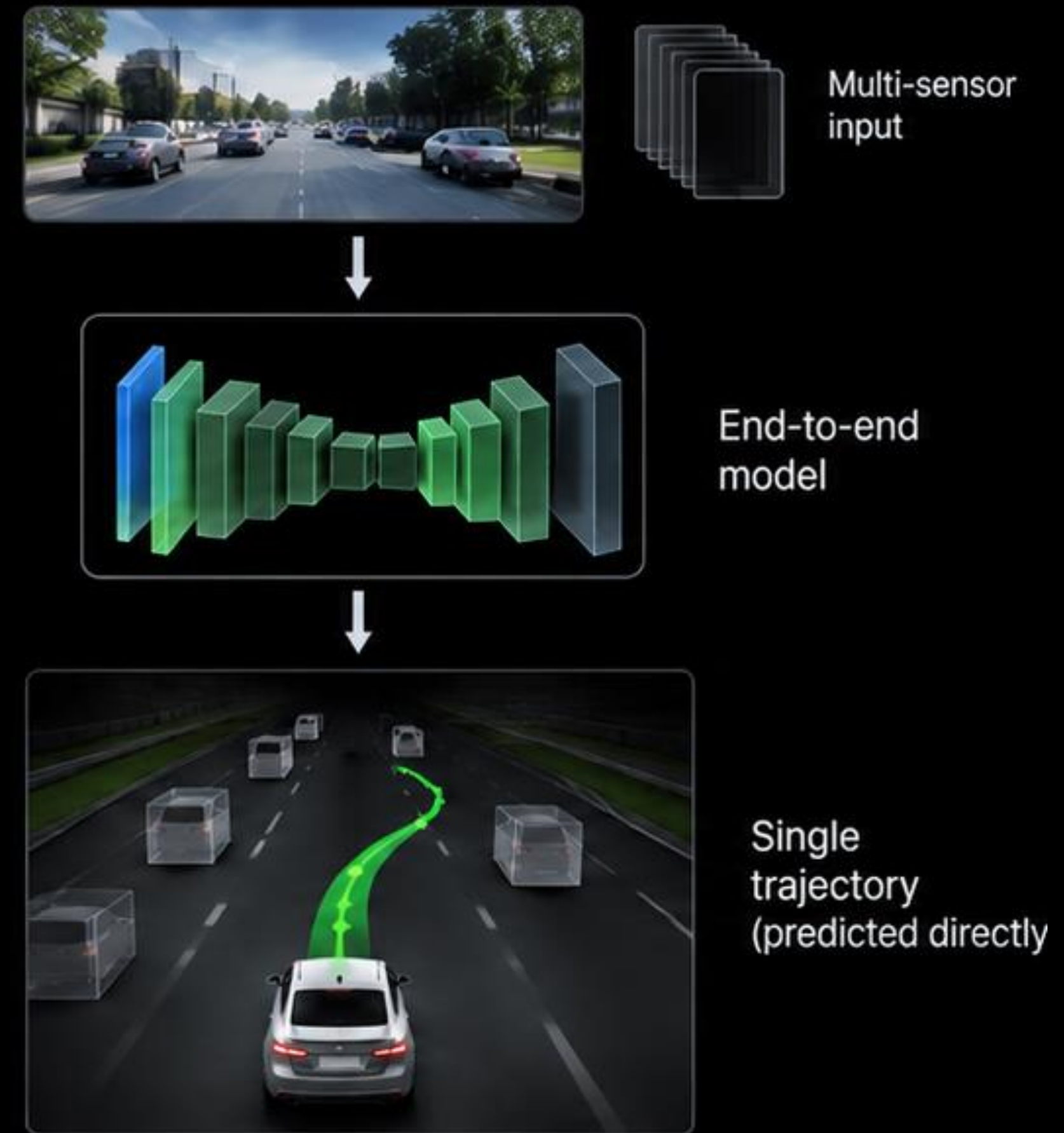
OmniDrive-Agent

If you follow the trajectory [PT, (+3.71, +0.25), (+7.27, +1.03), (+10.76, +2.37), (+13.95, +4.15), (+16.91, +6.37), (+19.67, +9.09)], what would happen?

Following this trajectory would lead to a **left turn**, which is not aligned with the current road layout and traffic signs indicating a right turn. This action would result in **moving out of the drivable area**, violating traffic rules and potentially causing a safety hazard. It's essential to adhere to the road signs and lane markings to ensure safety and compliance with traffic regulations.

End-to-End Models for Autonomous Driving

Basic E2E formulation



Predict a single trajectory from observation

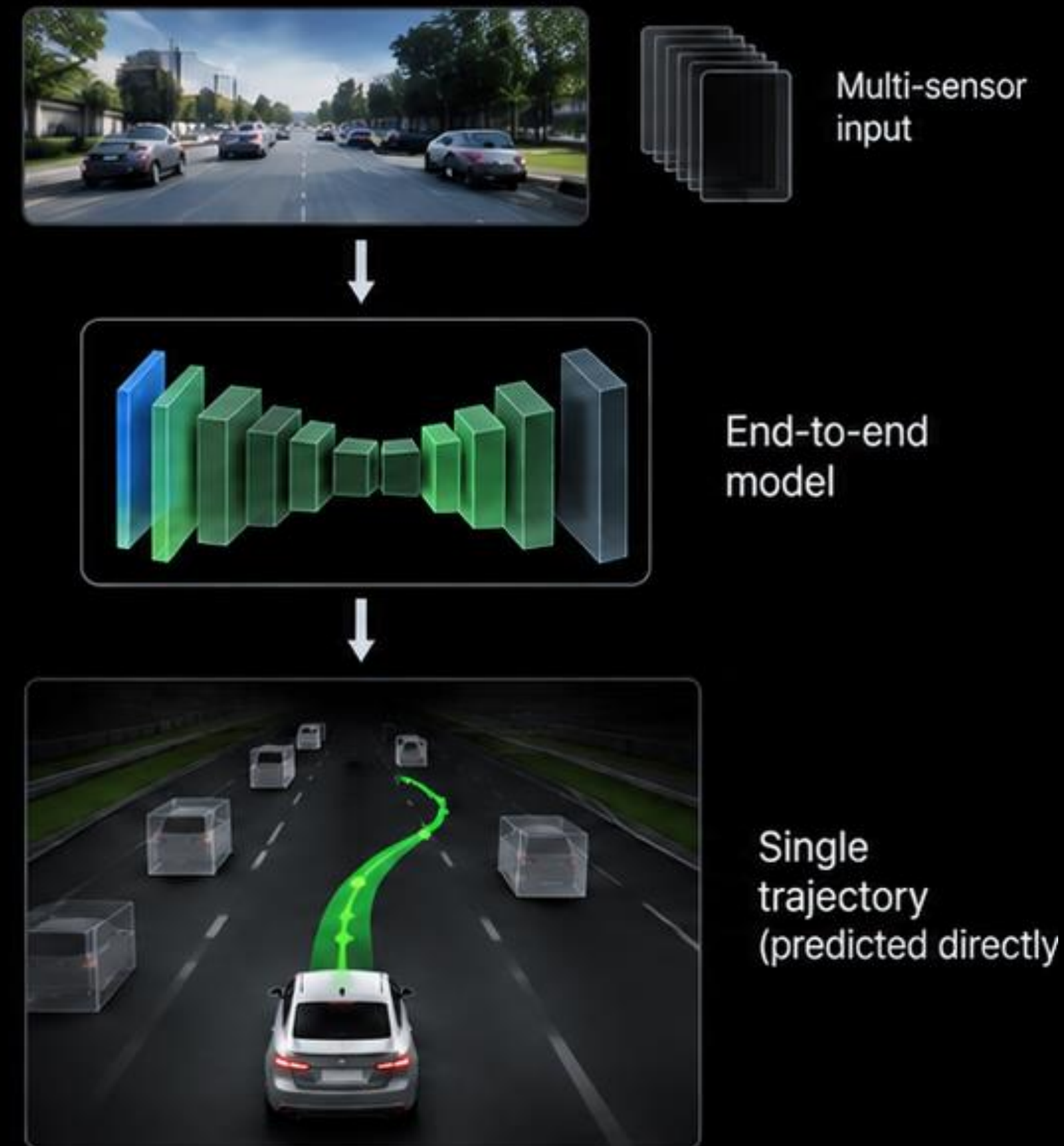
$$\pi_{\theta}(o_t) \rightarrow \tau_t$$

$$\theta^* = \arg \min_{\theta} D(\pi_{\theta}(o_t), \tau_t^{expert})$$

$$P(\tau|o) \neq \delta(\tau - \tau_{expert})$$

End-to-End Models for Autonomous Driving

Basic E2E formulation



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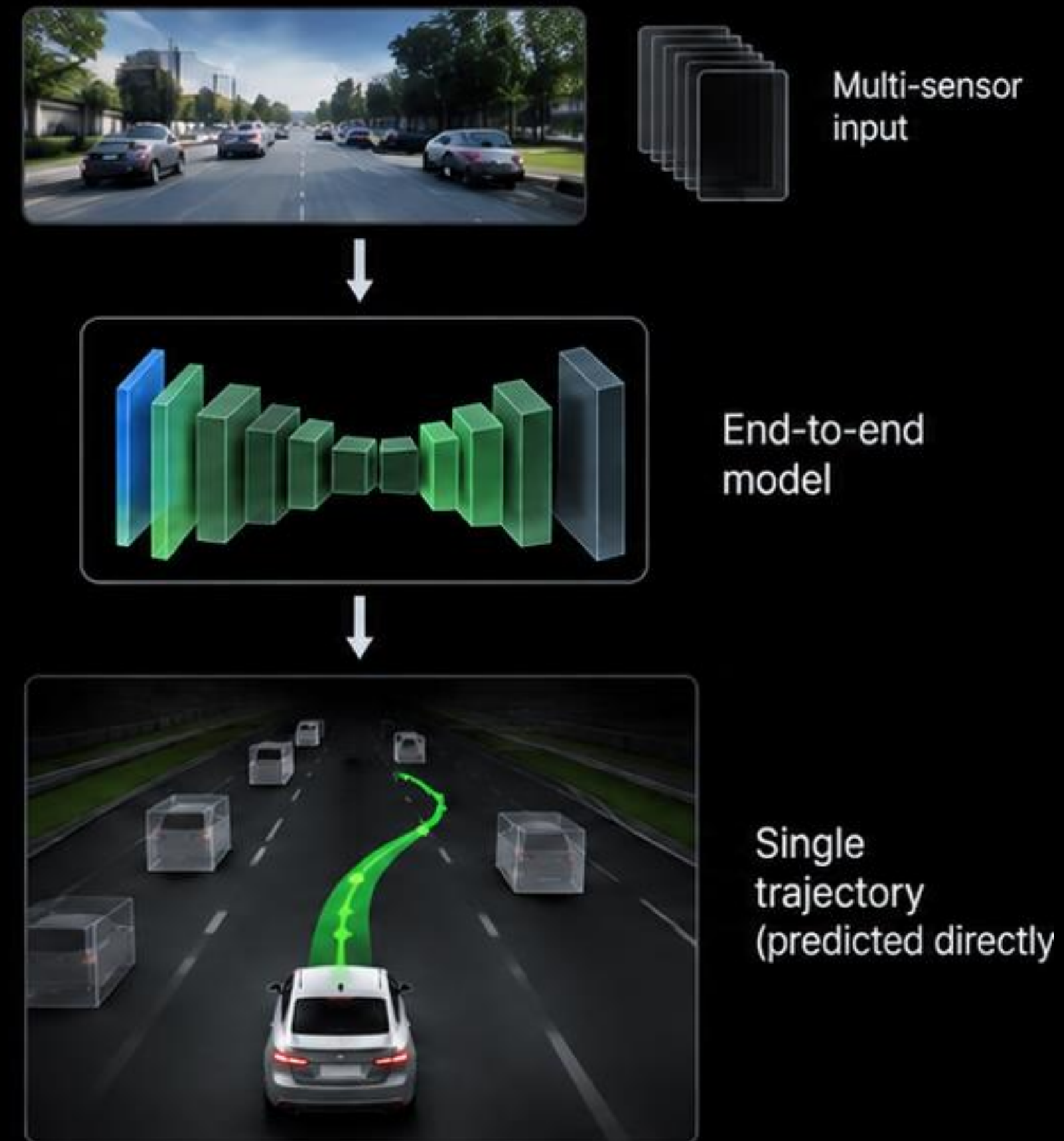
$$P(\tau|o) \neq \delta(\tau - \tau_{expert})$$

Key limitations:

- **Black-Box Nature and Poor Interpretability**
- **Covariate Shift and Error Accumulation:** Small initial prediction errors or states not seen during training can push the vehicle out of its safe distribution, compounding over time into catastrophic policy failure.
- **Imitation Learning Biases:** Models trained via behavioral cloning struggle to distinguish between actions that look human-like in data but are inherently unsafe, failing to capture long-horizon goals like comfort and precise safety.
- **Gradient Conflicts:** Jointly optimizing multi-task objectives (such as matching perception features while simultaneously planning smooth control) often leads to conflicting gradients and sub-optimal training dynamics.
- **Generalization and the Long-Tail Problem:** Rare road scenarios, complex agent interactions, and unexpected obstacles are hard to cover purely through historical expert datasets

End-to-End Models for Autonomous Driving

Next token prediction



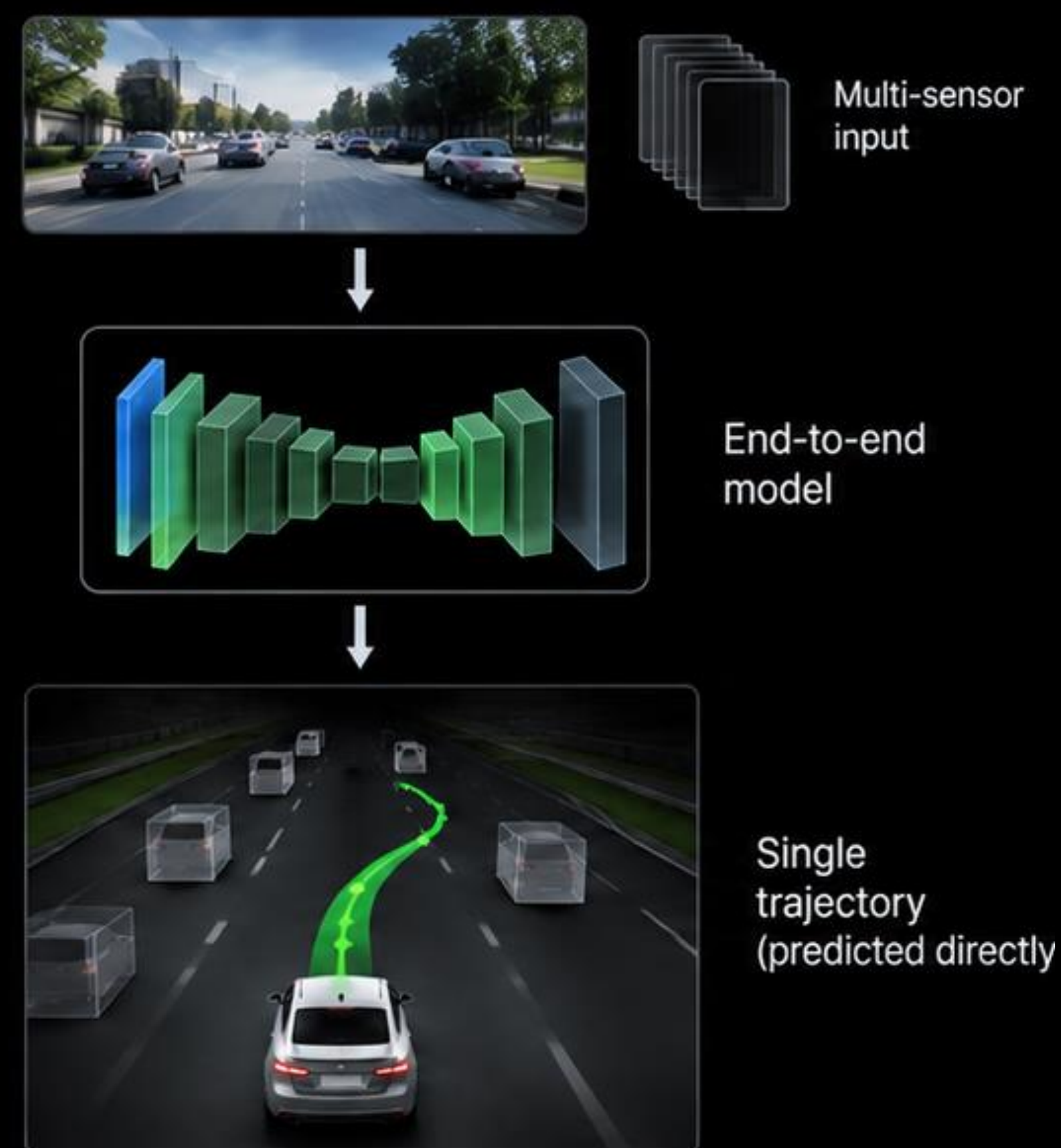
Idea: Generates the future path step-by-step, where the next coordinate depends on the previous coordinate

Predict a single trajectory from observation

$$\pi_{\theta}(o_t) \rightarrow \tau_t$$

End-to-End Models for Autonomous Driving

Next token prediction



Predict a single trajectory from observation

$$\pi_{\theta}(o_t) \rightarrow \tau_t$$

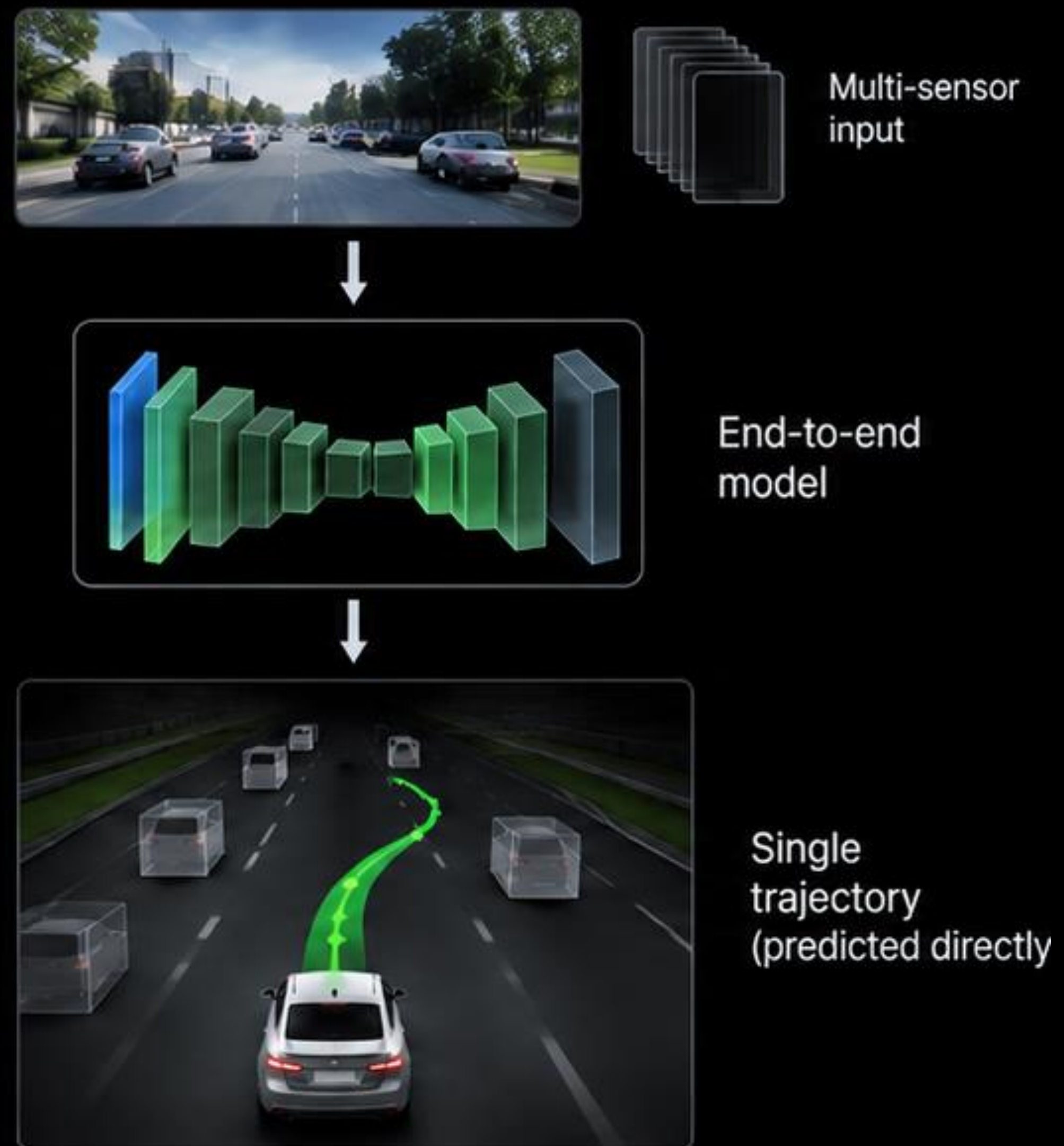
Idea: Generates the future path step-by-step, where the next coordinate depends on the previous coordinate

Key limitations of **autoregressive** prediction (next token prediction)

- **Error Compounding (Exposure Bias):** Autoregressive models predict future trajectory points or actions sequentially. If a small mistake happens early in the sequence, the model conditions future steps on this flawed state, pushing the vehicle out-of-distribution and compounding errors until a policy failure occurs.
- **Spatio-Temporal Misalignment:** Conditioning long-horizon future actions step-by-step on stale or fixed initial sensor observations causes a disconnect between how the environment evolves and how the ego-vehicle plans to react.
- **High Computational Latency:** Generating sequential or reasoning tokens autoregressively is inherently slow. This delay conflicts with the millisecond-level reaction times required for safe real-time driving maneuvers.
- **Probabilistic Hallucinations:** Because they function as statistical next-step guessers trained on human data, they can struggle with rare, long-tail edge cases where probabilistic likelihood fails to enforce rigid physical safety boundaries.

Outline

Direct regression / Autoregressive prediction

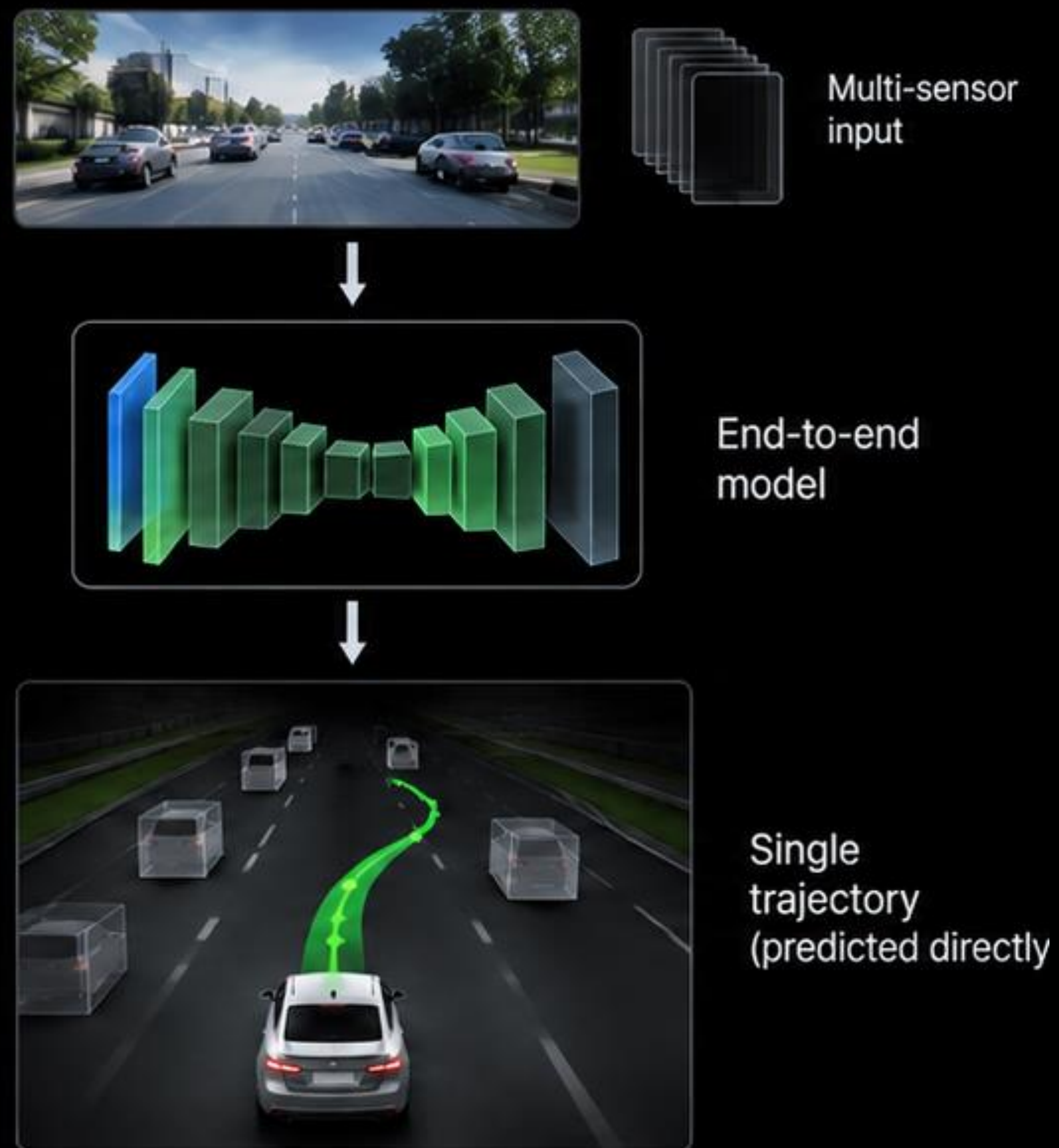


Predict a single trajectory from observation

Outline

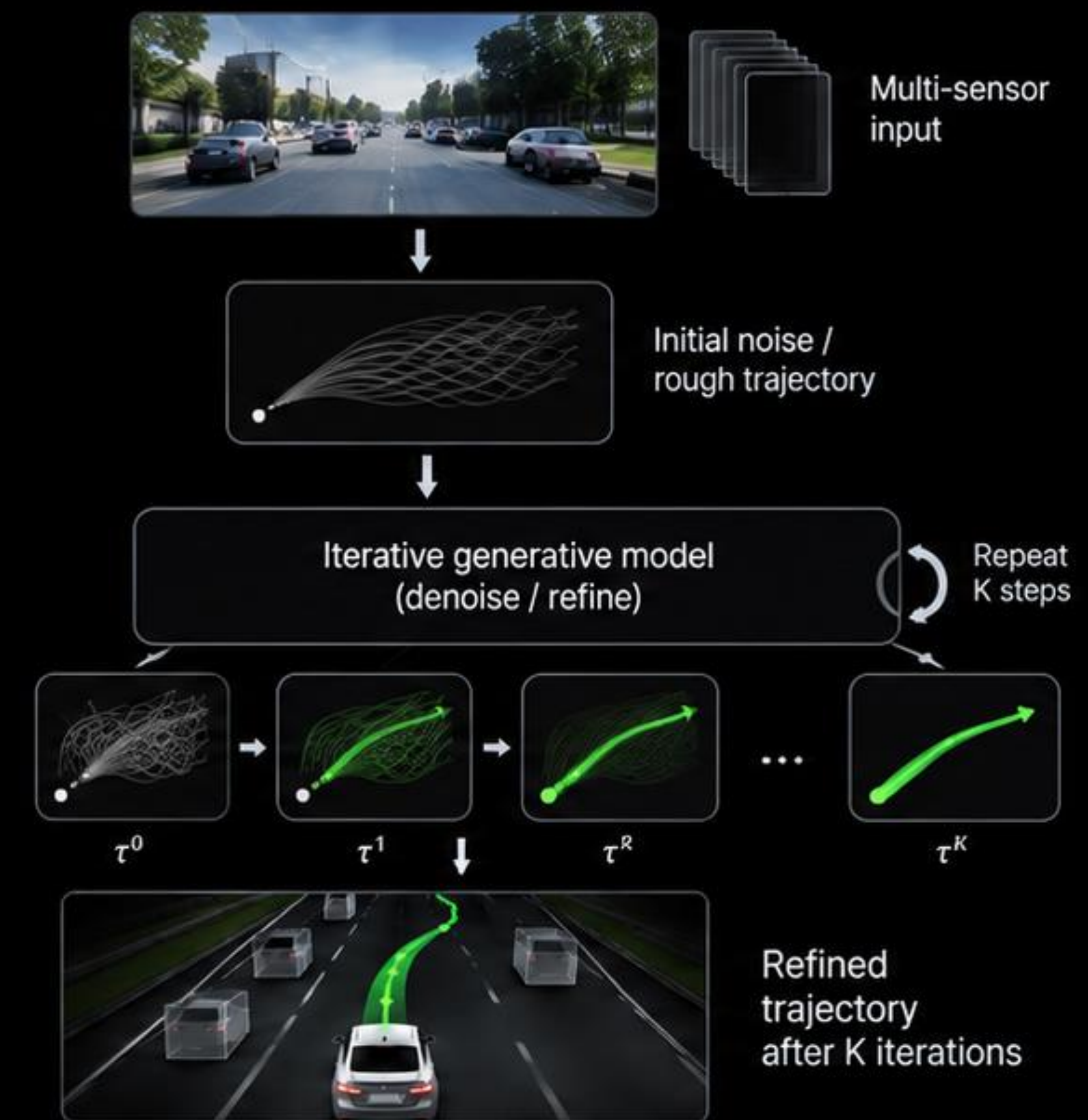
From prediction... ..to deliberation

Direct regression



Predict a single trajectory from observation

Iterative Generative Planning

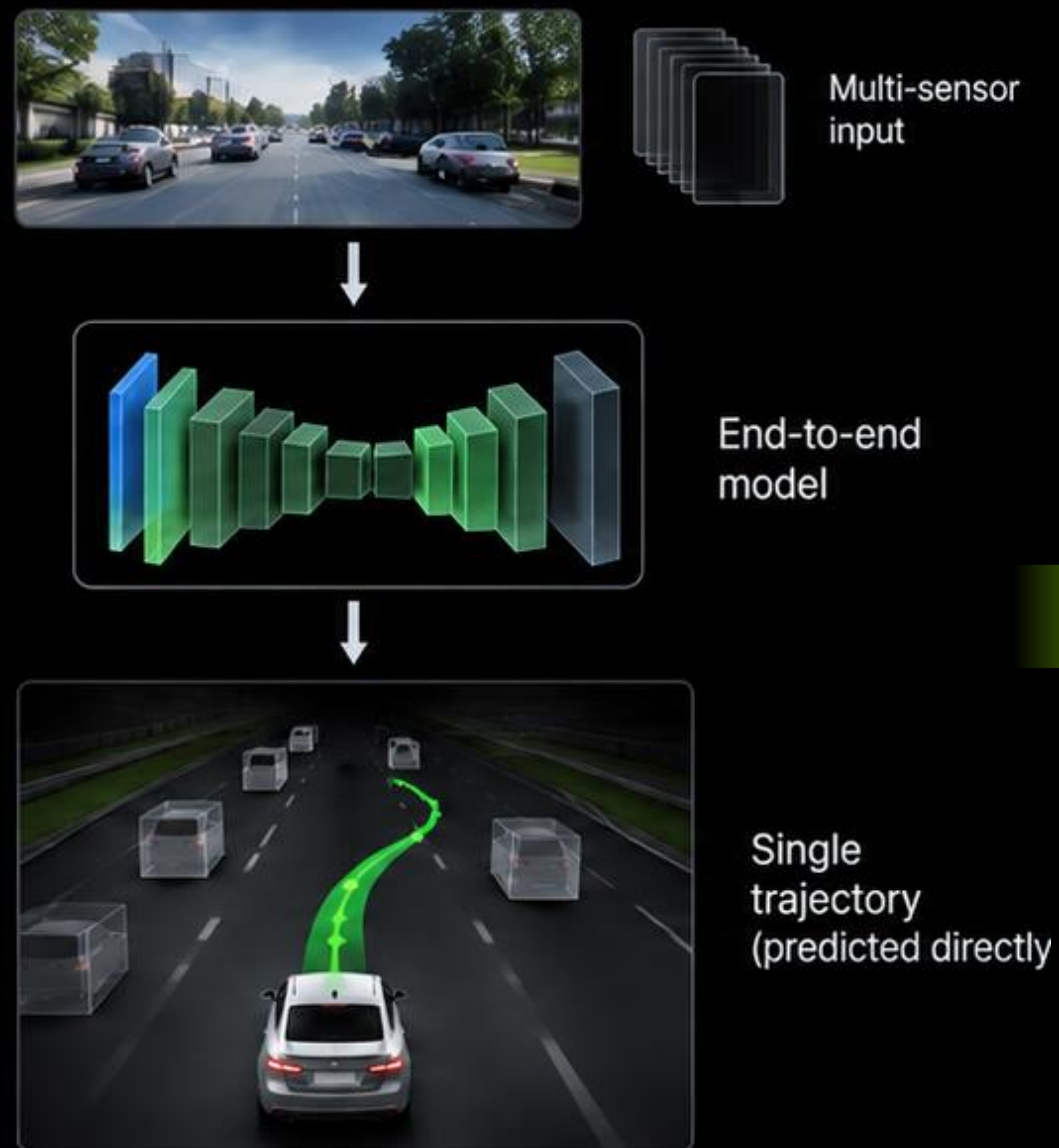


Iterative search and refine trajectories at test-time

Outline

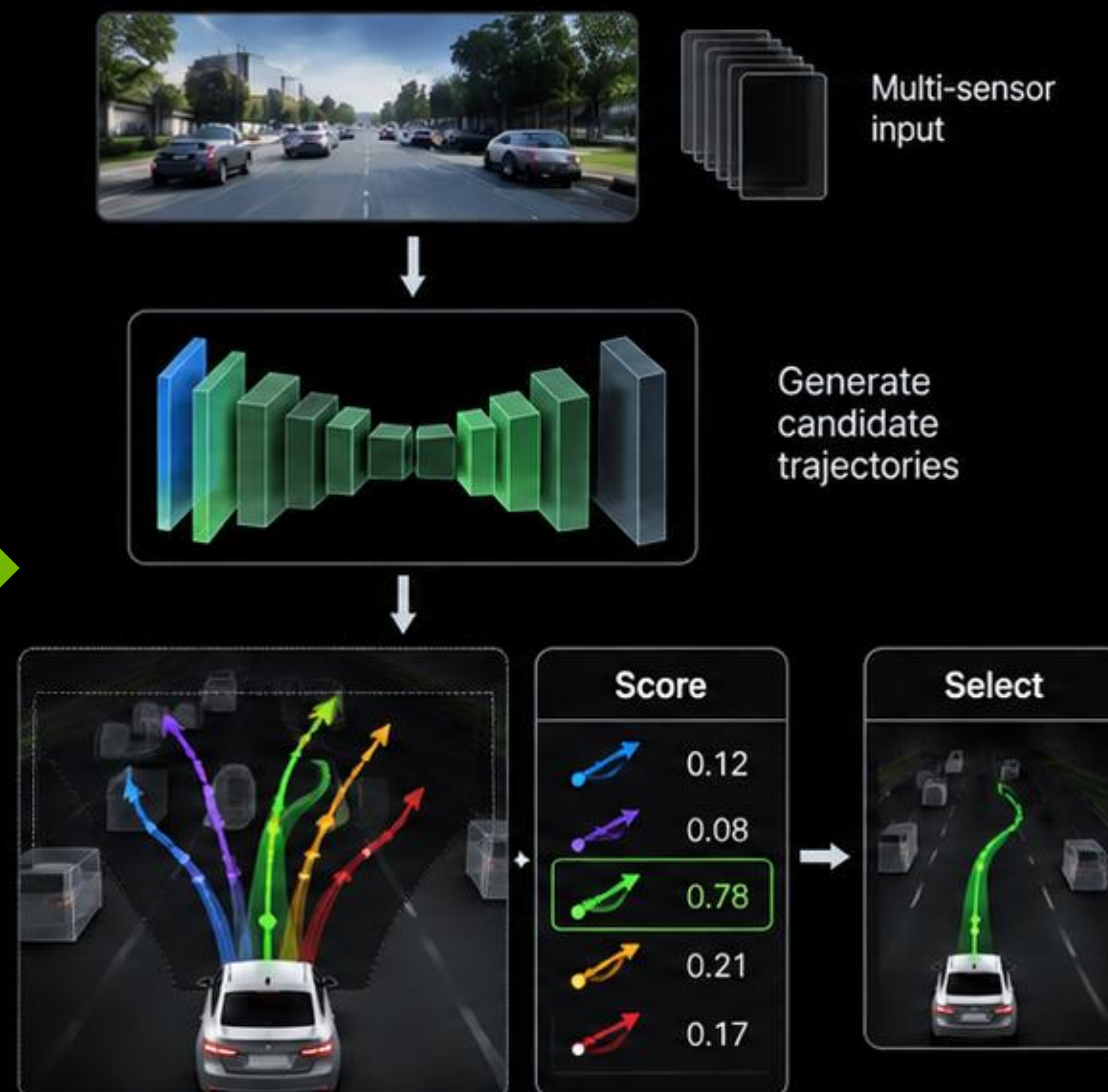
Single-shot prediction → Explicit alternatives → Test-time computation

Direct regression



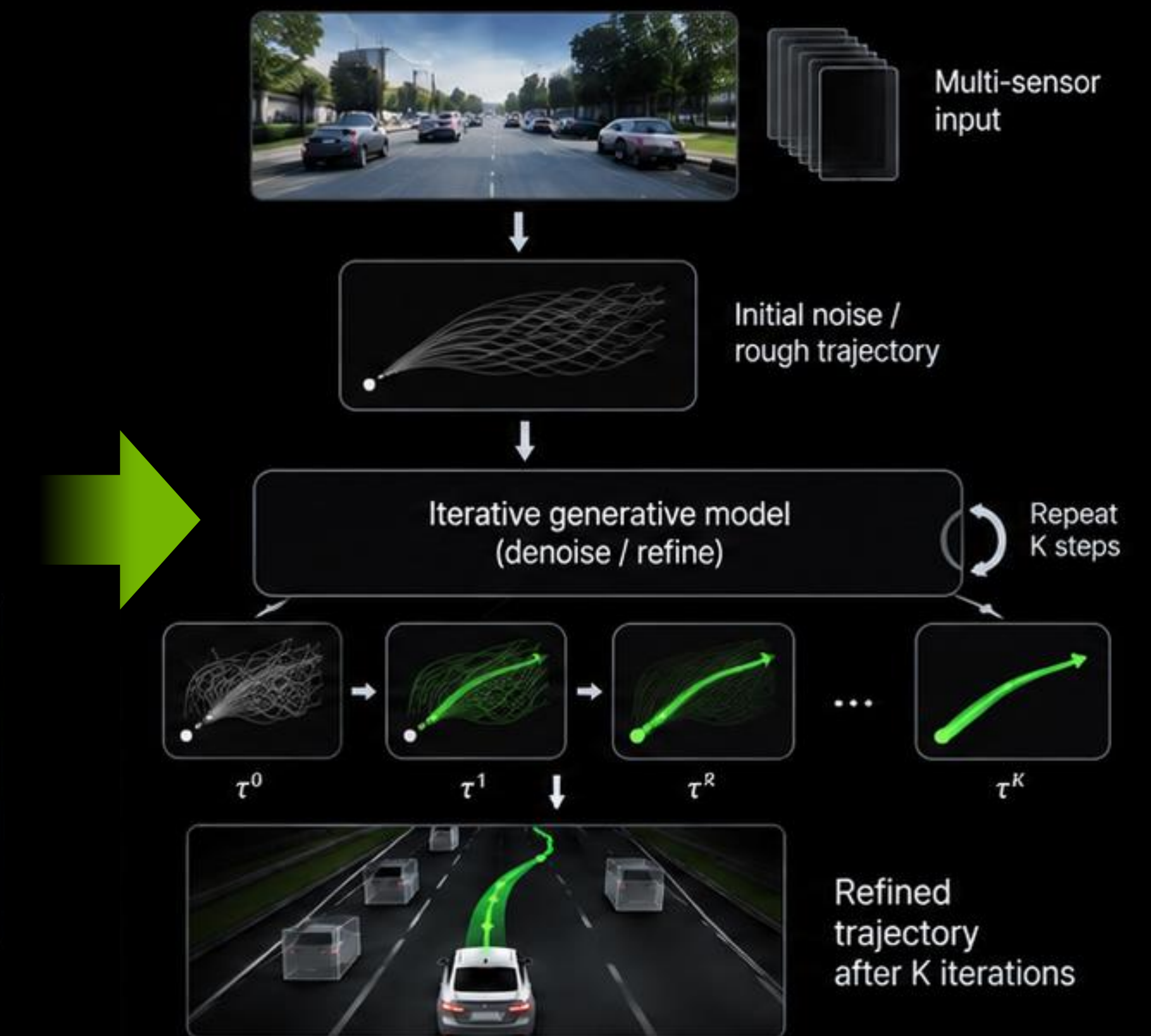
Predict a single trajectory from observation

Generate / score / select



Generate multiple plausible trajectories, score them and select one

Iterative Generative Planning

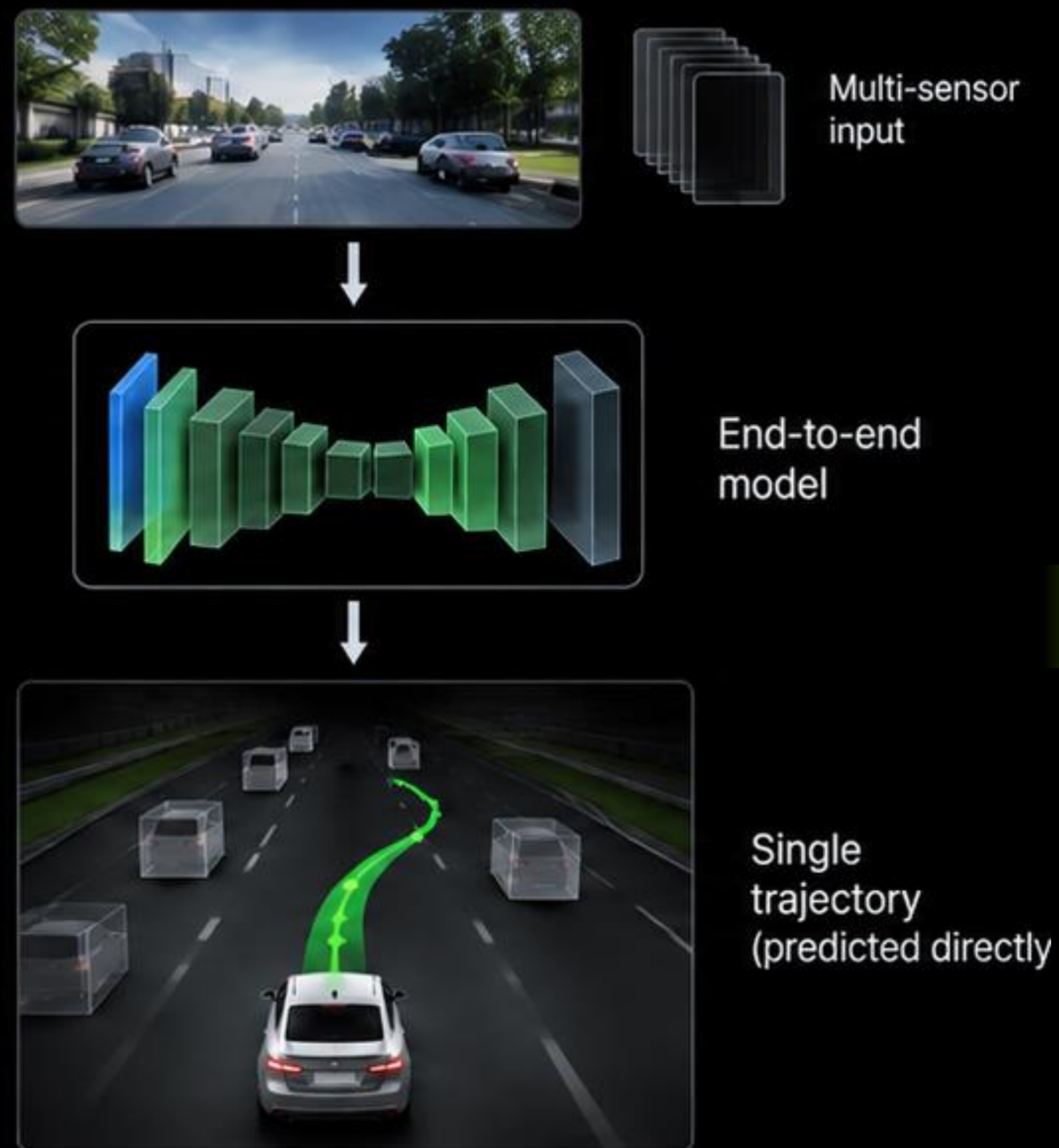


Iterative search and refine trajectories at test-time

Outline

Single-shot prediction → Explicit alternatives → Test-time computation

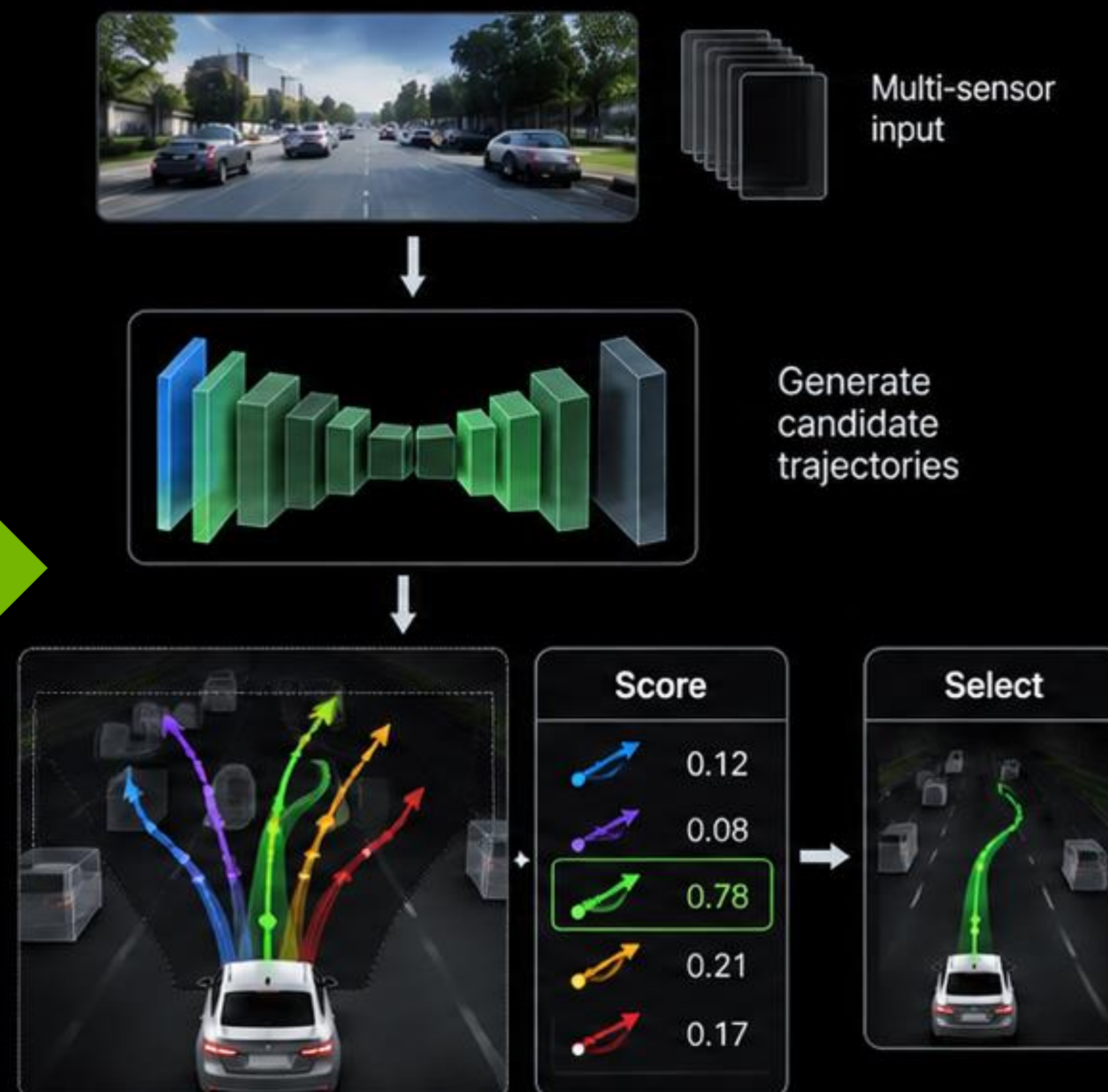
Direct regression



Predict a single trajectory from observation

1st Generation

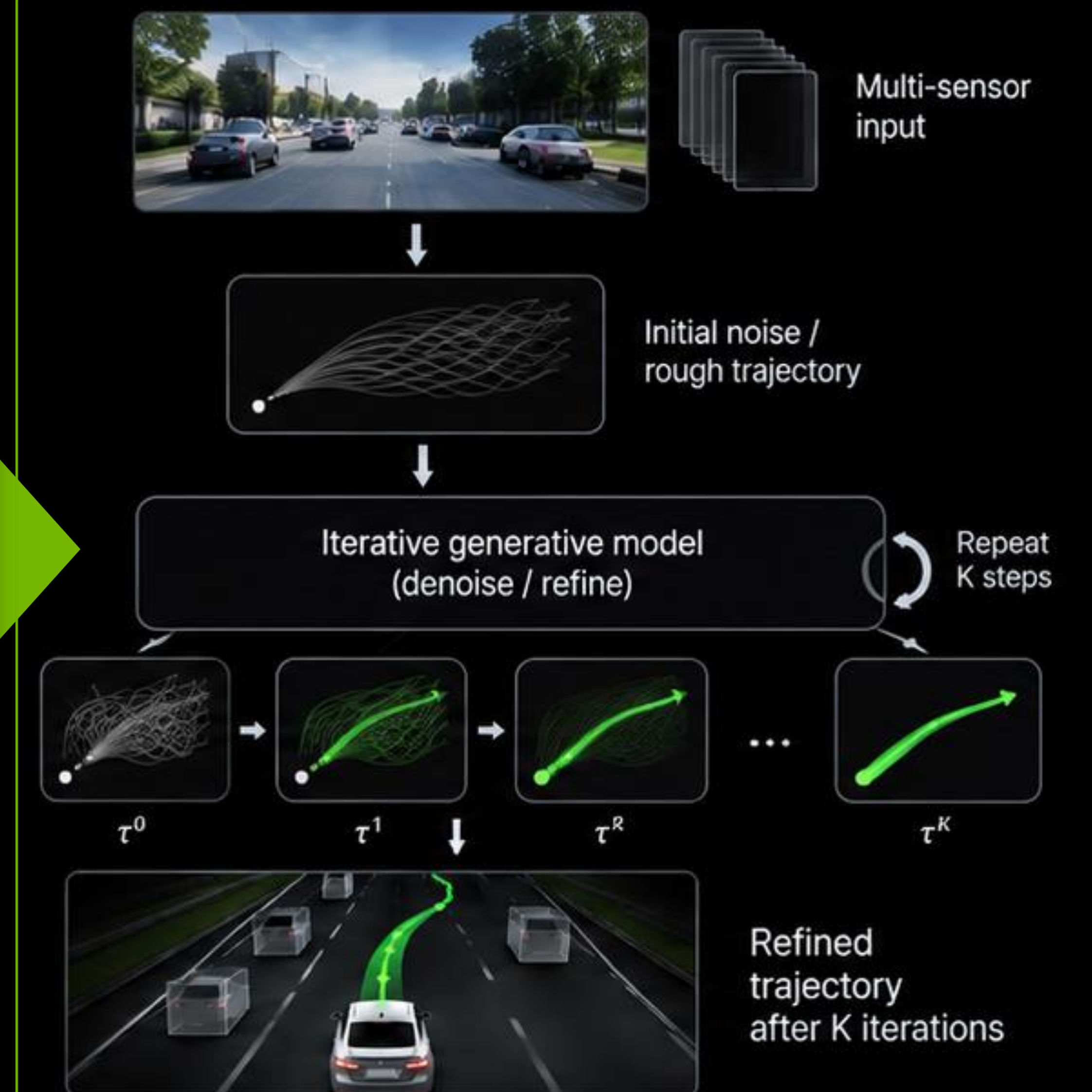
Generate / score / select



Generate multiple plausible trajectories, score them and select one

2nd Generation

Iterative Generative Planning



Iterative search and refine trajectories at test-time

3rd Generation

Outline

References and guidelines

Part	Reference
Direct prediction	Background
Multimodal planning	Hydra-MDP, CVPR 2024 (e2e challenge)
Closing the control loop	Hydra-NeXt, ICCV 2025
Generate & evaluate	GTRS, CVPR 2025 (e2e challenge)
Selection as the bottleneck	DriveSuprim, AAAI 2026
Beyond imitation	ZTRS, ECCV 2026
Deliberation	HAD, ECCV 2026
Edge deliberation	Synthesis

2nd Generation

Generate multiple plausible trajectories, score them and select one

2nd Generation: Multimodal Planners

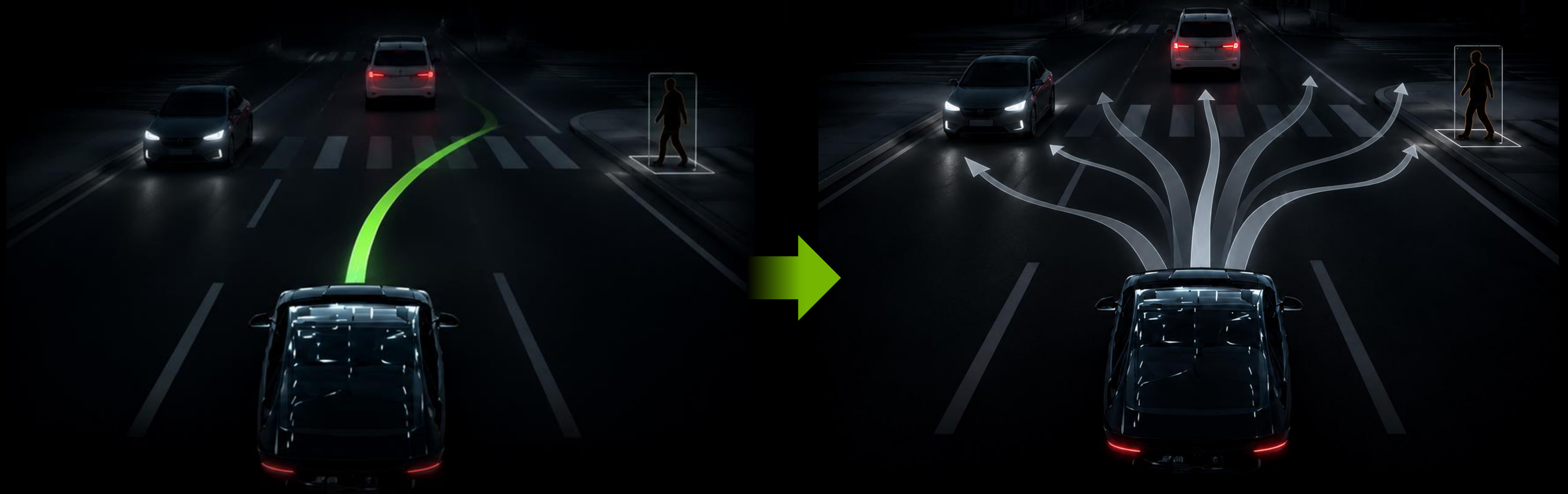
Hydra-MDP models produce multiple feasible trajectories



Single Prediction
One trajectory, one future.

2nd Generation: Multimodal Planners

Hydra-MDP models produce multiple feasible trajectories



Single Prediction
One trajectory, one future.

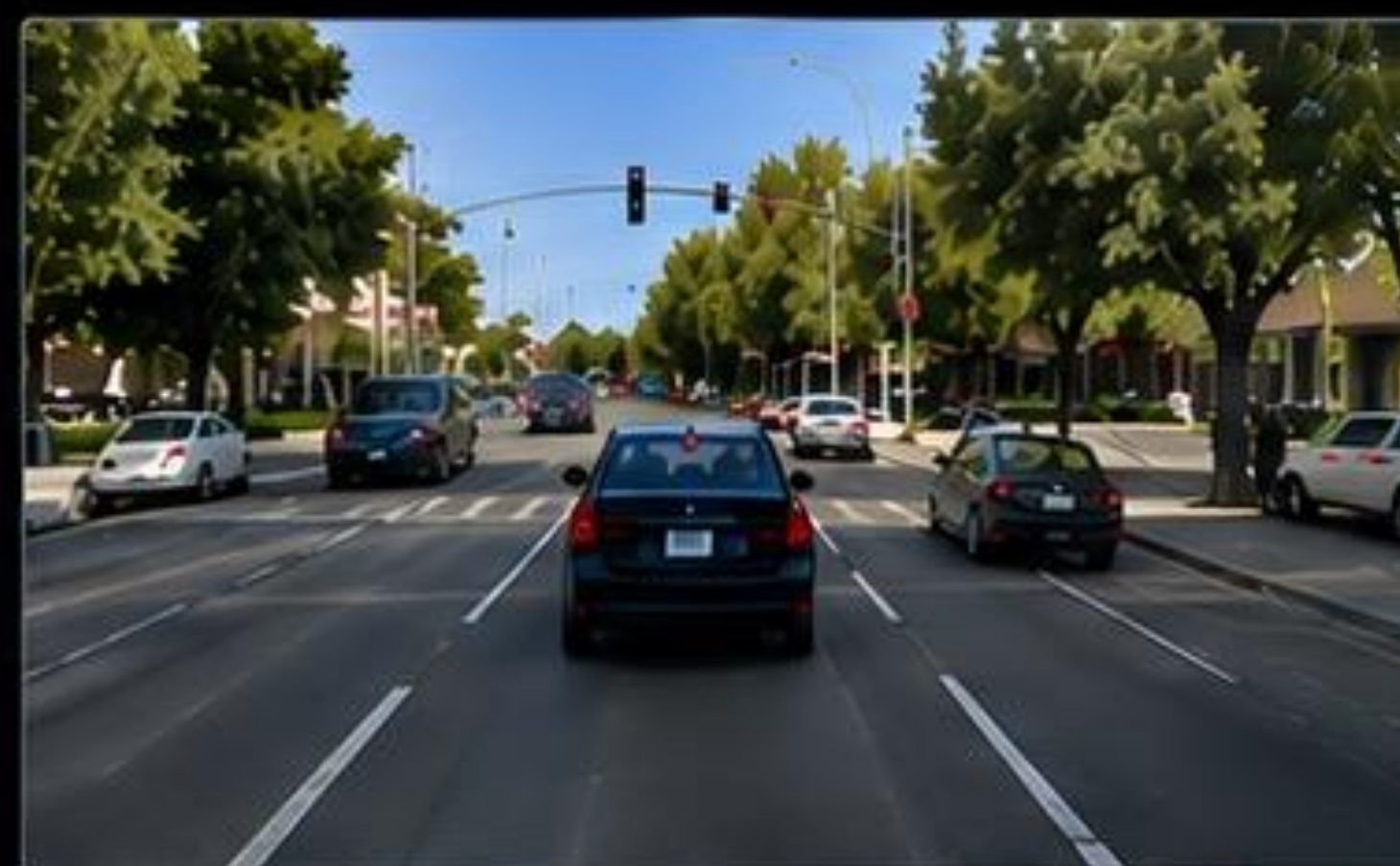
Multimodal Prediction
Multiple plausible futures.

From Regression to Classification for Multi-Modal Planning

Discretize the future into a finite set of modes, classify over them, and select the optimum

Trajectory Regression

Predict a continuous future trajectory



$$\hat{\tau} = f_{\theta}(x_t)$$

$$\hat{\tau} = \{(x_1, y_1), \dots, (x_T, y_T)\} \in \mathbb{R}^{2T}$$

Training (Regression Loss)

$$\mathcal{L}_{reg} = \frac{1}{T} \sum_{t=1}^T \|\hat{p}_t - p_t^*\|_2^2$$

where $p_t = (x_t, y_t)$ and p_t^* is the ground-truth position.

Direct regression in a continuous space
is multi-modal and tends to average
over plausible behaviors.

From Regression to Classification for Multi-Modal Planning

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Direct regression in a continuous space is multi-modal and tends to average over plausible behaviors.

Discretize future
→

Candidate Trajectory Set

Generate a finite set of plausible trajectories (modes)



$$\mathcal{T} = \{\tau_1, \tau_2, \dots, \tau_K\}$$

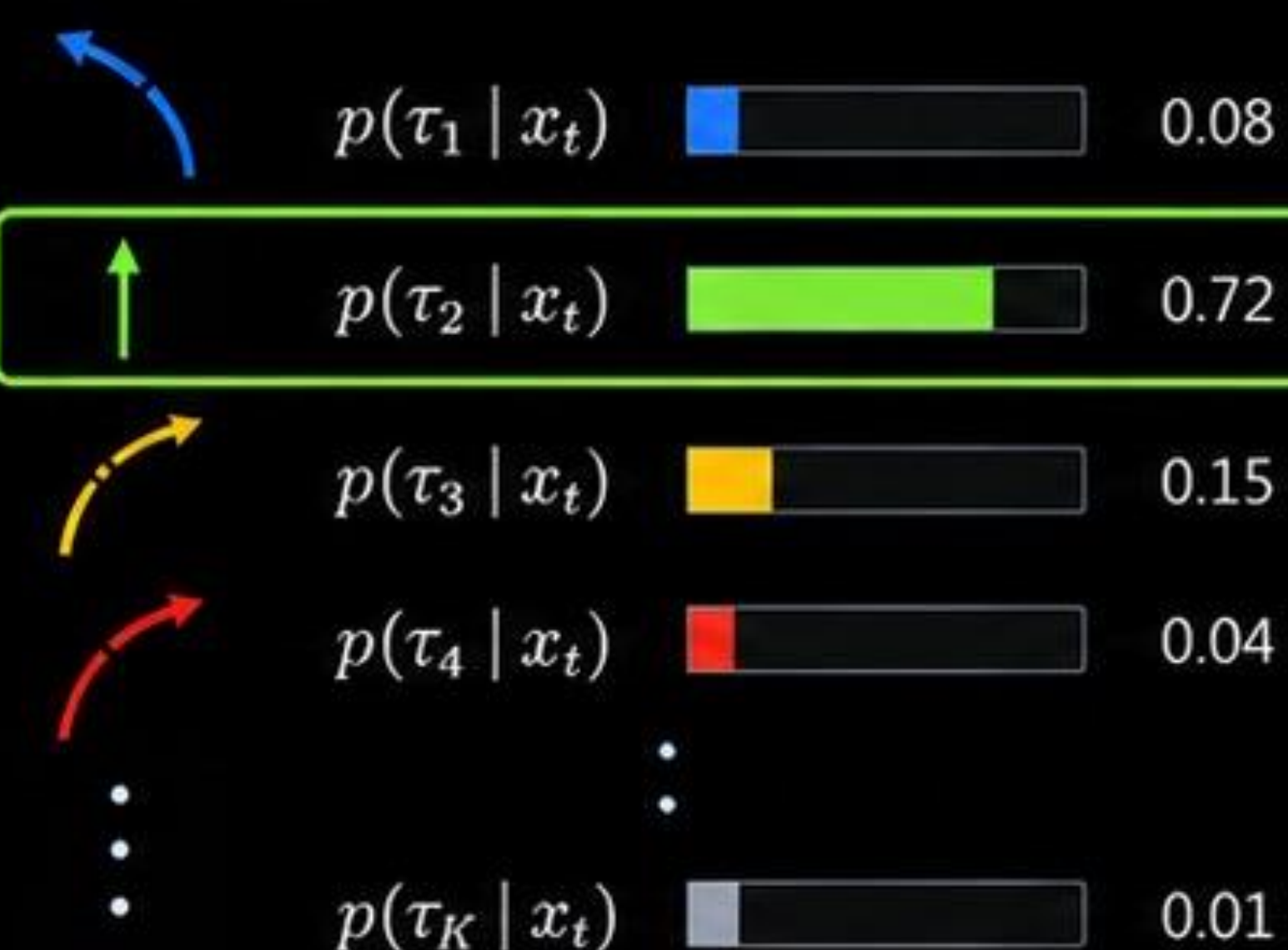


Cover the multi-modal future with a structured, discrete set of candidates.

Classify
→

Classification over Modes

Predict a probability for each candidate



$$p(\tau_i | x_t) = \frac{\exp(z_i)}{\sum_{j=1}^K \exp(z_j)} \quad z = g_{\theta}(x_t, \tau_i)$$

Training (Cross-Entropy Loss)

$$\mathcal{L}_{CE} = - \sum_{i=1}^K y_i \log p(\tau_i | x_t)$$

where y_i is the one-hot label of the ground-truth mode.

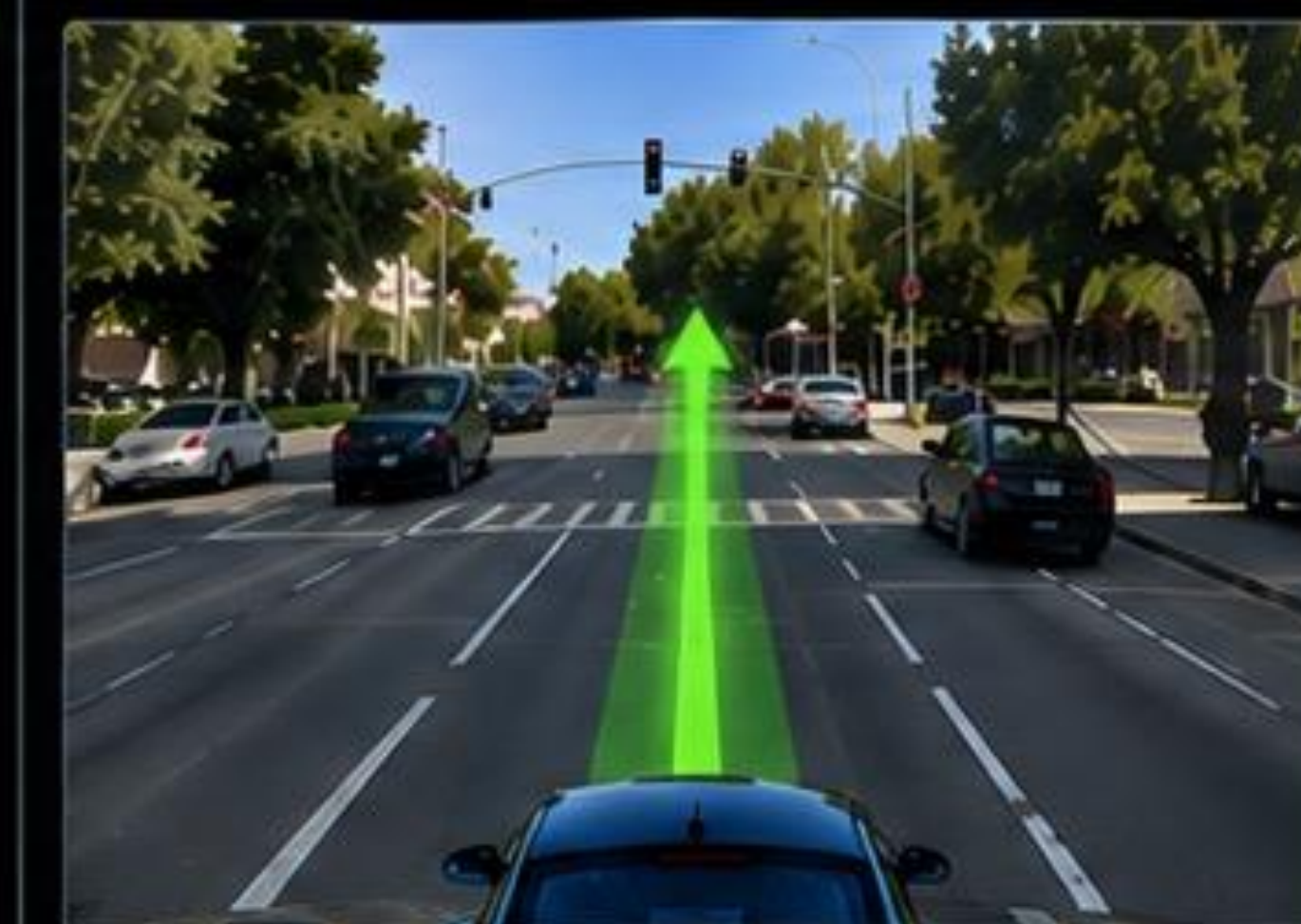
Select (argmax)
→

Select the Optimal Mode

Choose the most likely trajectory with argmax

$$\hat{i} = \arg \max_{i \in \{1, \dots, K\}} p(\tau_i | x_t)$$

$$\hat{\tau} = \tau_{\hat{i}}$$

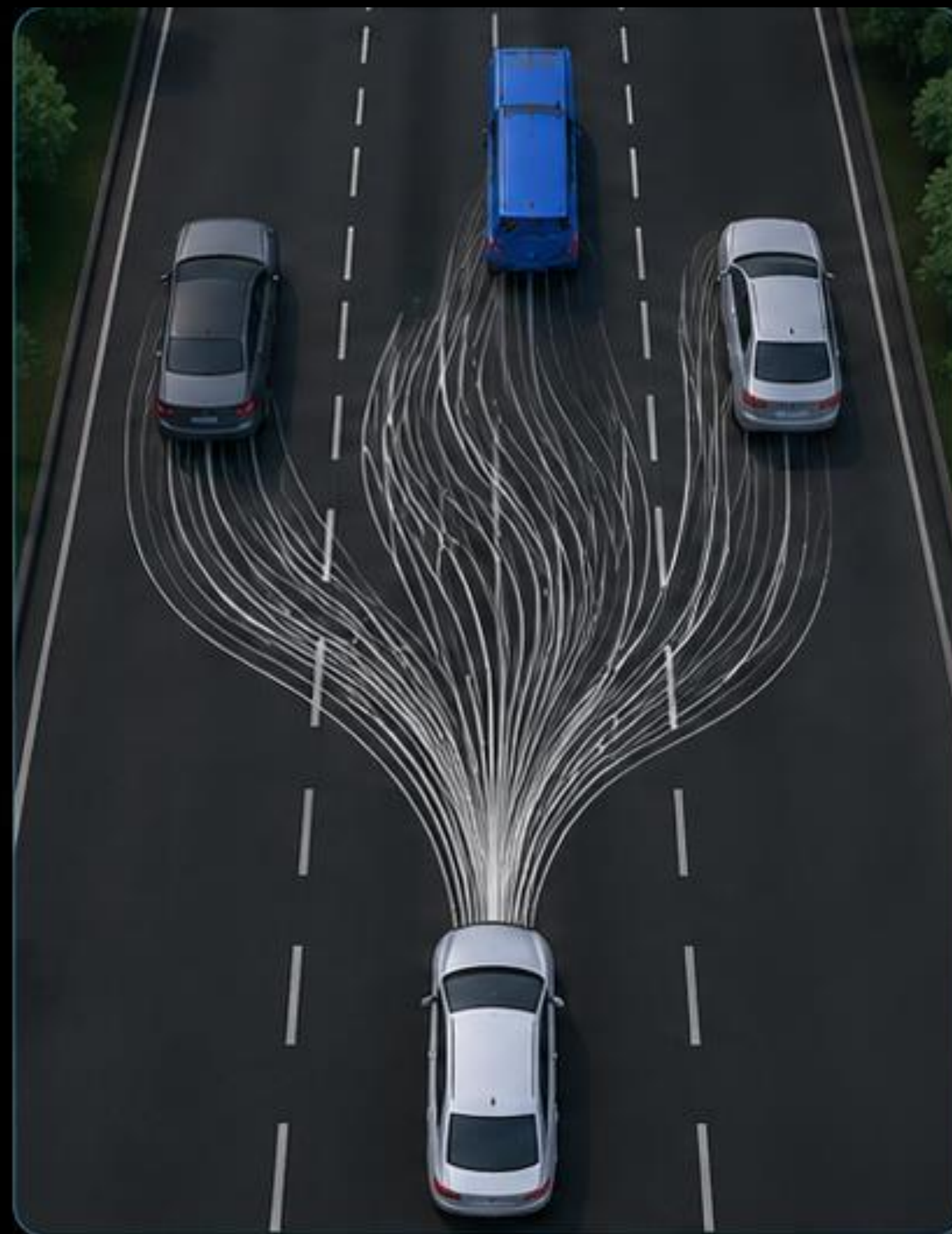


Execute the selected trajectory (or use it to initialize a motion planner).

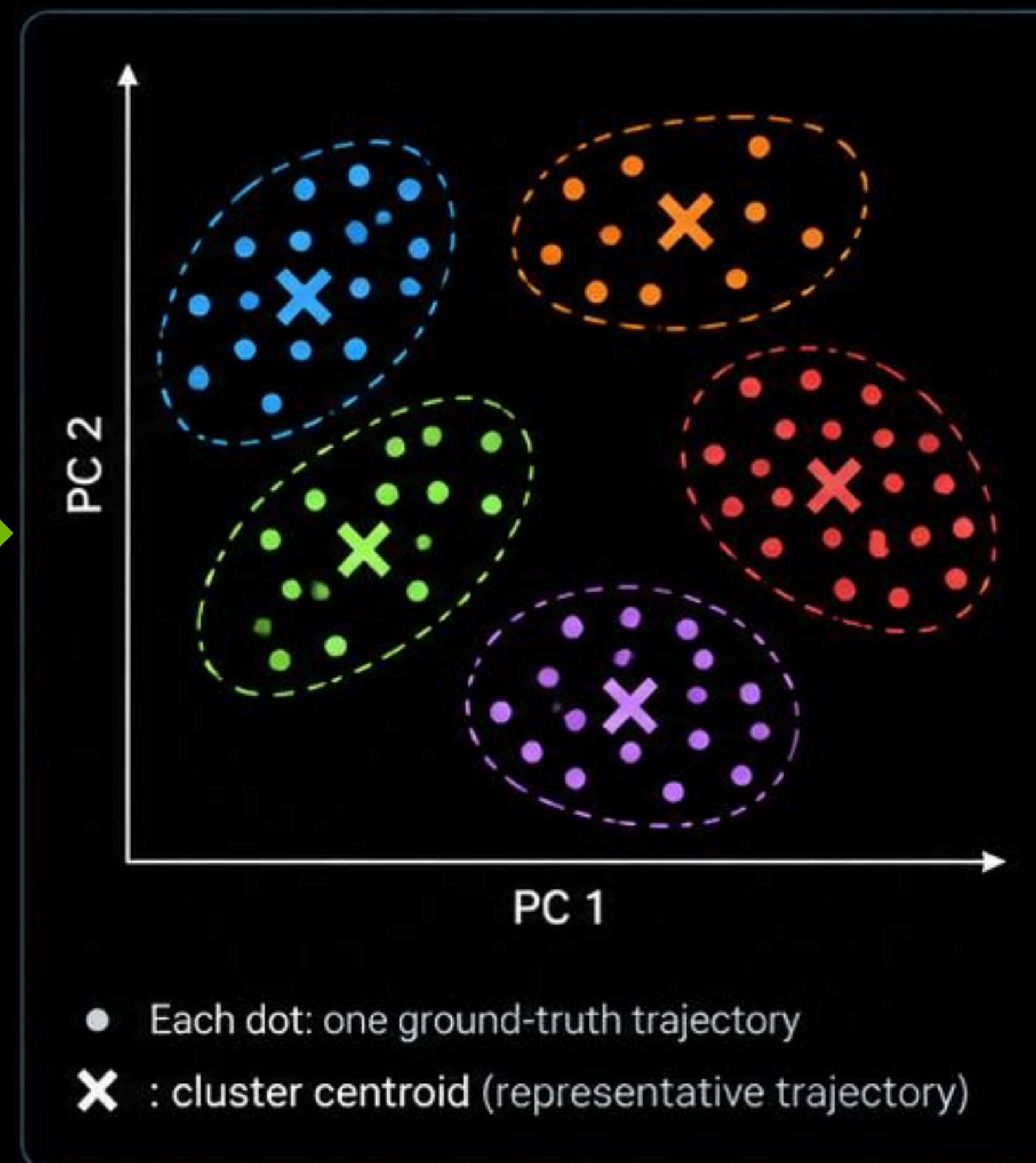
Trajectory Set

Discretize the future into a finite set of modes

GT Trajectories



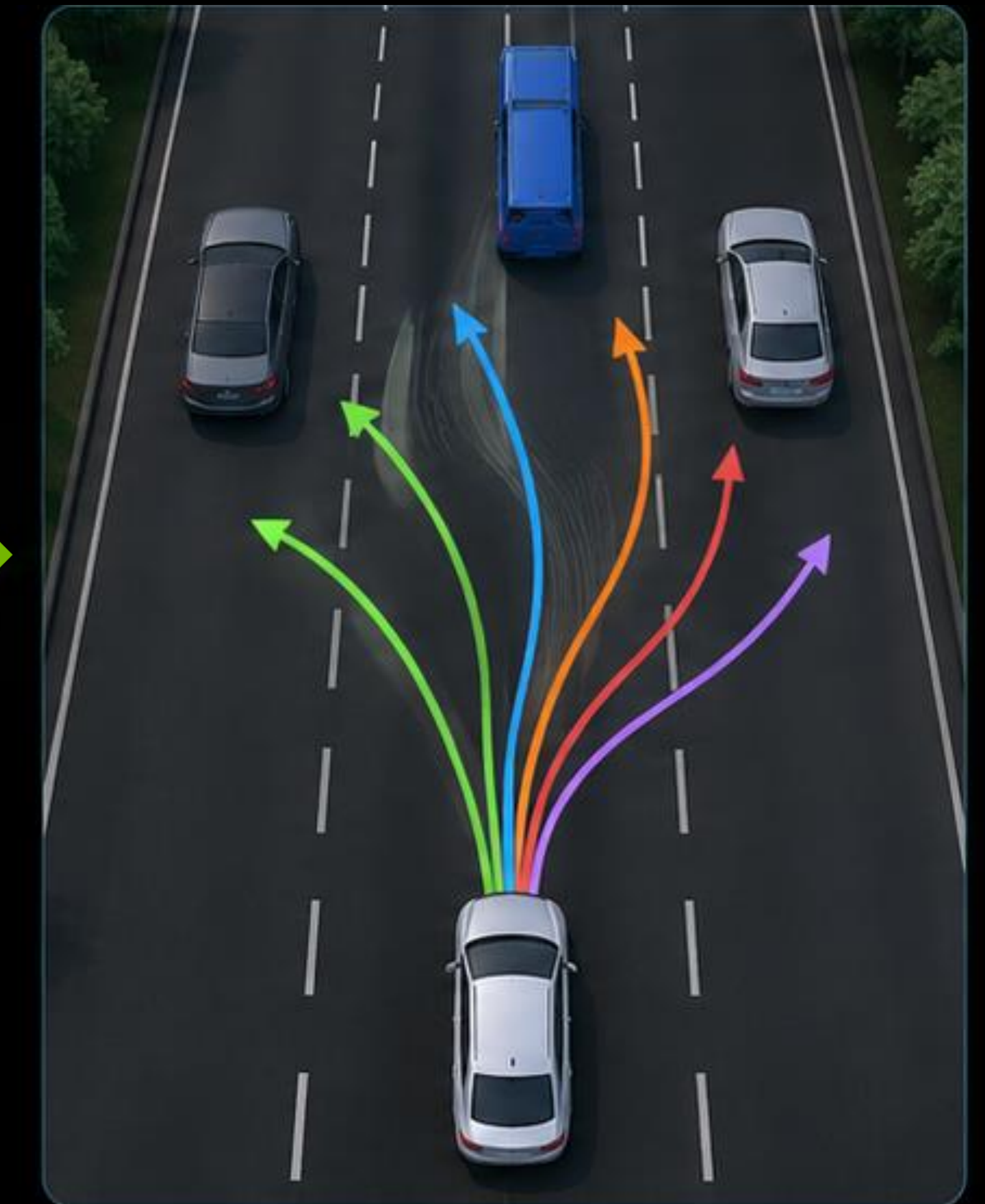
K-means clustering



Centroids

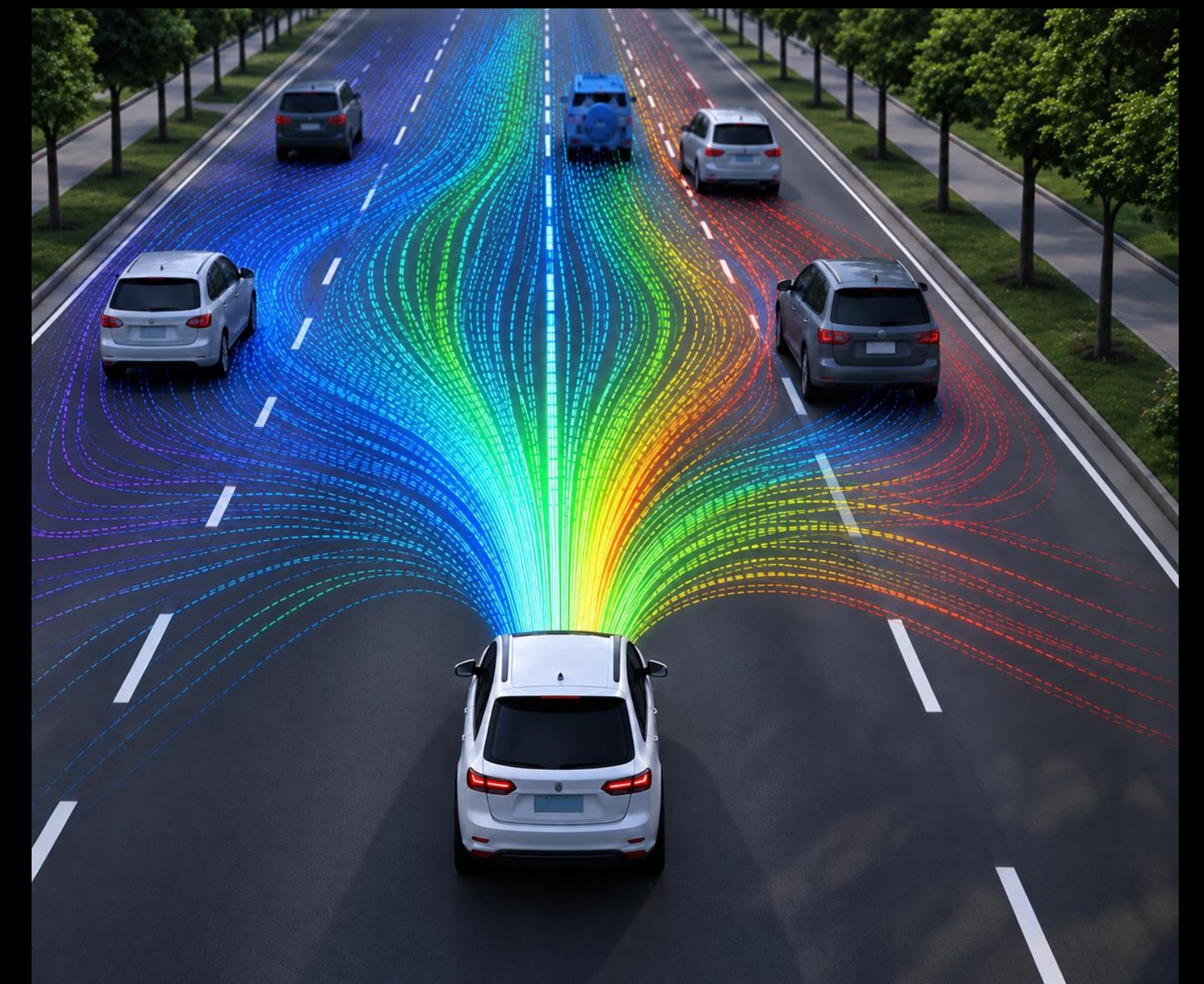
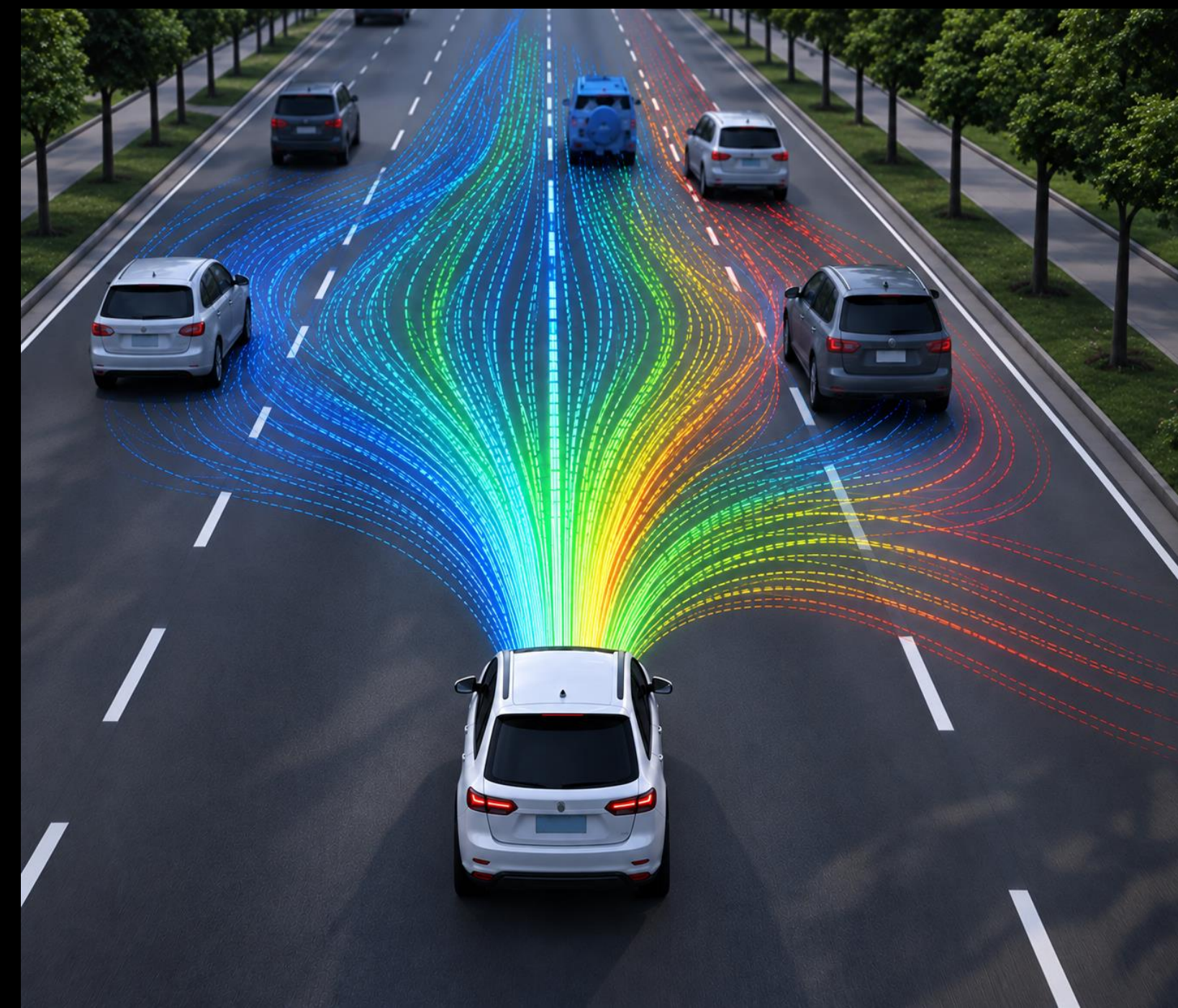
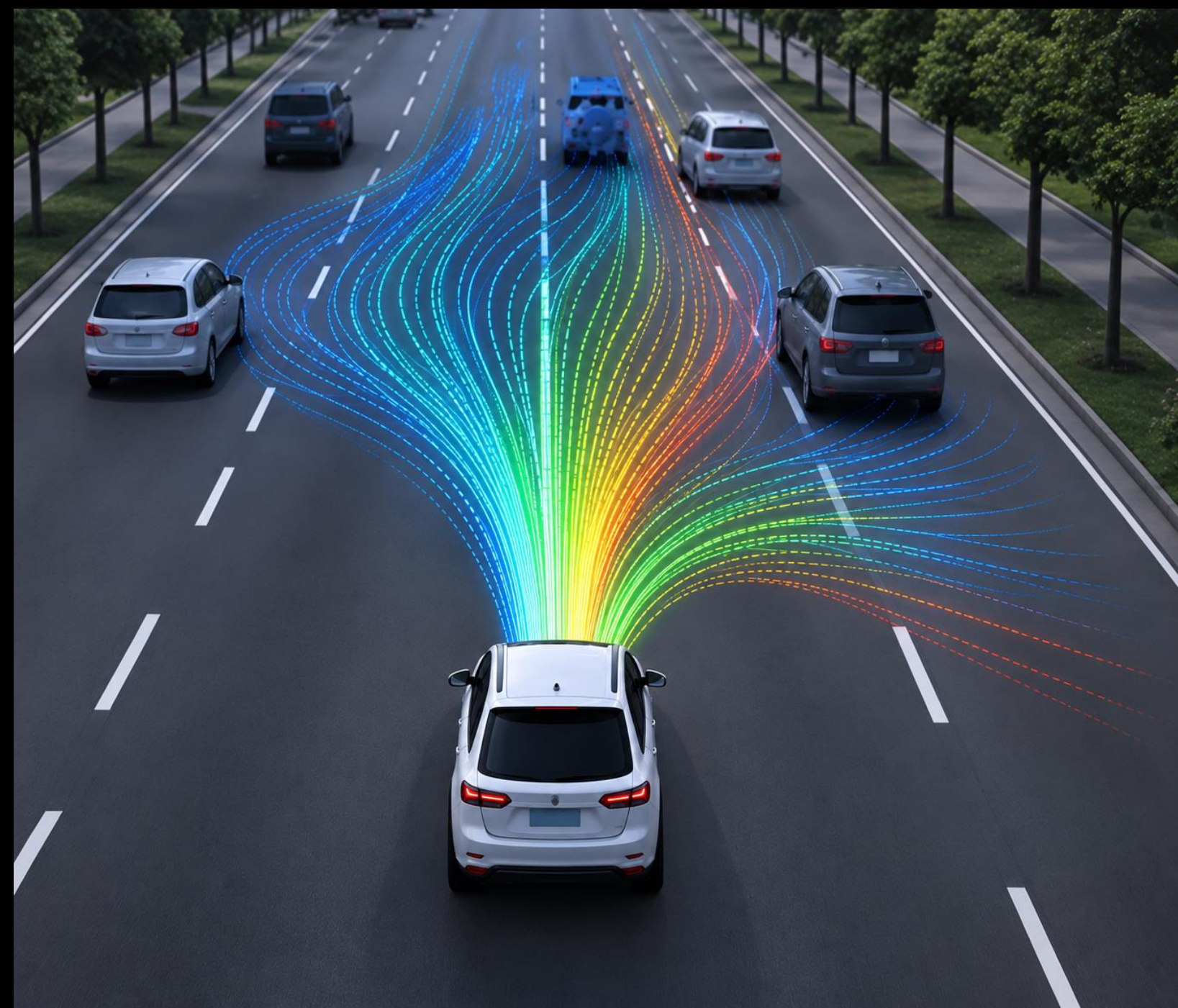


Candidate Trajectories



Trajectory Set

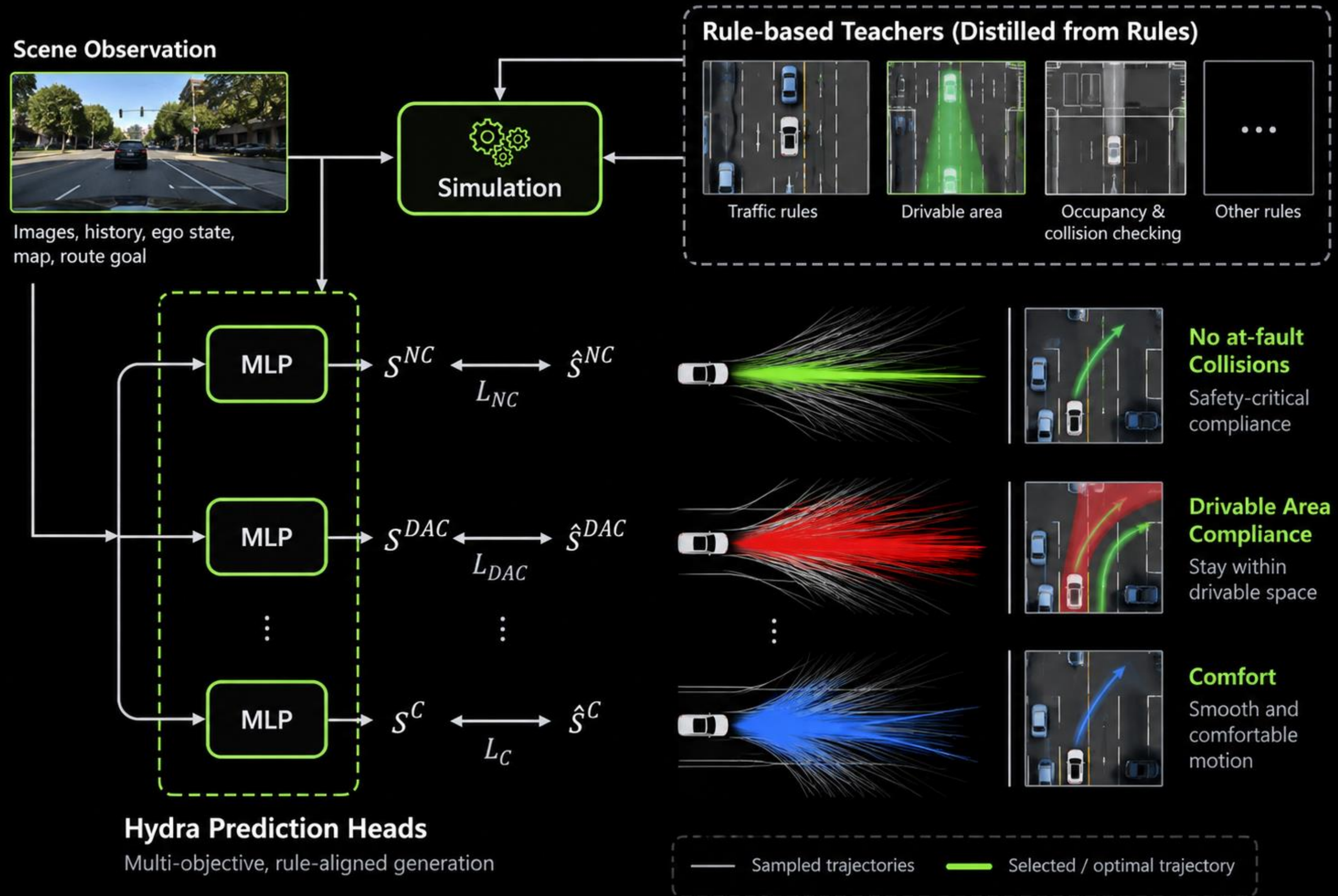
Discretize the future into a finite set of modes



Coverage and diversity are critical

Beyond Imitation Learning

Rule-based distillation based on EPDMS (Extended Predictive Driver Model Score)



Optimal Mode Selection

Three different strategies to aggregate sub-scores

Balanced Aggregation

$$s(\tau_i) = \frac{1}{M} \sum_{m=1}^M s_i^{(m)}$$



- ✓ Equal contribution from all criteria
- ✓ Simple and robust baseline
- ✓ Works well when sub-scores are calibrated

Hand-designed prior (PDM—based weights)

$$s(\tau_i) = \sum_{m=1}^M w_m^{PDM} \cdot s_i^{(m)}$$



- ✓ Weights set from PDM/heuristics
- ✓ Encodes expert knowledge / priorities (e.g., safety > progress > comfort)
- ✓ Interpretable and stable

Learned Aggregation

$$s(\tau_i) = f_{\phi}(s_i^{(1)}, \dots, s_i^{(M)})$$



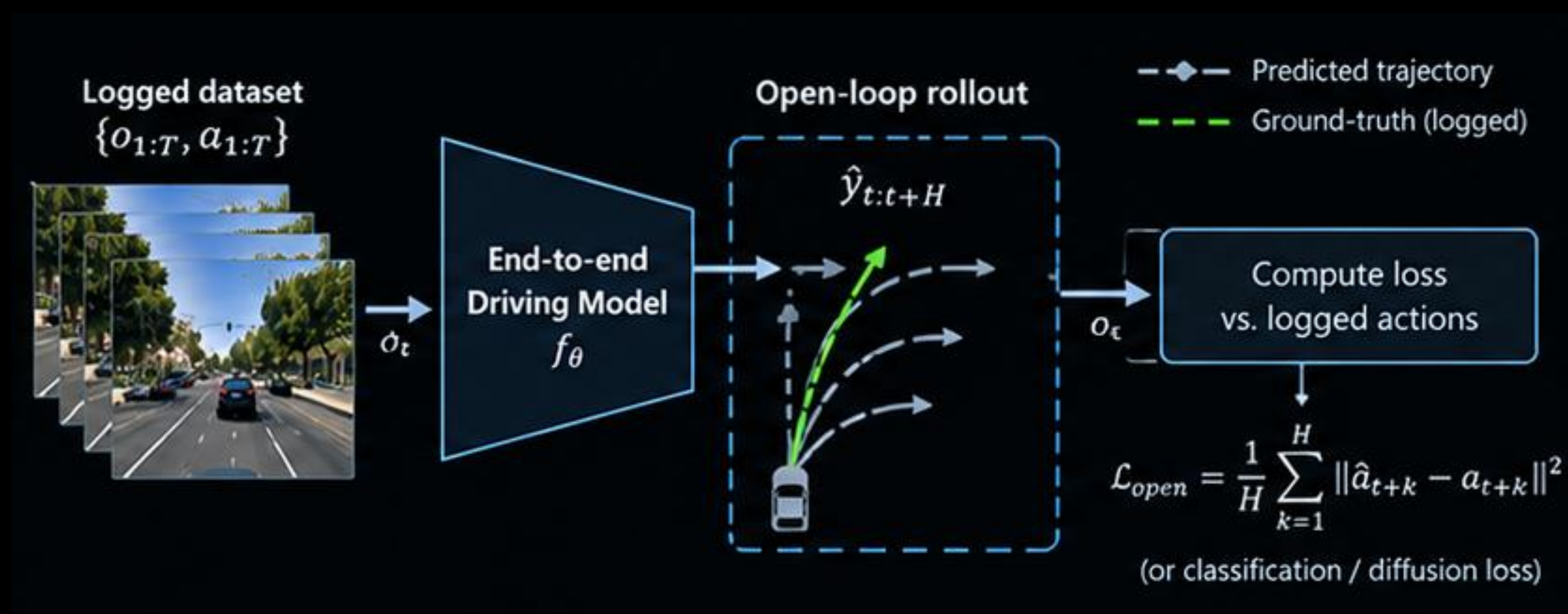
- ✓ Data-driven, adaptive weighting
- ✓ Can capture complex trade-offs
- ✓ Typically higher performance, more flexible

Bridging Open-to-closed-loop Gap

Directly optimizing what matters, but still trained in open loop

Open-loop

Trained with logged data and open-loop rollouts



Characteristics

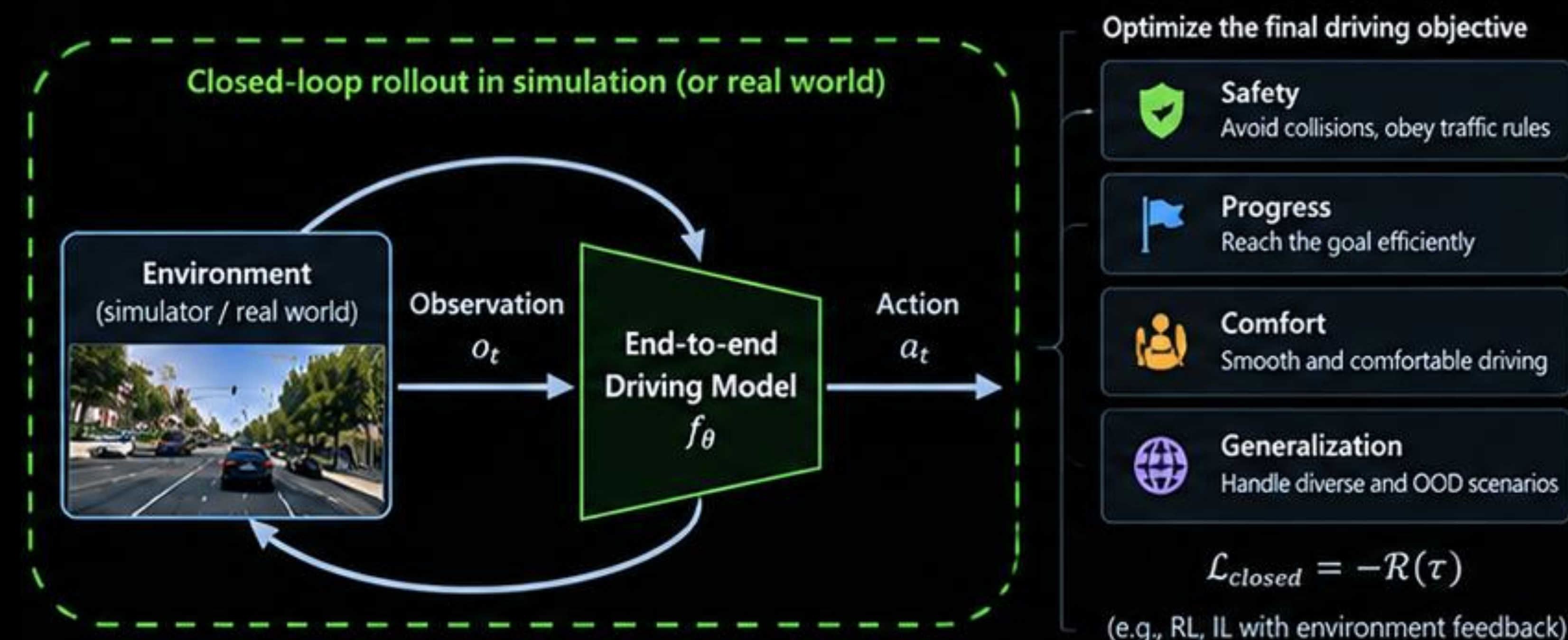
- Uses pre-collected human driving logs
- Teacher-forcing (conditioning on ground-truth history)
- Open-loop rollout (no interaction with the environment)
- Optimizes imitation of logged actions (proxy for the end goal)

Limitations

- No feedback from the environment
- Exposure bias and compounding errors
- Does not optimize the final driving objective (e.g., safety, progress, comfort)
- Limited ability to handle out-of-distribution scenarios

Closed-loop

Trained by interacting with the environment and optimizing what matters



Key Benefits

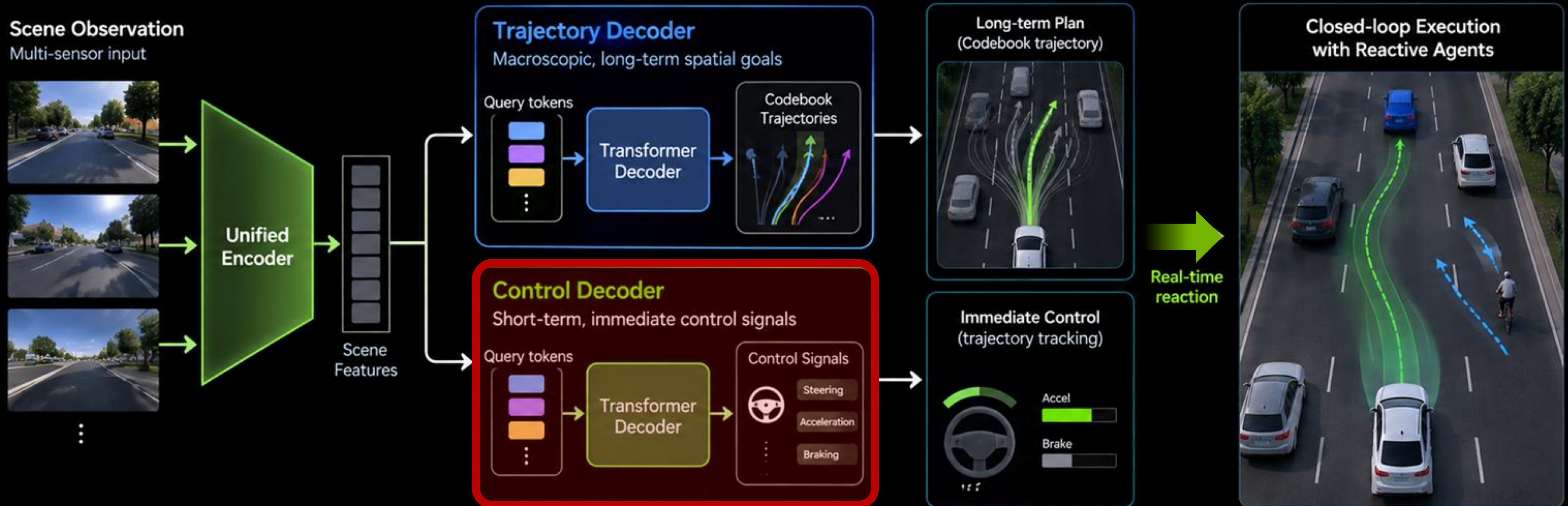
- Learns from interaction with the environment
- Optimizes what truly matters (task-level objectives)
- Accounts for multi-agent reactions and long-horizon consequences
- More robust and better aligned with real-world deployment

Challenges

- Sample efficiency and training stability
- Requires high-fidelity simulators (and/or real-world data)
- Multi-agent interactions and non-stationarity
- Safety during training

Bridging Open-to-closed-loop Gap

Unified multi-branch planning for closed-loop, reactive, and kinematically feasible driving



Bridges open-to-closed-loop gap

Direct control outputs for immediate execution



Handles reactive agents

Multi-branch planning with adaptive control



Respects kinematic constraints

Feasible and executable trajectories and controls



Robust planning framework

Unifies long-term goals and short-term control

From Regression to Classification for Multi-Modal Planning

Discretize the future into a finite set of modes, classify over them, and select the optimum

Candidate Trajectory Set

Generate a finite set of plausible trajectories (modes)

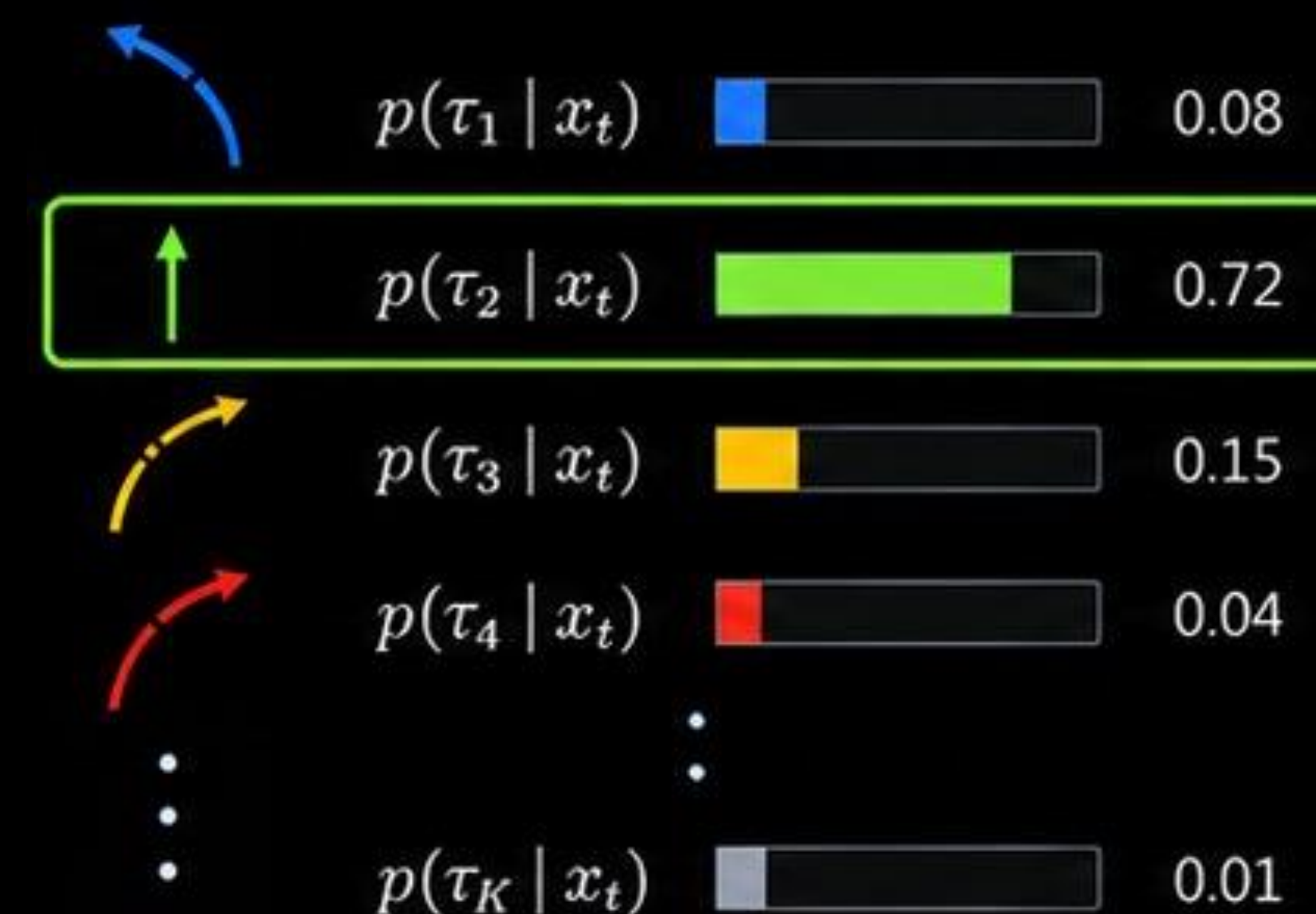


$$\mathcal{T} = \{\tau_1, \tau_2, \dots, \tau_K\}$$



Classification over Modes

Predict a probability for each candidate



Planning Quality $\approx$ Coverage of candidates $\times$ Quality of evaluator

Do I have the
right candidate?



Planning Quality $\approx$ Coverage of candidates $\times$ Quality of evaluator

Do I have the
right candidate?



Planning Quality $\approx$ Coverage of candidates $\times$ Quality of evaluator



Can I recognize the
right candidate?

Do I have the
right candidate?



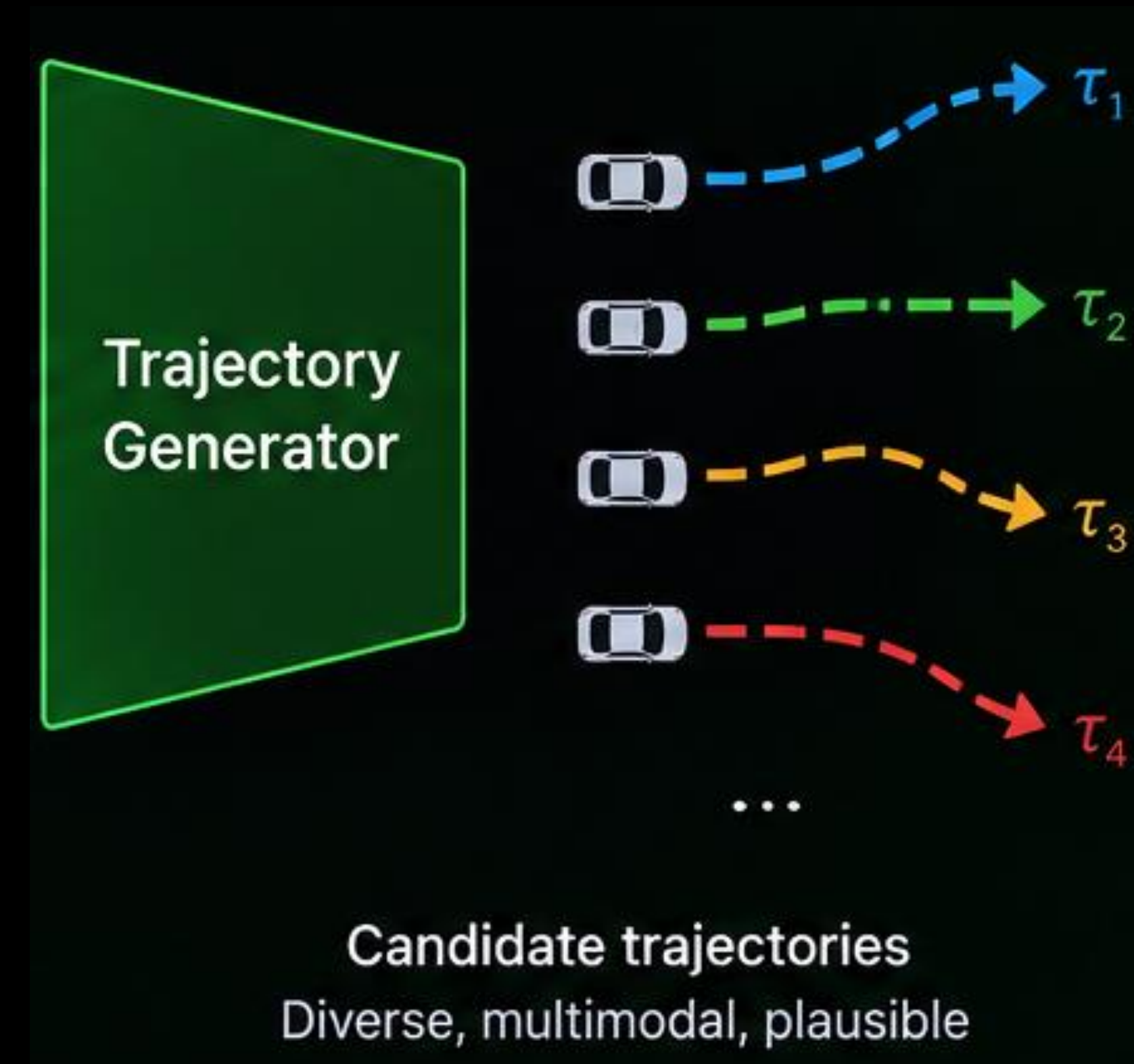
Planning Quality $\approx$ Coverage of candidates $\times$ Quality of evaluator

GTRS: Generalized Trajectory Scoring

Explicitly decouples action proposal from action judgment

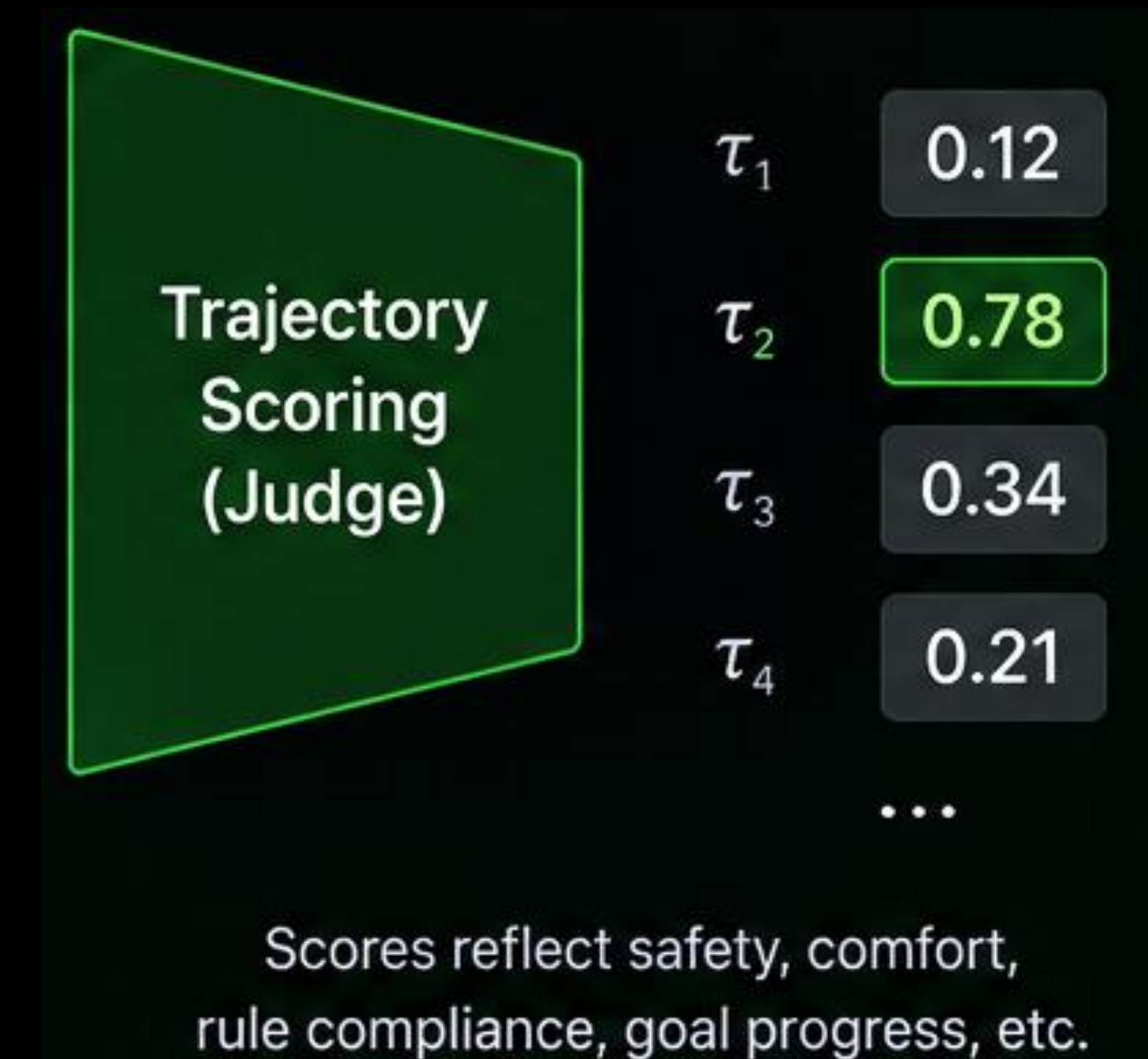
Propose Actions

Generate multiple candidate trajectories
conditioned on the current scene



Judge Actions

Score and evaluate each candidate using
a learned / world-model based evaluator

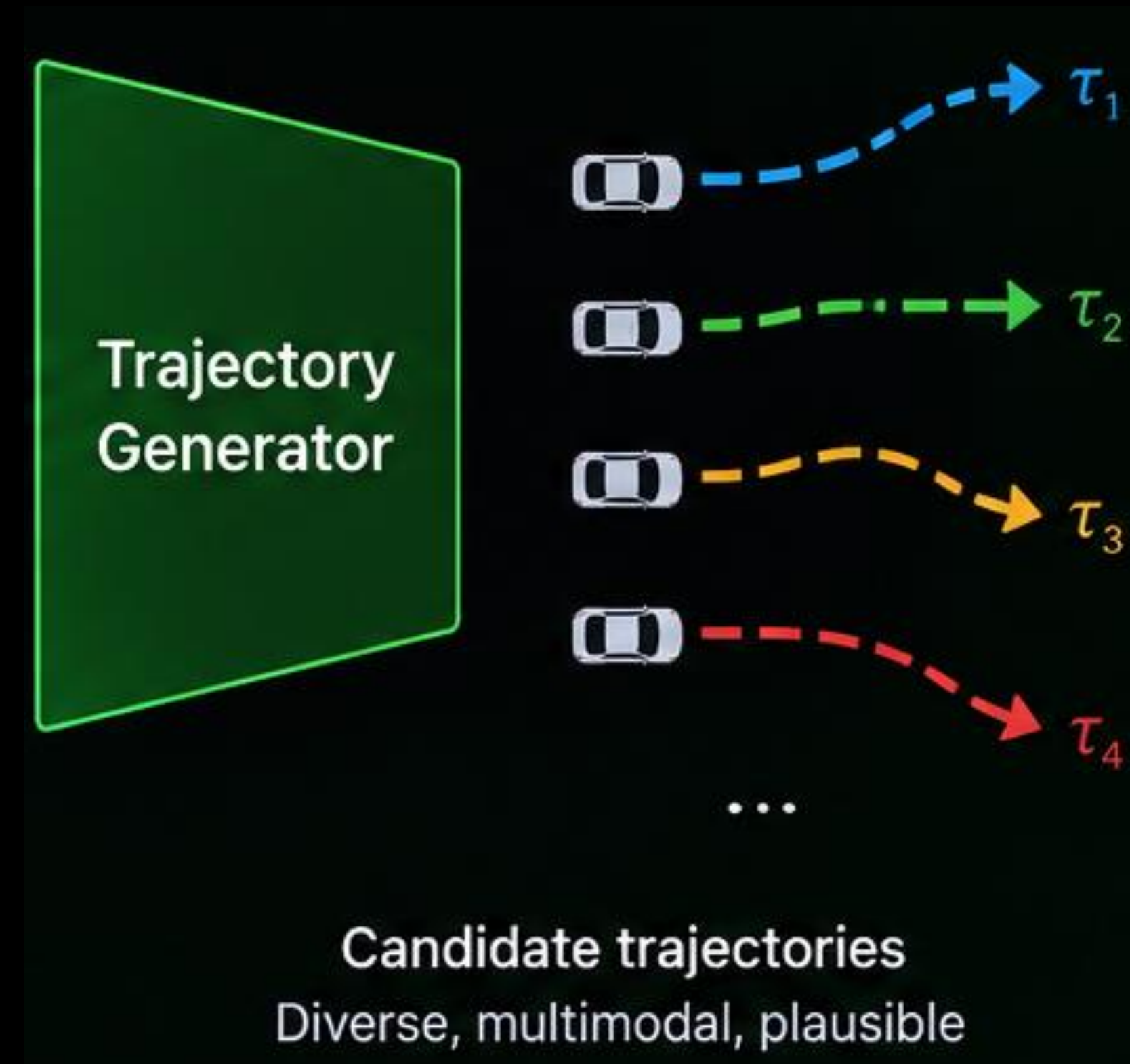


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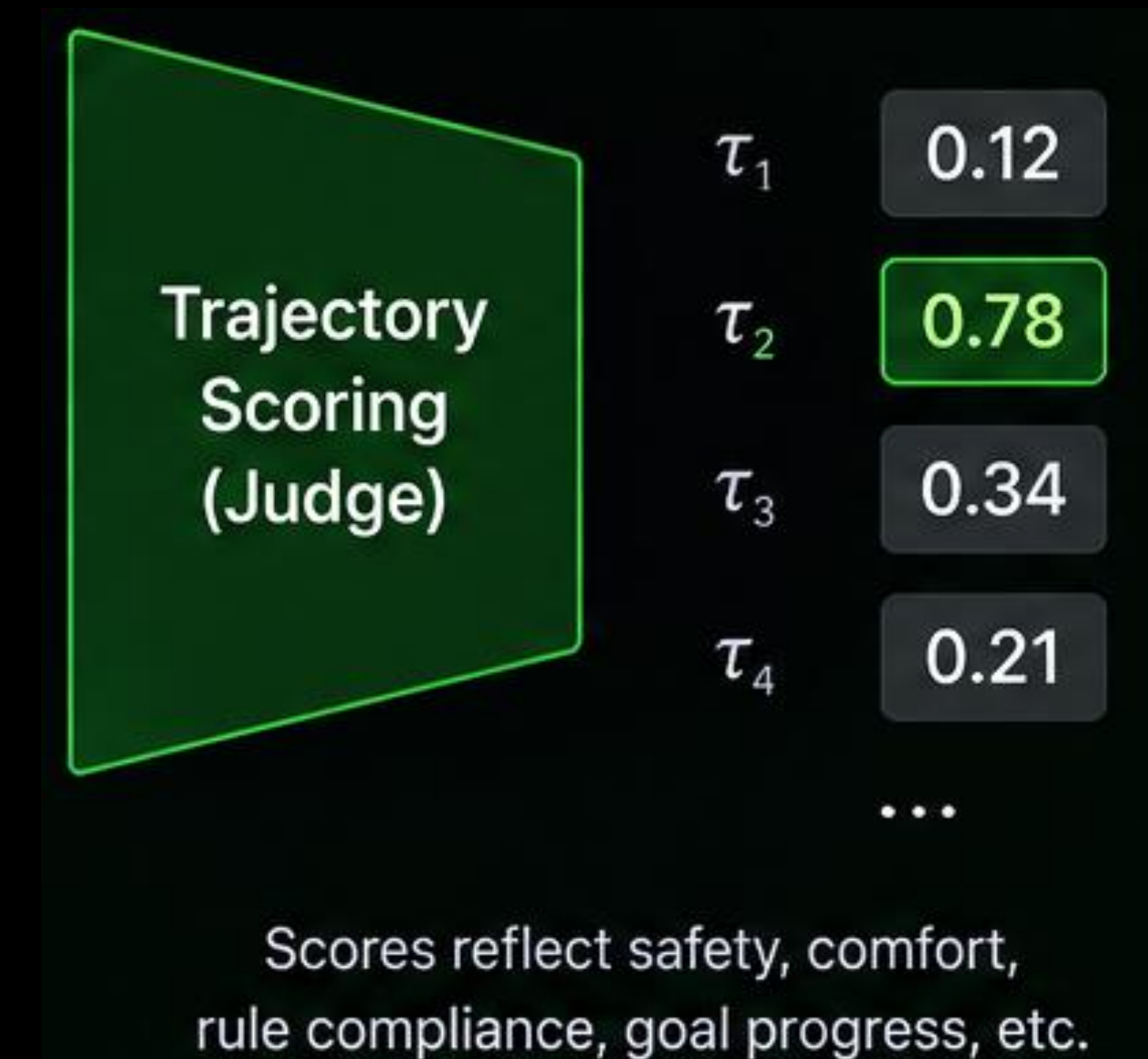
Propose Actions

Generate multiple candidate trajectories
conditioned on the current scene



Judge Actions

Score and evaluate each candidate using
a learned / world-model based evaluator



- Enables more flexible, scalable, and safer decision-making.
- Enables deliberation: The system considers alternatives before committing to an action.

GTRS: Generalized Trajectory Scoring

Explicitly decouples action proposal from action judgment

Dynamic Trajectory Candidates



Static Trajectory Candidates



Top-K Trajectory Candidates



Final Trajectory



Planning Quality $\approx$ Coverage of candidates $\times$ Quality of evaluator



Can I recognize the
right candidate?

Small Differences, Big Consequences

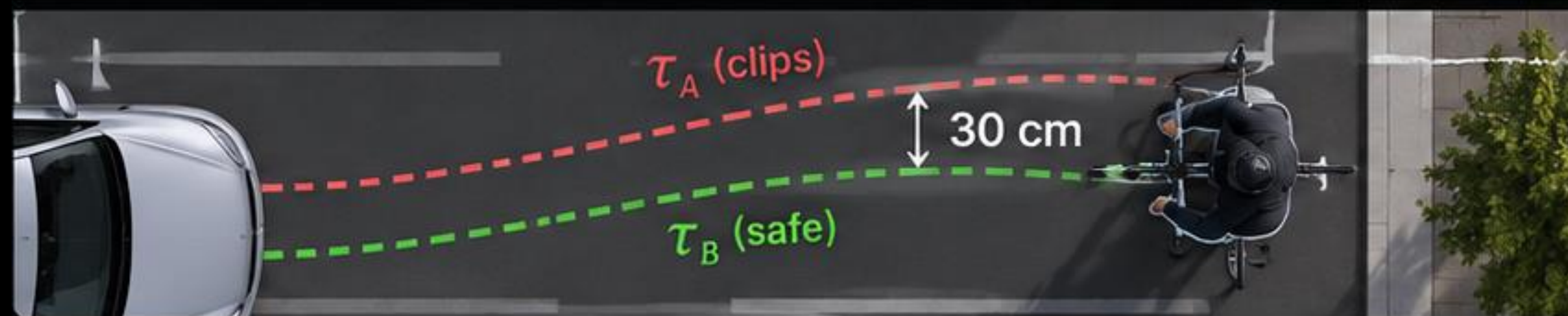
Candidate A - Unsafe

Hits the cyclist



Candidate B - Safe

Passes the cyclist safely



Zoomed view of the lateral difference

Small Differences, Big Consequences

Candidate A - Unsafe

Hits the cyclist



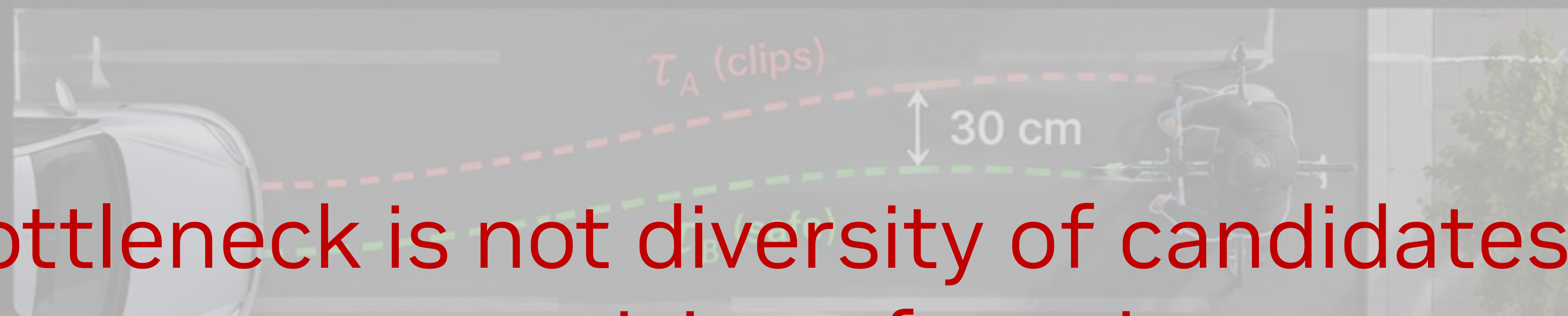
Candidate B - Safe

Passes the cyclist safely



Can our scorer tell the difference?

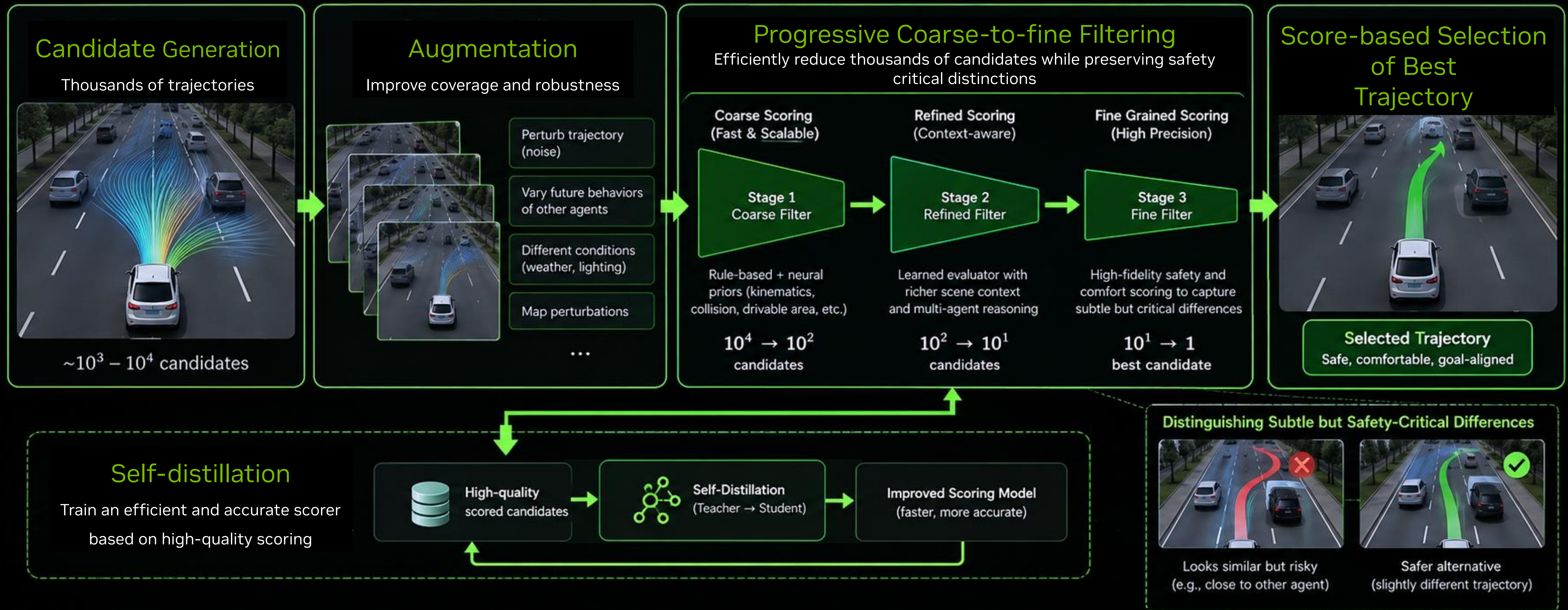
Bottleneck is not diversity of candidates, it is precision of scoring



Zoomed view of the lateral difference

DriveSuprim: Scalable and Precise Trajectory Scoring

Progressive coarse-to-fine filtering with augmentation and self-distillation to select the optimal trajectory among thousands of candidates.

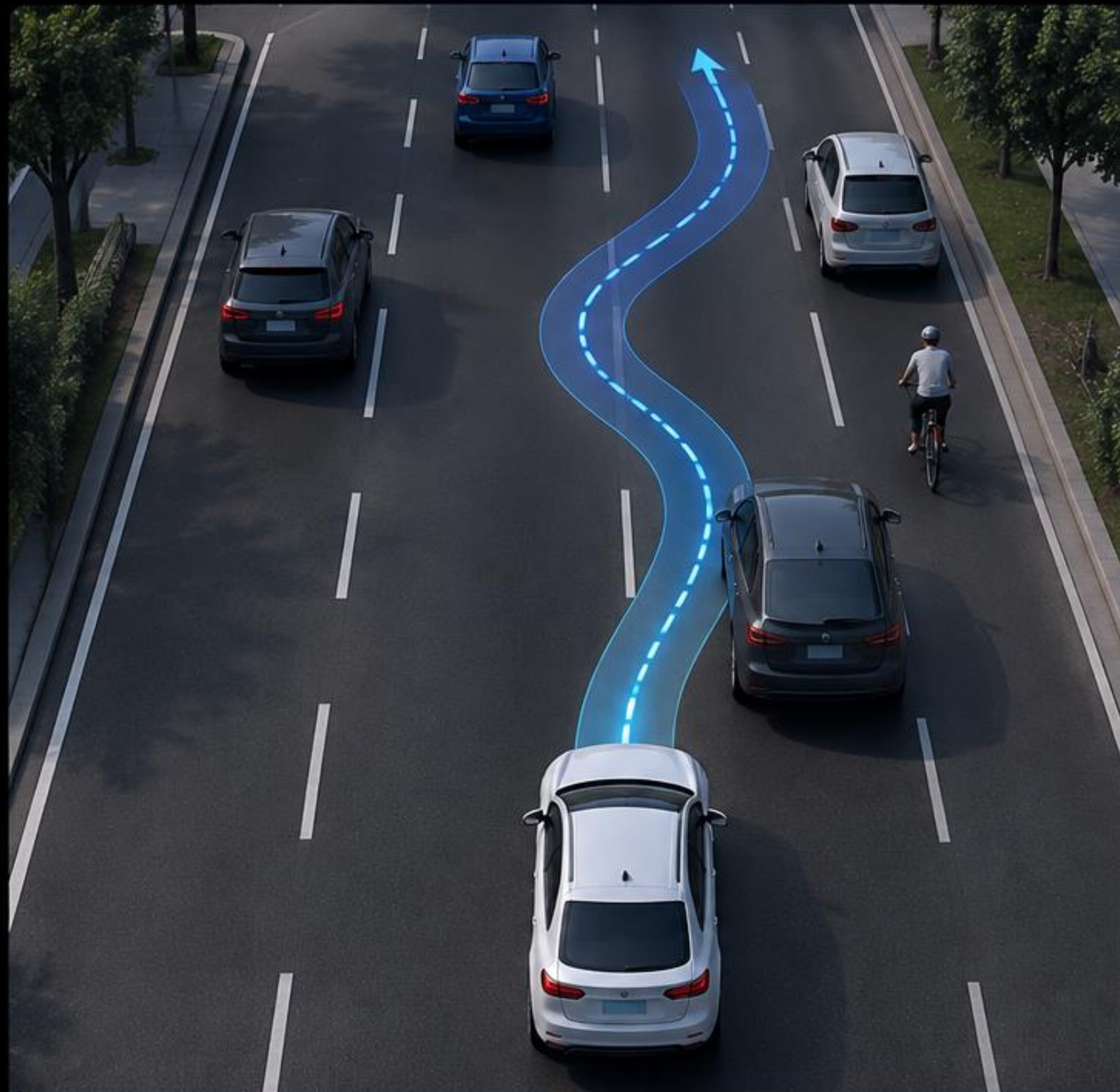


Enables more flexible, scalable, and safer decision-making.

Should we Imitate the Human at All?

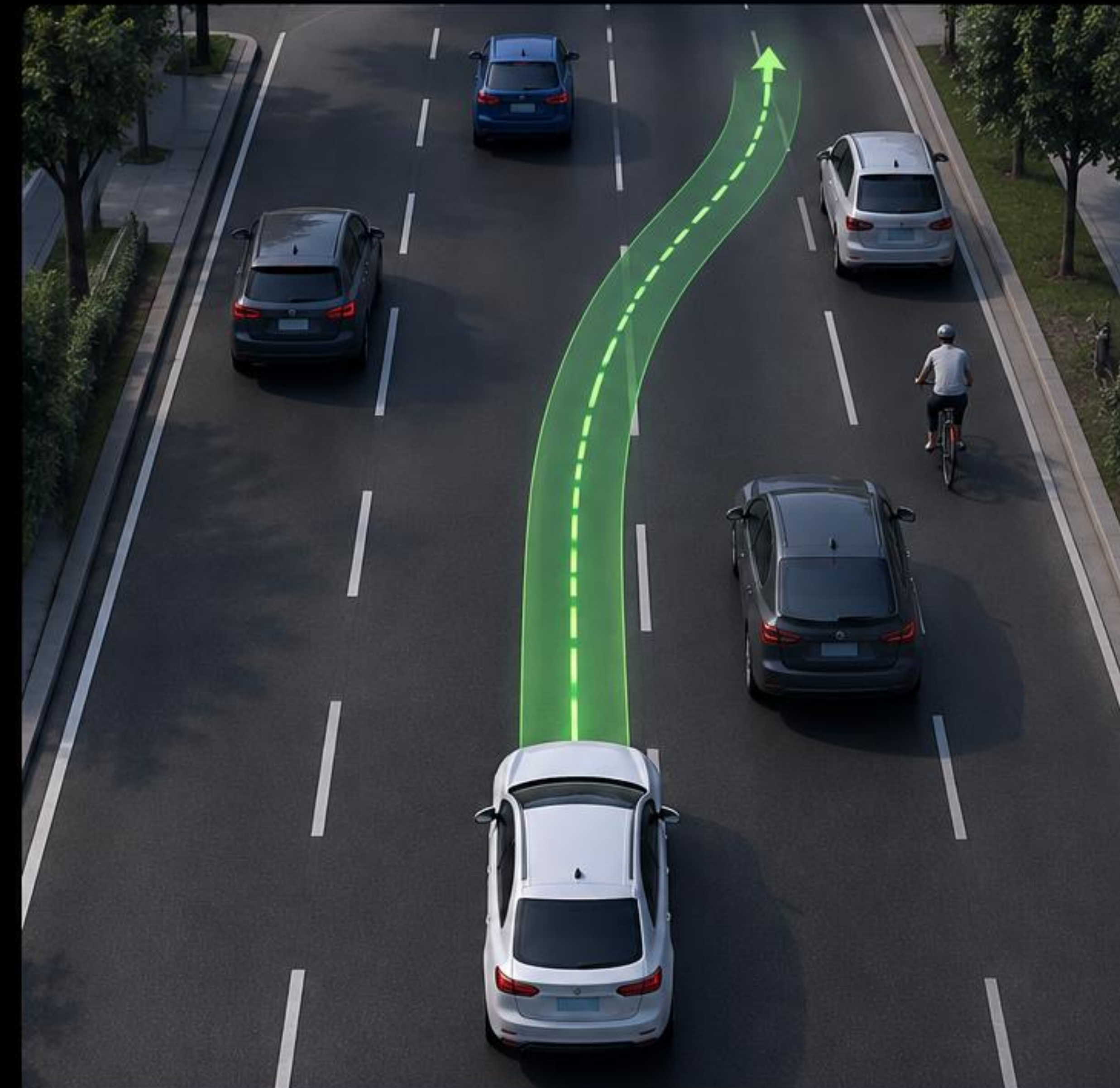
Should we Imitate the Human at All?

Human Trajectory
Safe, but suboptimal



- ✓ Safe (no collision, respects constraints)
- ✗ Slightly suboptimal (larger deviation, longer path)
- ✗ More conservative (higher steering effort)

Alternative Trajectory
Also safe, smoother, more efficient

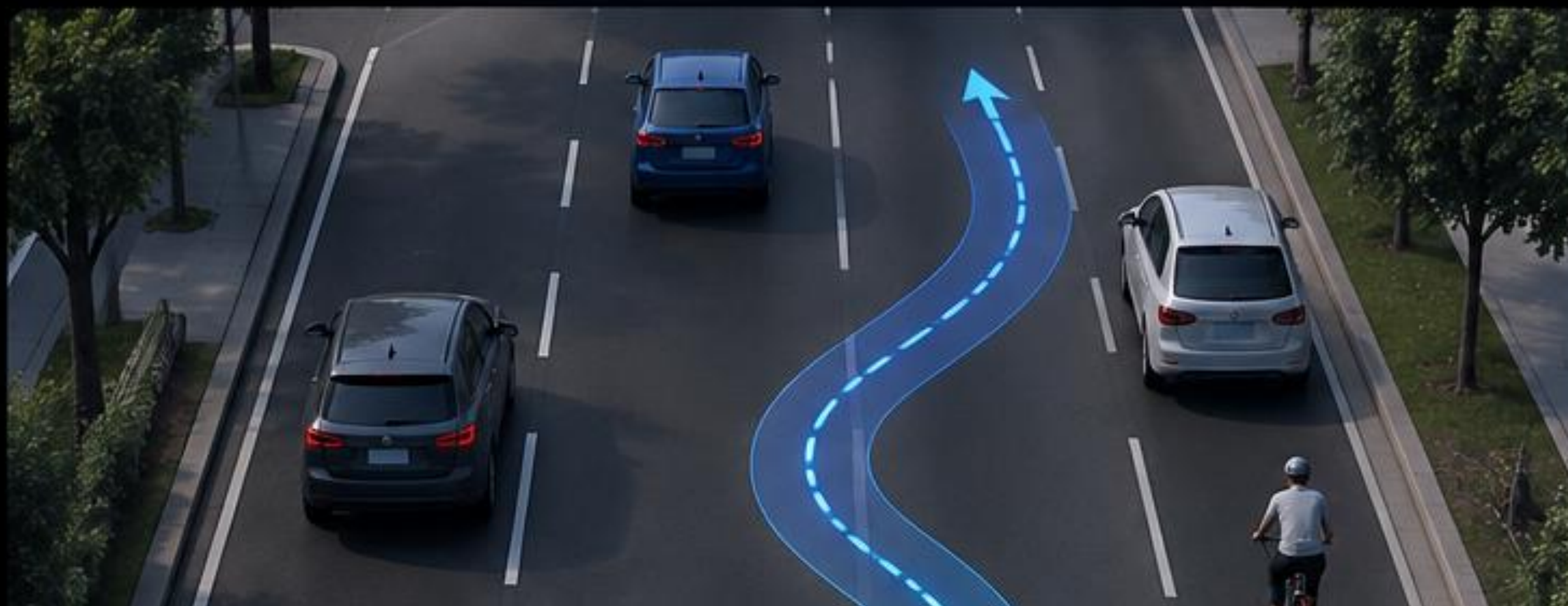


- ✓ Safe (no collision, respects constraints)
- ✓ Smoother (lower curvature, less steering effort)
- ✓ More efficient (shorter path, closer to nominal route)

Should we Imitate the Human at All?

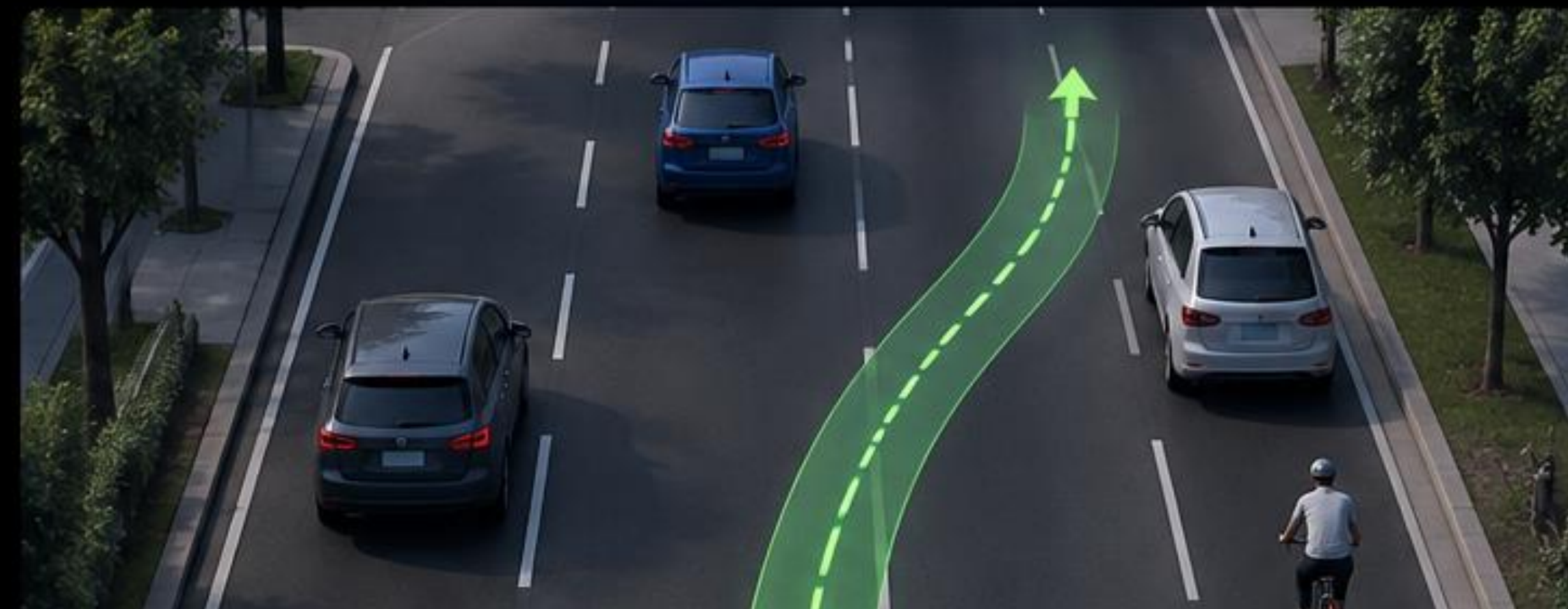
Human Trajectory

Safe, but suboptimal



Alternative Trajectory

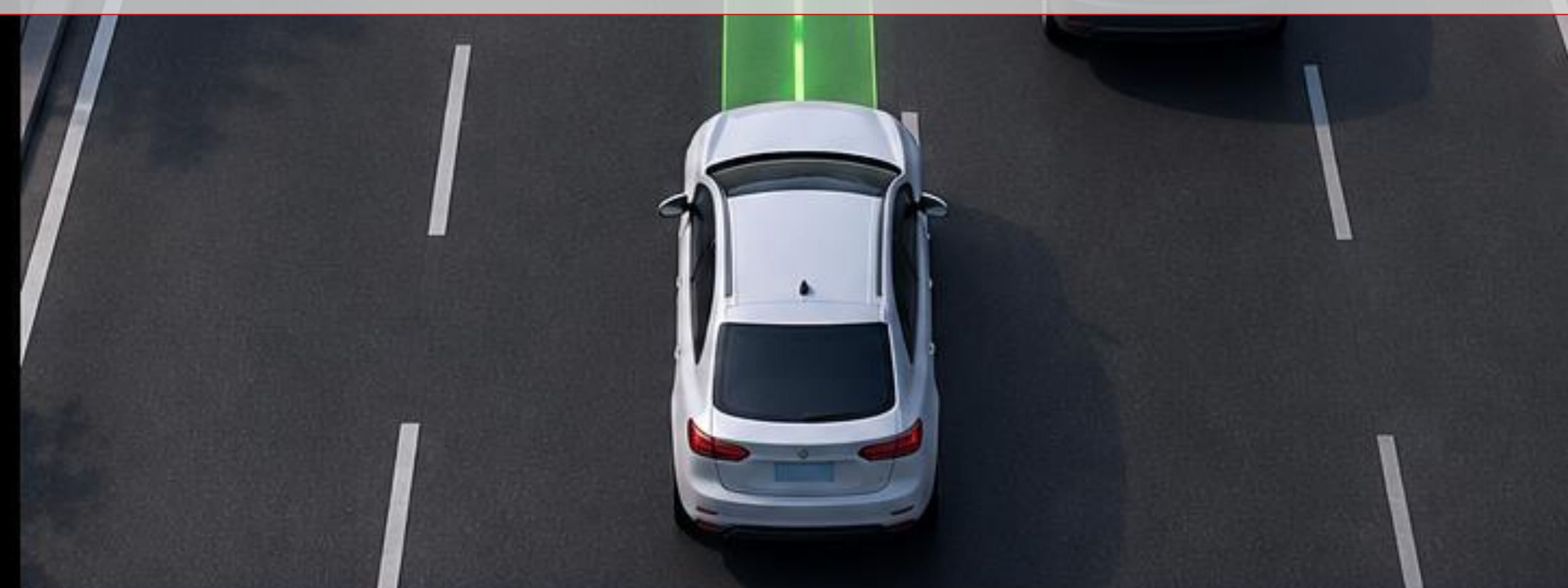
Also safe, smoother, more efficient



Why should the human trajectory have probability 1 and everything else probability 0?



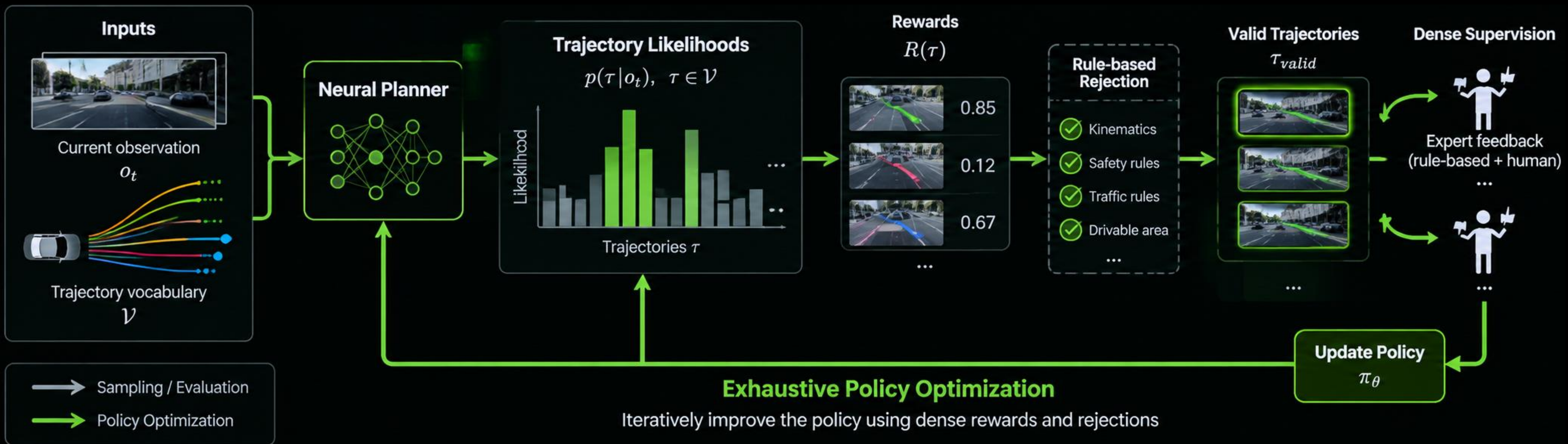
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- ✓ Safe (no collision, respects constraints)
- ✓ Smoother (lower curvature, less steering effort)
- ✓ More efficient (shorter path, closer to nominal route)

ZTRS

Zero-shot trajectory rejection sampling

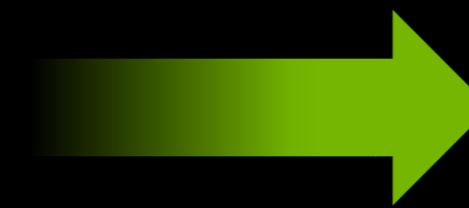


ZTRS

Learning paradigm change

Imitation learning

$$\tau^* = \arg \min_{\tau} D(\tau, \tau_{human})$$



Reward optimization

$$\tau^* = \arg \max_{\tau} R(o, \tau)$$

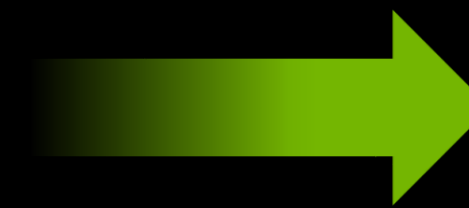
This changes the role of evaluation from something we do after training into something that can drive learning itself.

ZTRS

Learning paradigm change

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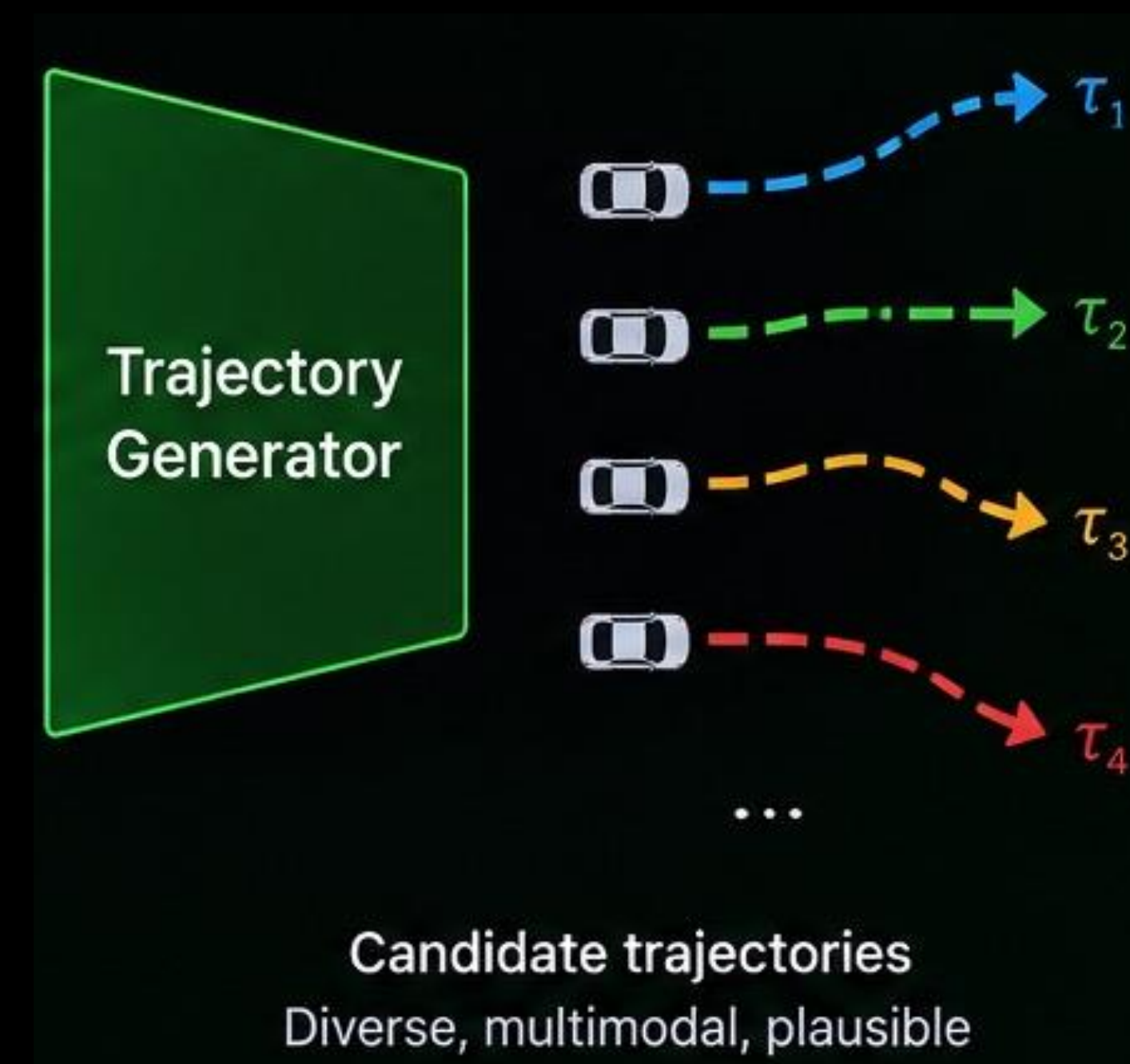
Another major step toward deliberative systems.

3rd Generation

Real-time adaptation without replanning from scratch to sudden and unexpected agent maneuver

3rd Generation: Search Optimization

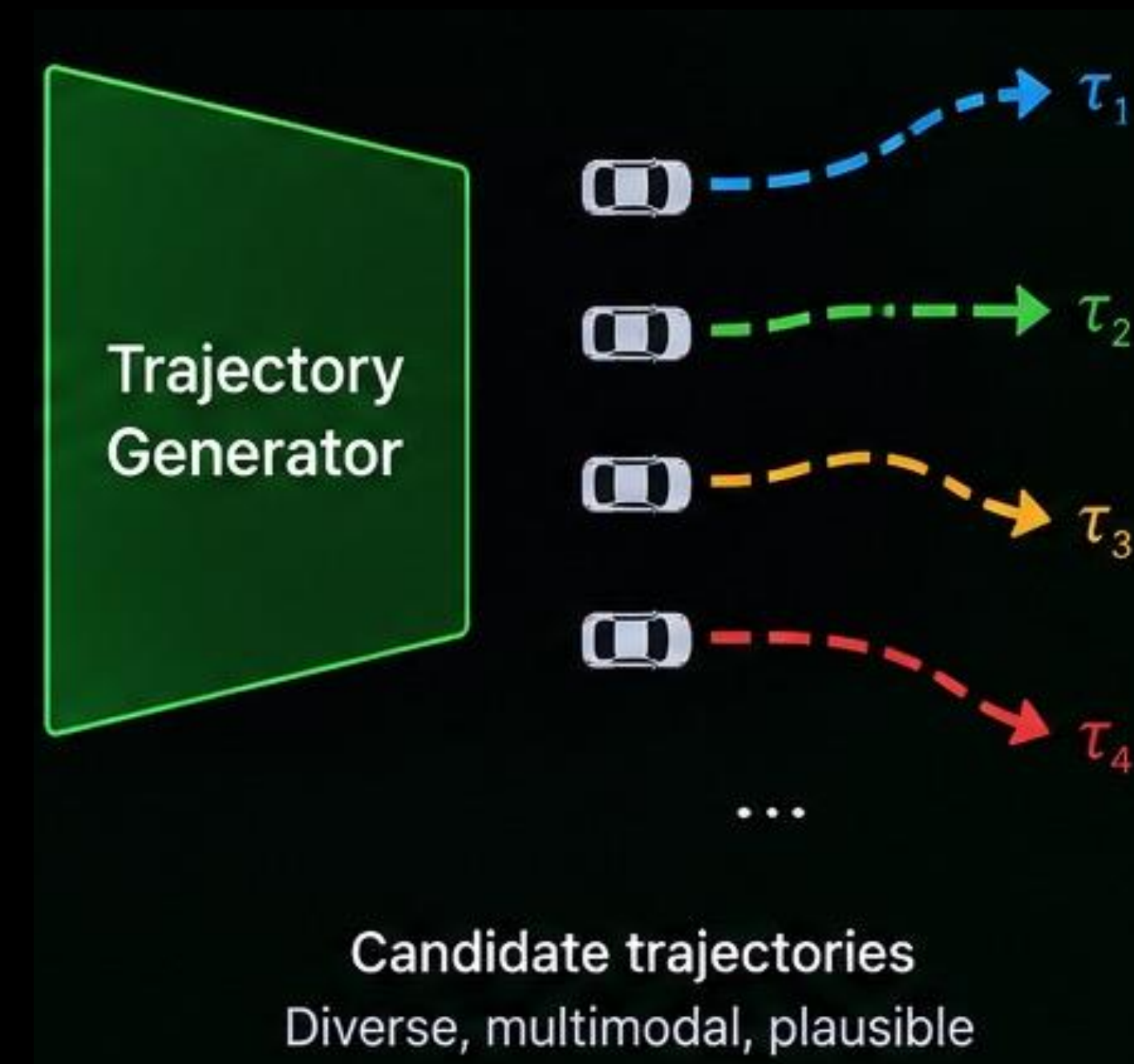
Why evaluate thousands of unrelated trajectories independently?



A better planner should progressively concentrate compute on promising regions of action space.

3rd Generation: Search Optimization

Why evaluate thousands of unrelated trajectories independently?

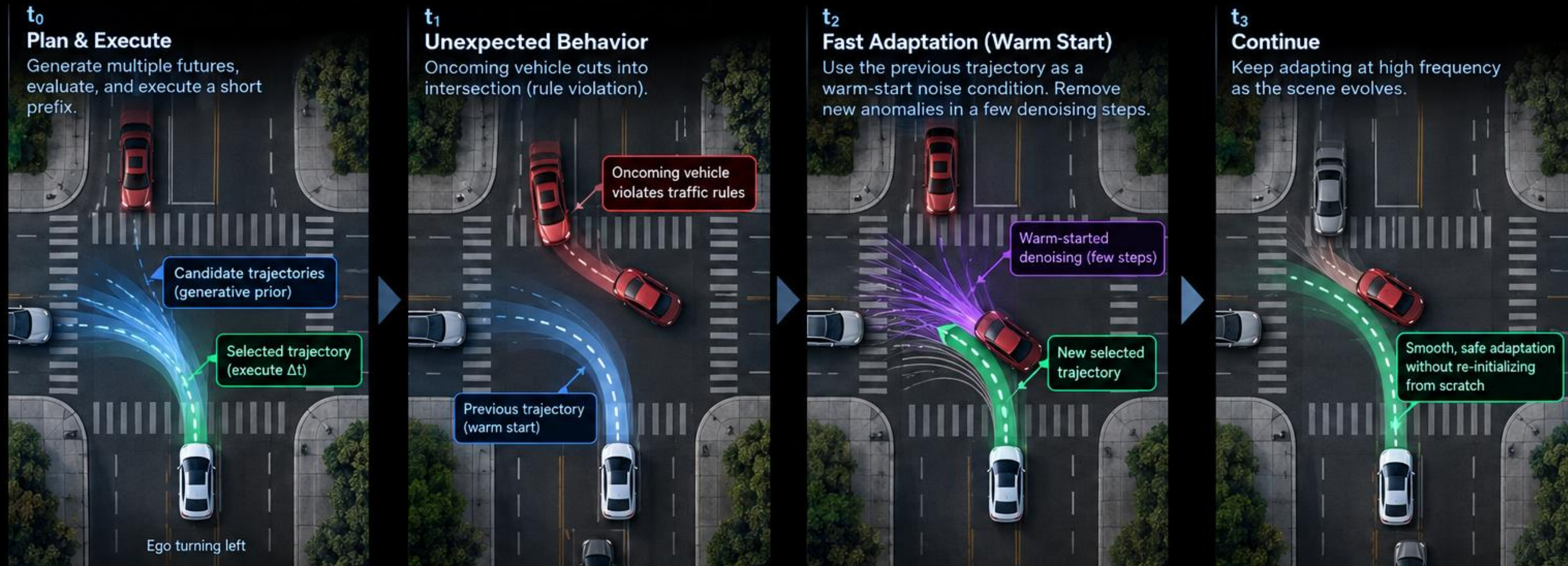


A better planner should progressively concentrate compute on promising regions of action space.

Generate → Evaluate → Execute a Little → Observe Again → Adapt

3rd Generation: Search Optimization

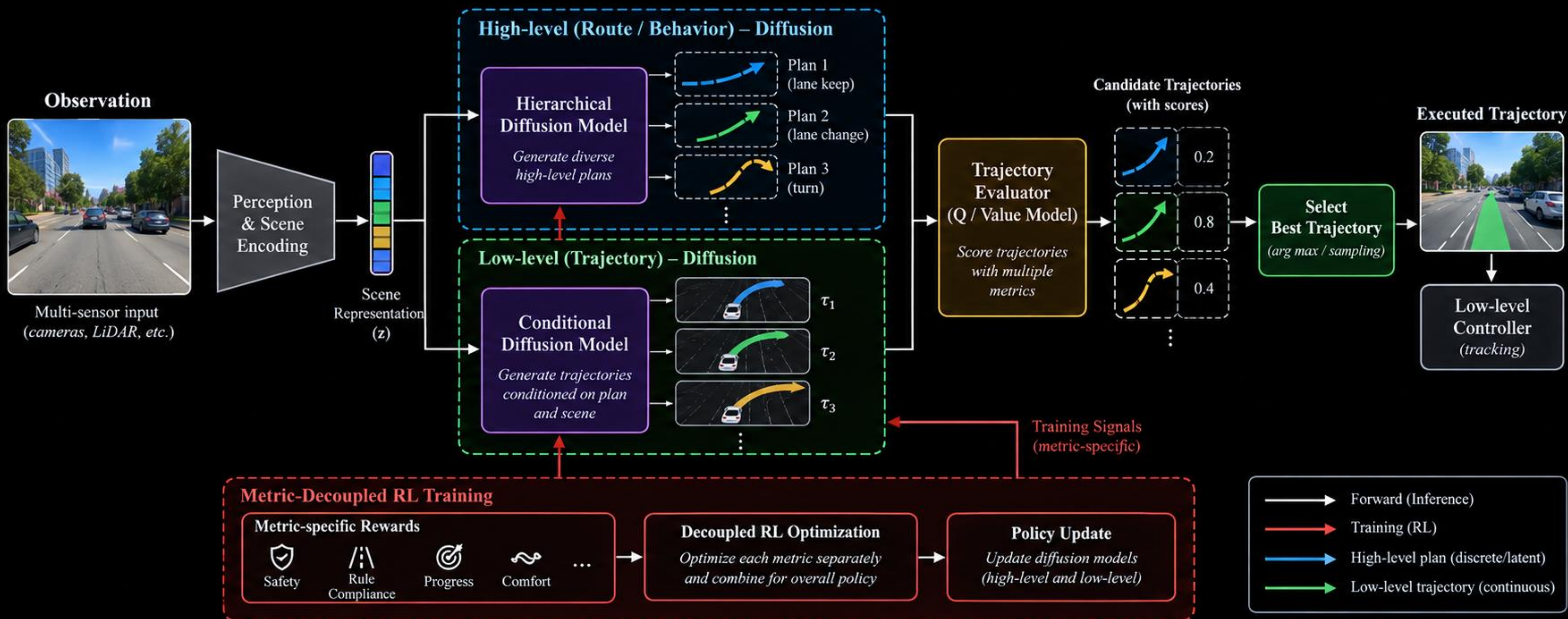
Real-time adaptation without replanning from scratch to sudden and unexpected agent maneuver



- ✓ Near-instantaneous reaction to erratic agents
- ✓ No need to re-initialize a new trajectory
- ✓ Warm-started denoising strips new noise anomalies
- ✓ Leverages temporal consistency for smooth control
- ✓ Enables safe, natural, and rule-aware behavior

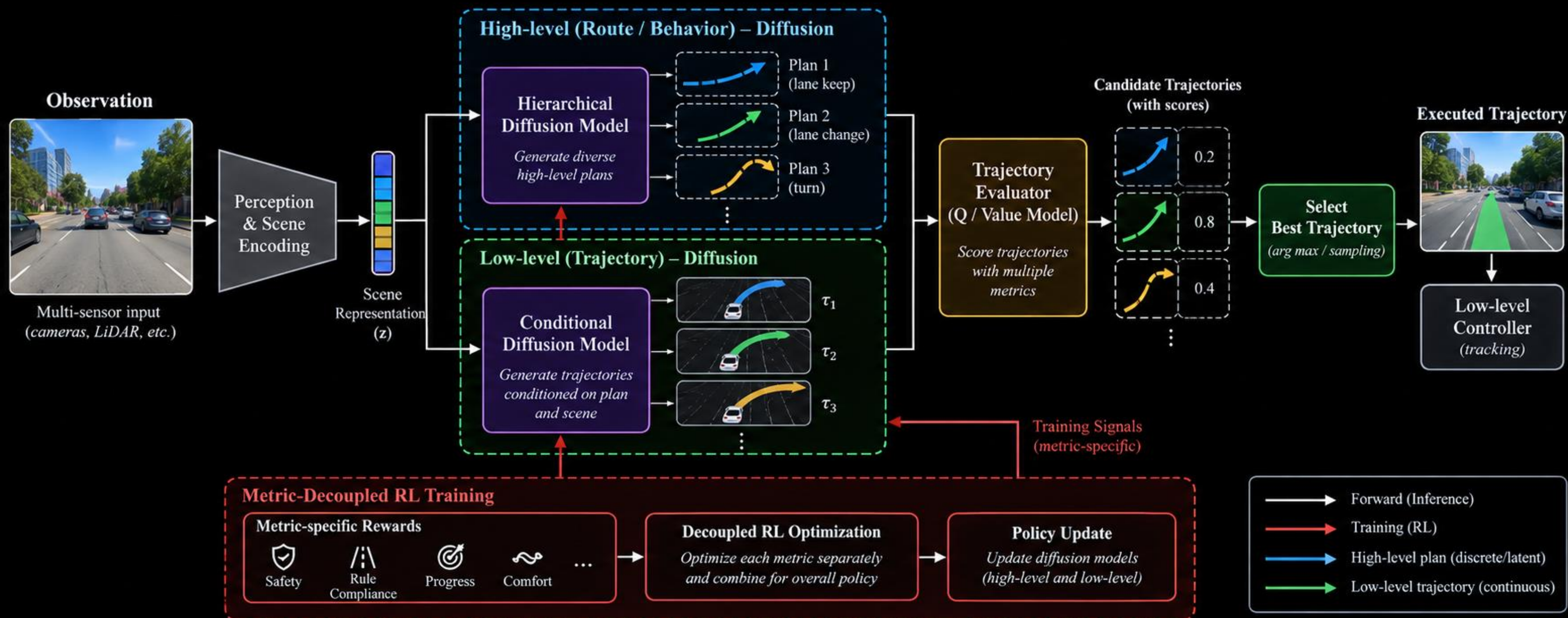
3rd Generation: Search Optimization

Hierarchical generative policy with metric-decoupled RL



3rd Generation: Search Optimization

Hierarchical generative policy with metric-decoupled RL



- Hierarchical diffusion for diverse, structured decision-making,
- Metric-decoupled RL for robust policy learning

Hierarchical Diffusion with Polar Noise

Structured denoising for diverse and physically plausible trajectories



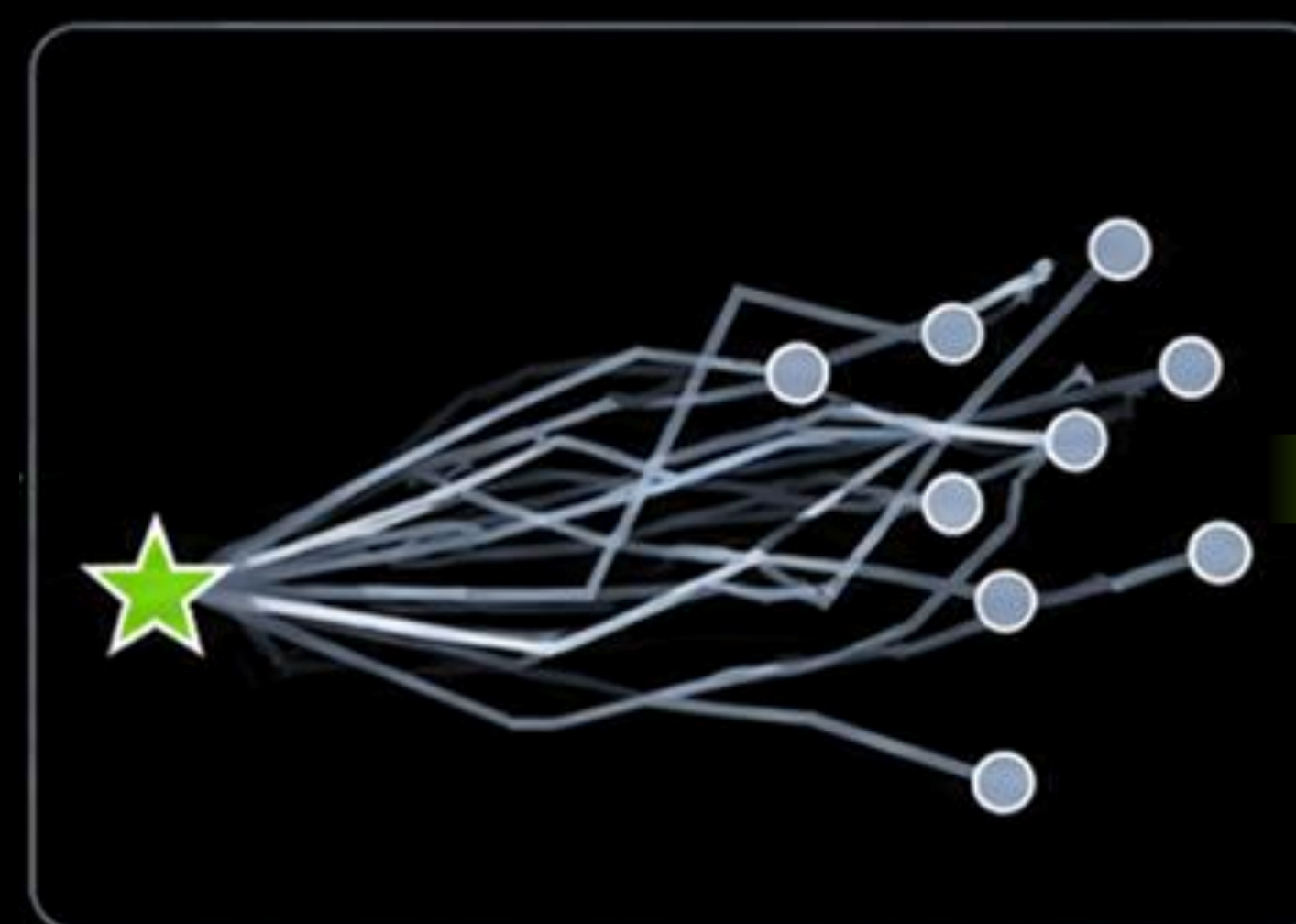
Origin Trajectory
(conditioning / warm start)

Hierarchical Diffusion with Polar Noise

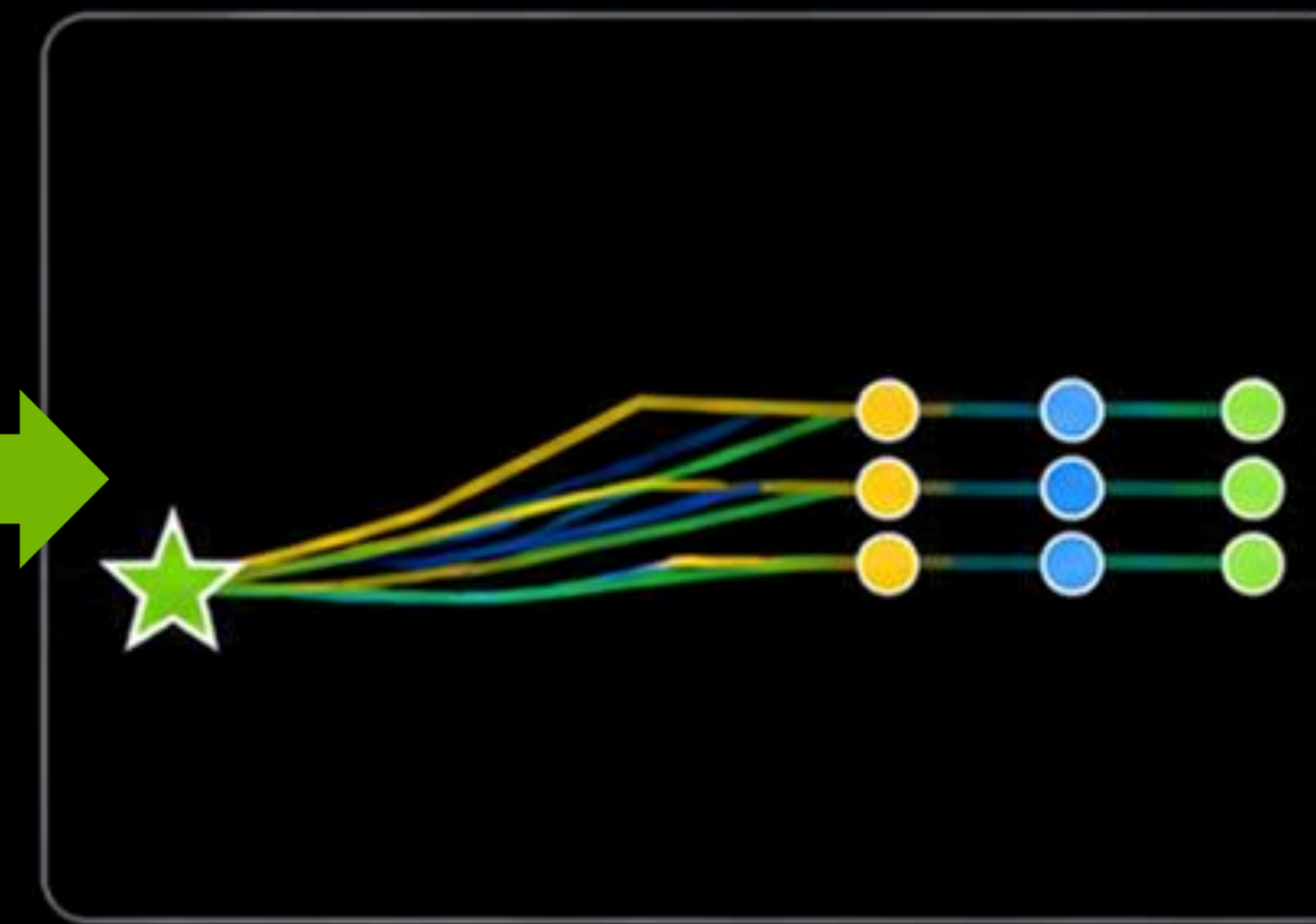
Structured denoising for diverse and physically plausible trajectories



Origin Trajectory
(conditioning / warm start)



Random Noise
(isotropic in Cartesian space)



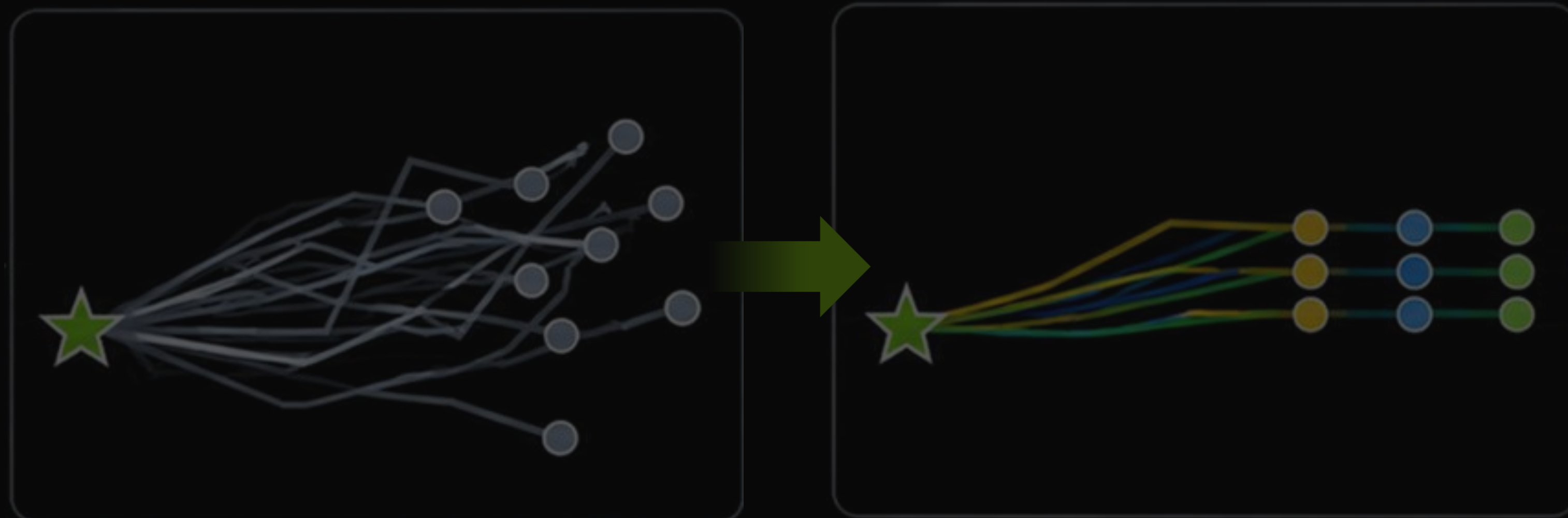
XY Expand
(denoise / refine in Cartesian space)

Hierarchical Diffusion with Polar Noise

Structured denoising for diverse and physically plausible trajectories

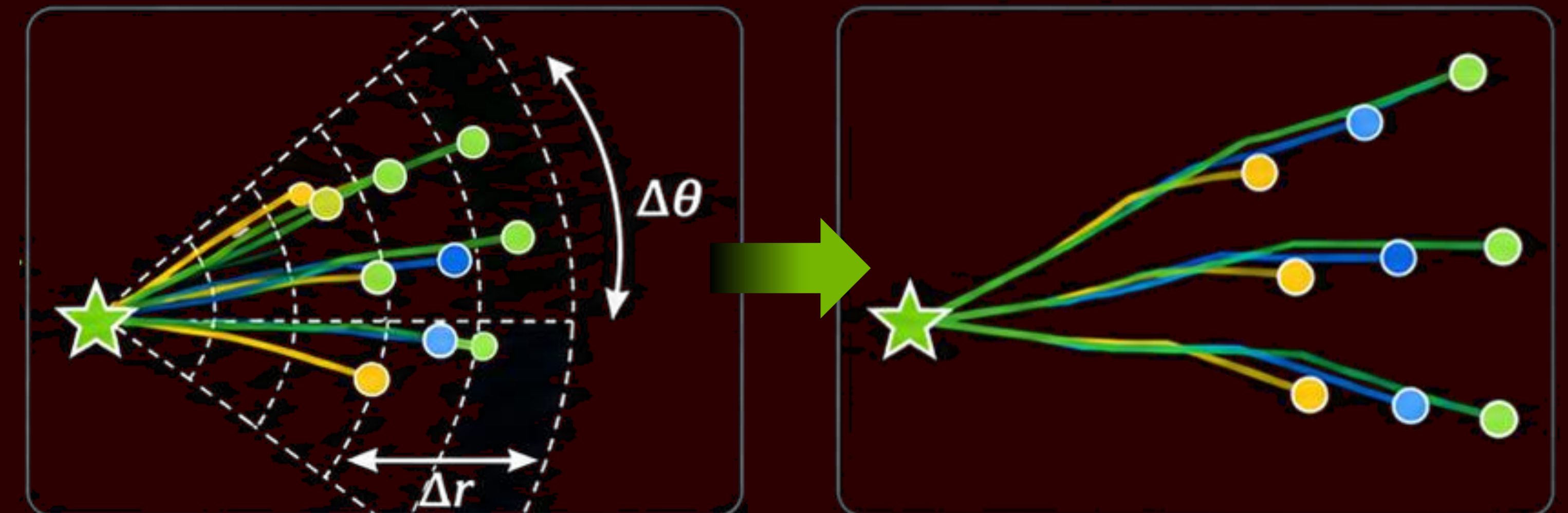


Origin Trajectory
(conditioning / warm start)



Random Noise
(isotropic in Cartesian space)

XY Expand
(denoise / refine in Cartesian space)

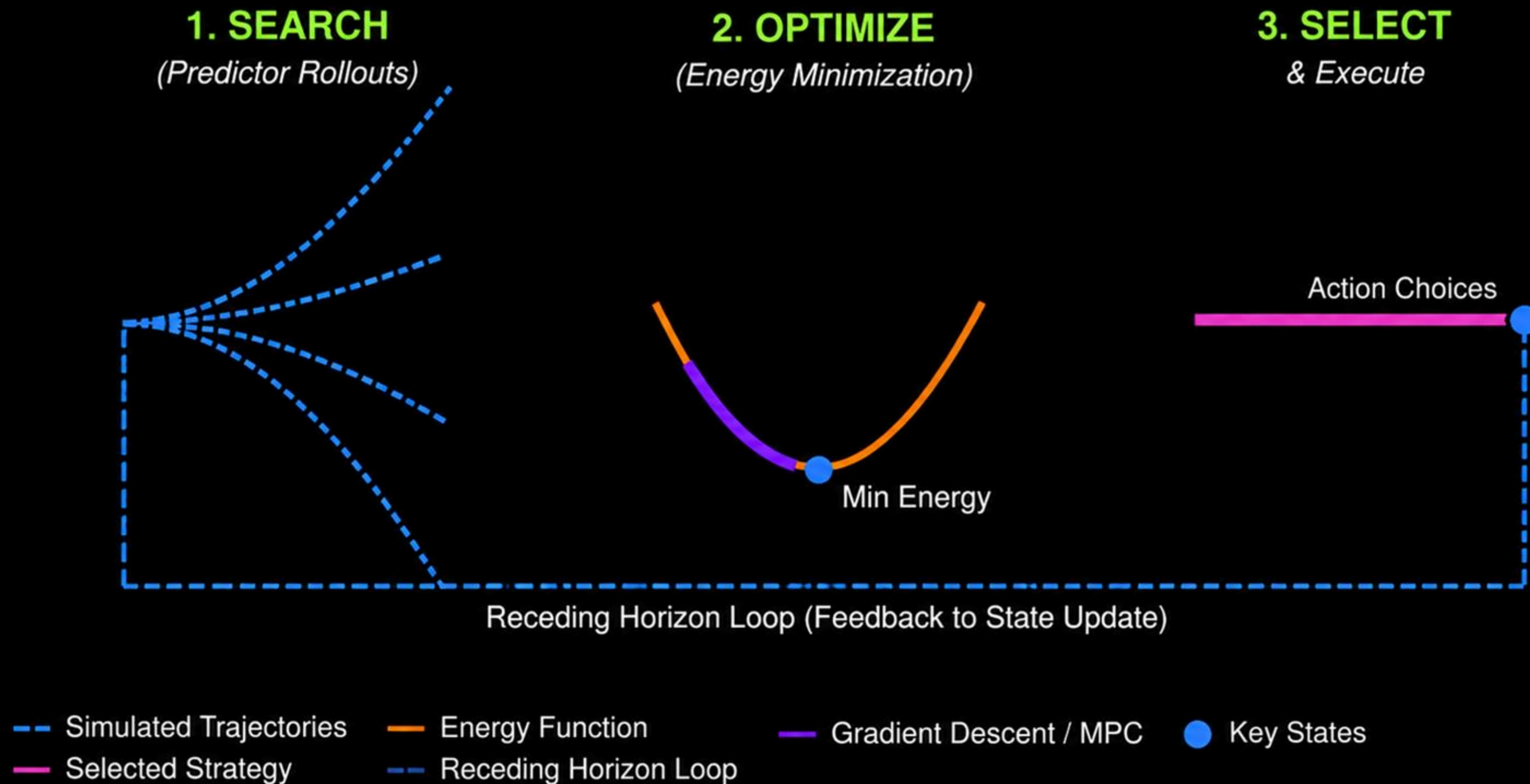


Polar Noise
(structured sampling
in polar coordinates)

Polar Expand
(refine with polar noise)

Counterfactual Deliberation on the Edge

[LeCun's] World Model Loop: Search, Optimize, and Select

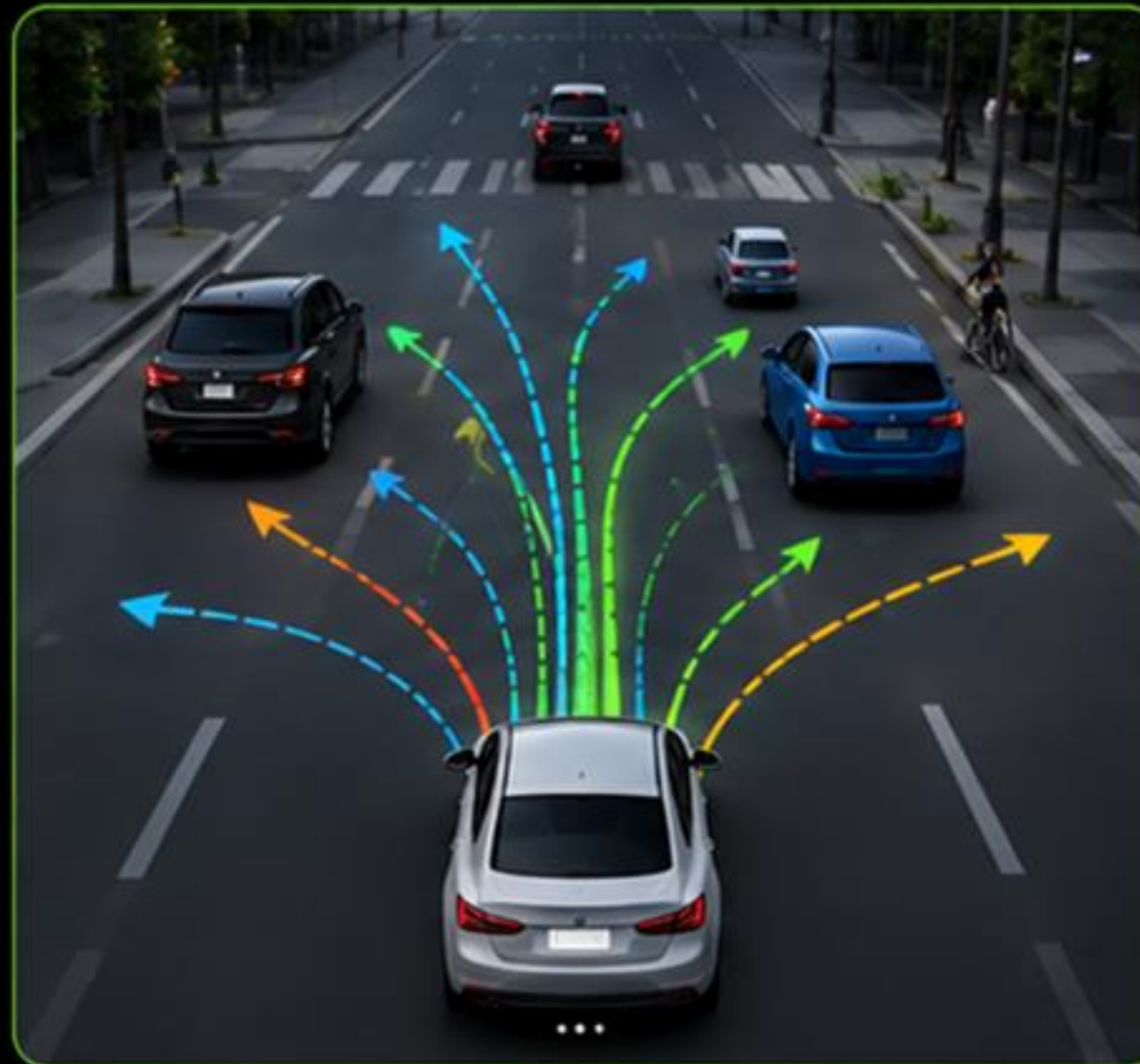


Deliberate on the Edge

Search, Optimize, and Select

Search

Generate diverse candidate futures and actions



Sample multiple plausible trajectories and action sequences

Optimize

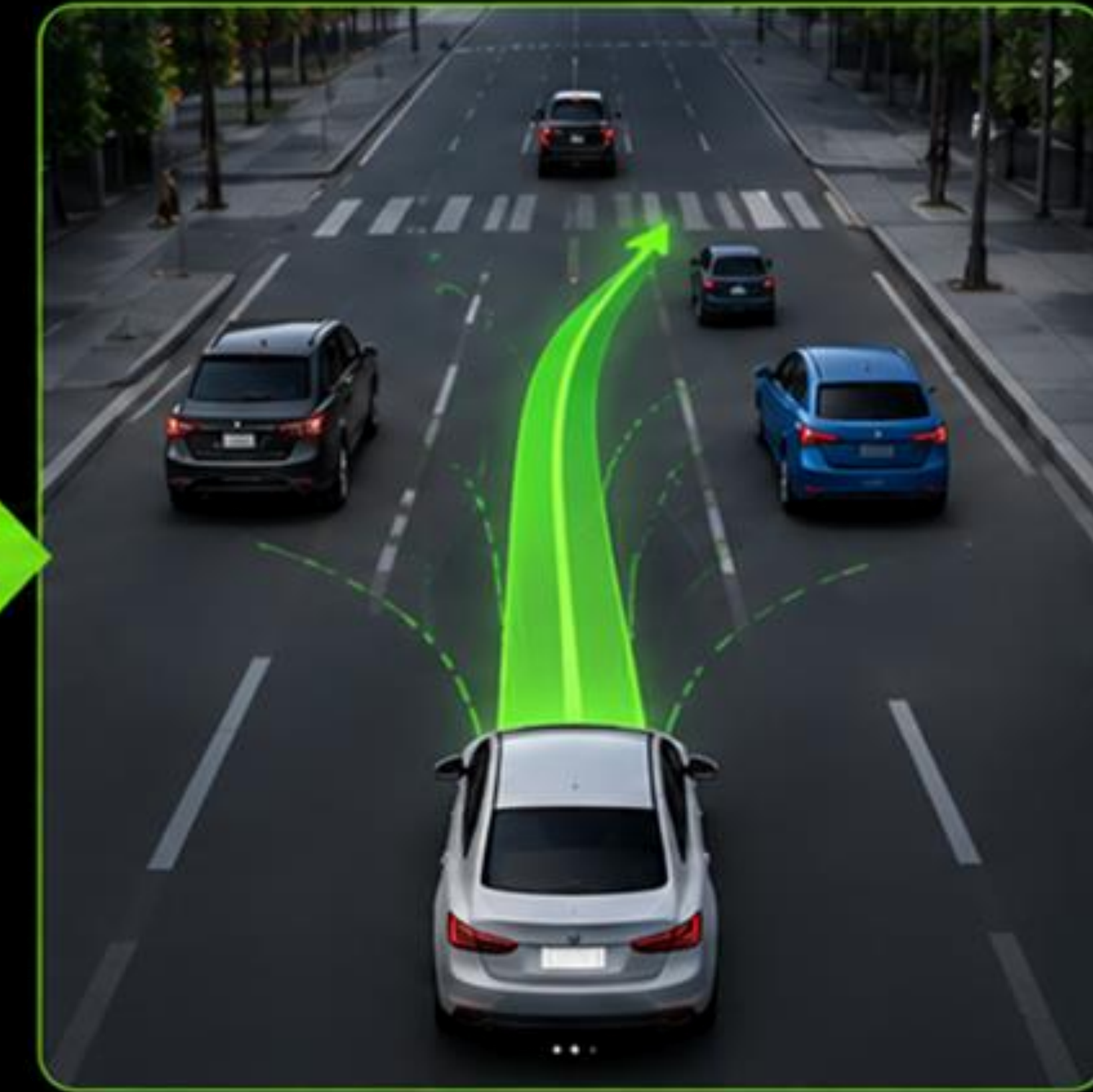
Evaluate and refine with objectives and constraints



Compute rewards, check constraints, and refine (e.g., gradient-based, search-based)

Select

Choose the best plan and execute



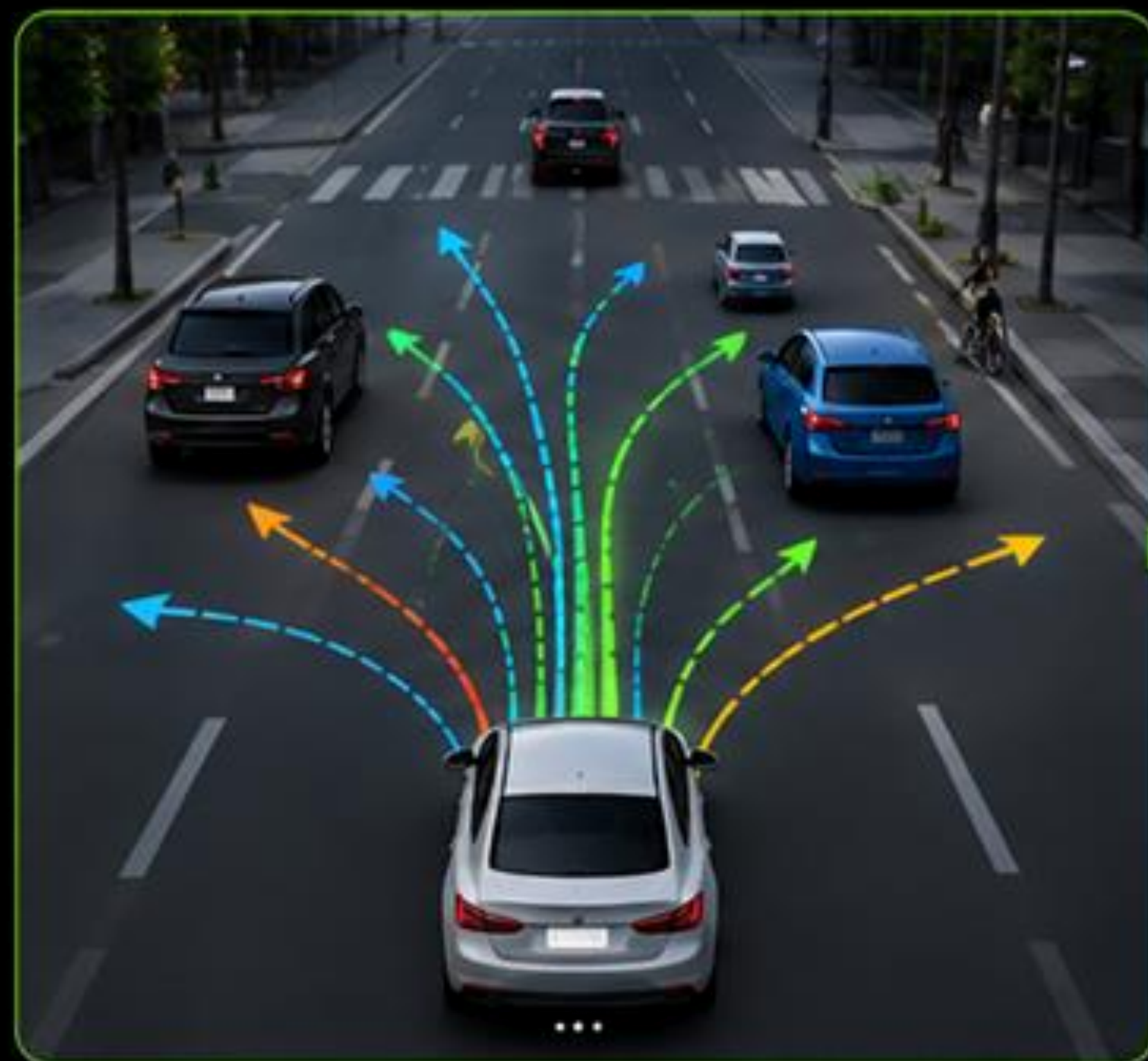
Pick the optimal trajectory and execute in the real world

Deliberate on the Edge

Search, Dream, Optimize, and Select

Search

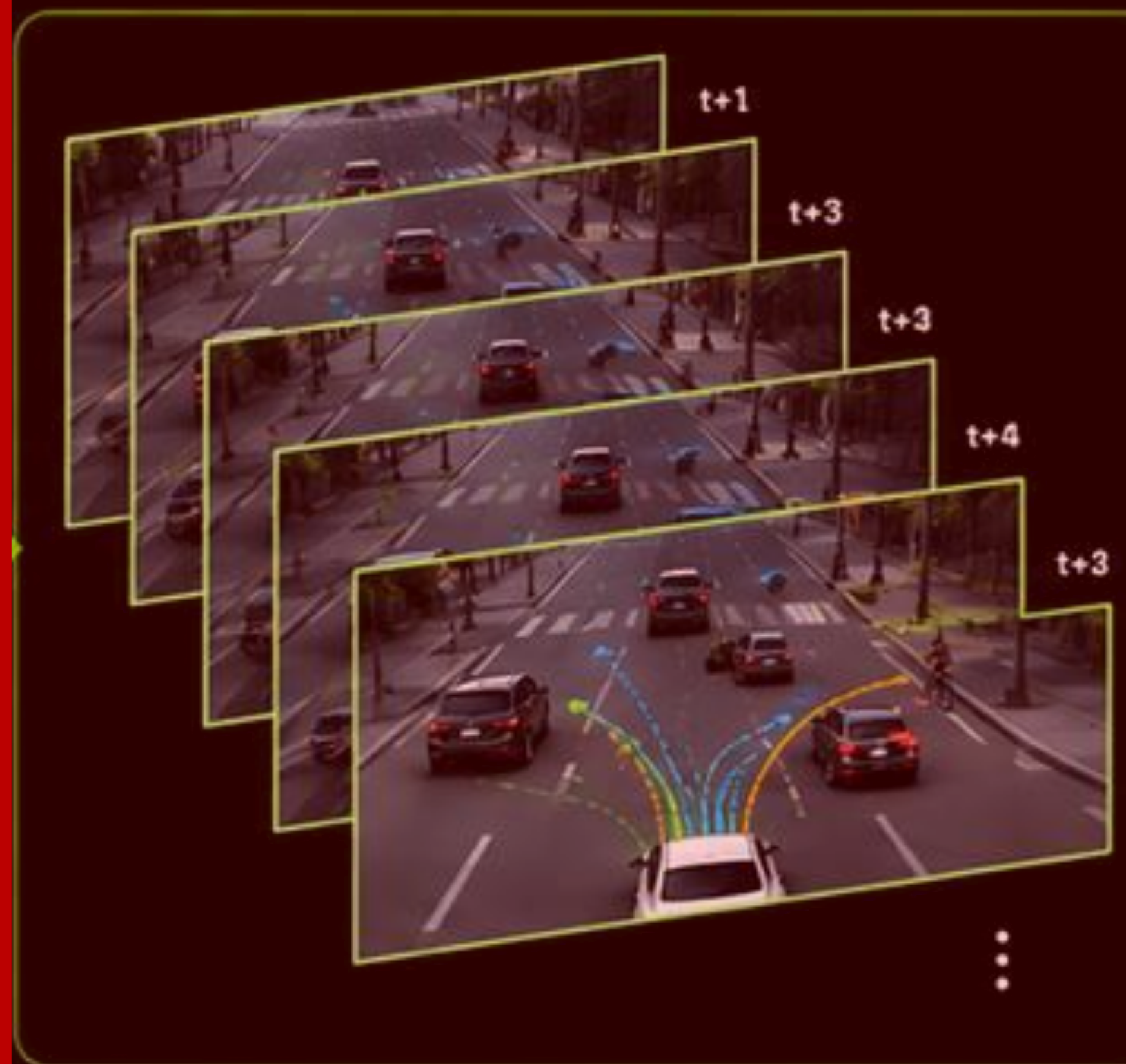
Generate diverse candidate futures and actions



Sample multiple plausible trajectories and action sequences

Dream

Simulate and imagine possible outcomes



Use a world model to roll out each candidate and predict what matters (not every pixel)

Optimize

Evaluate and refine with objectives and constraints



Compute rewards, check constraints, and refine (e.g., gradient-based, search-based)

Select

Choose the best plan and execute



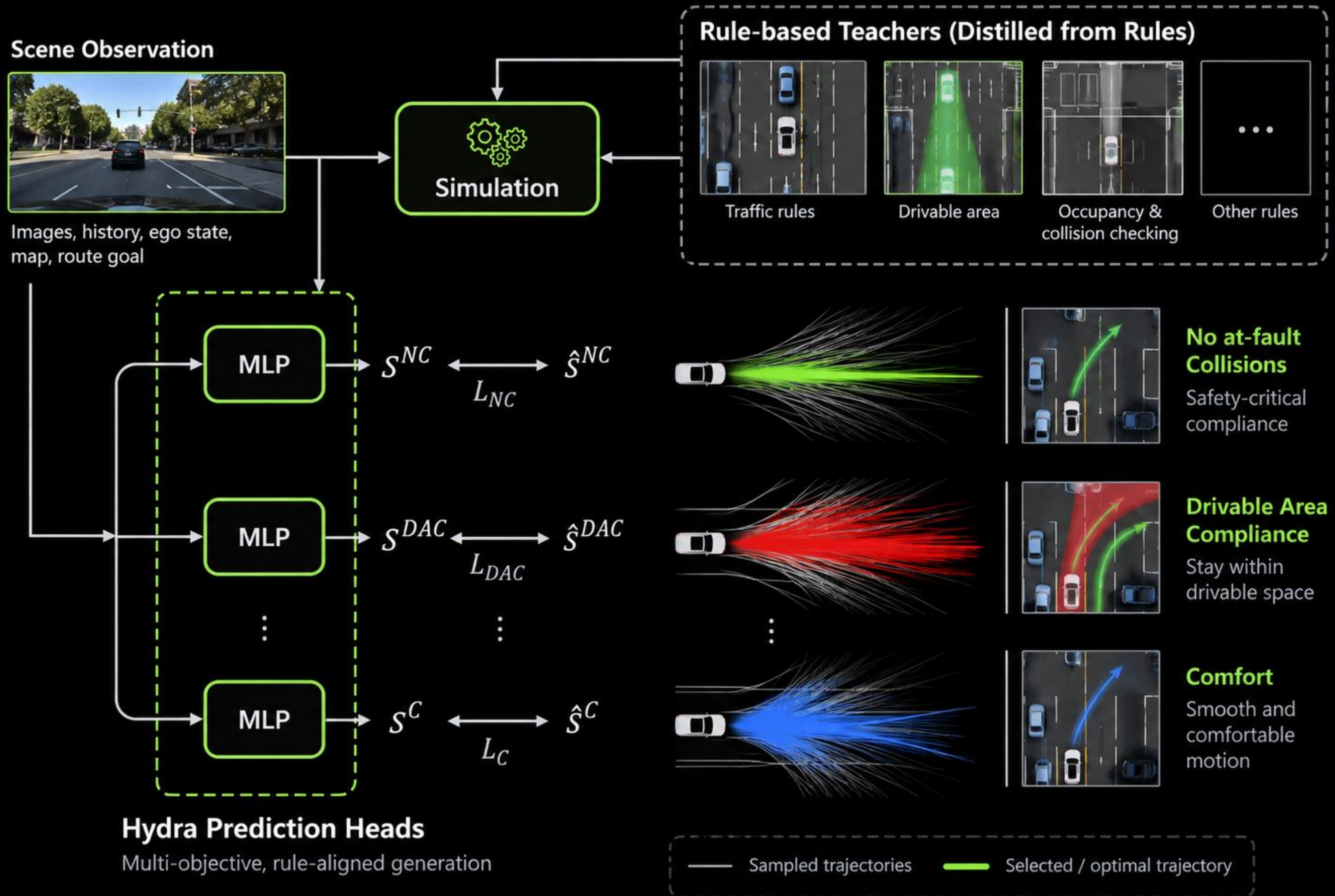
Pick the optimal trajectory and execute in the real world

Closed-loop continuous improvement

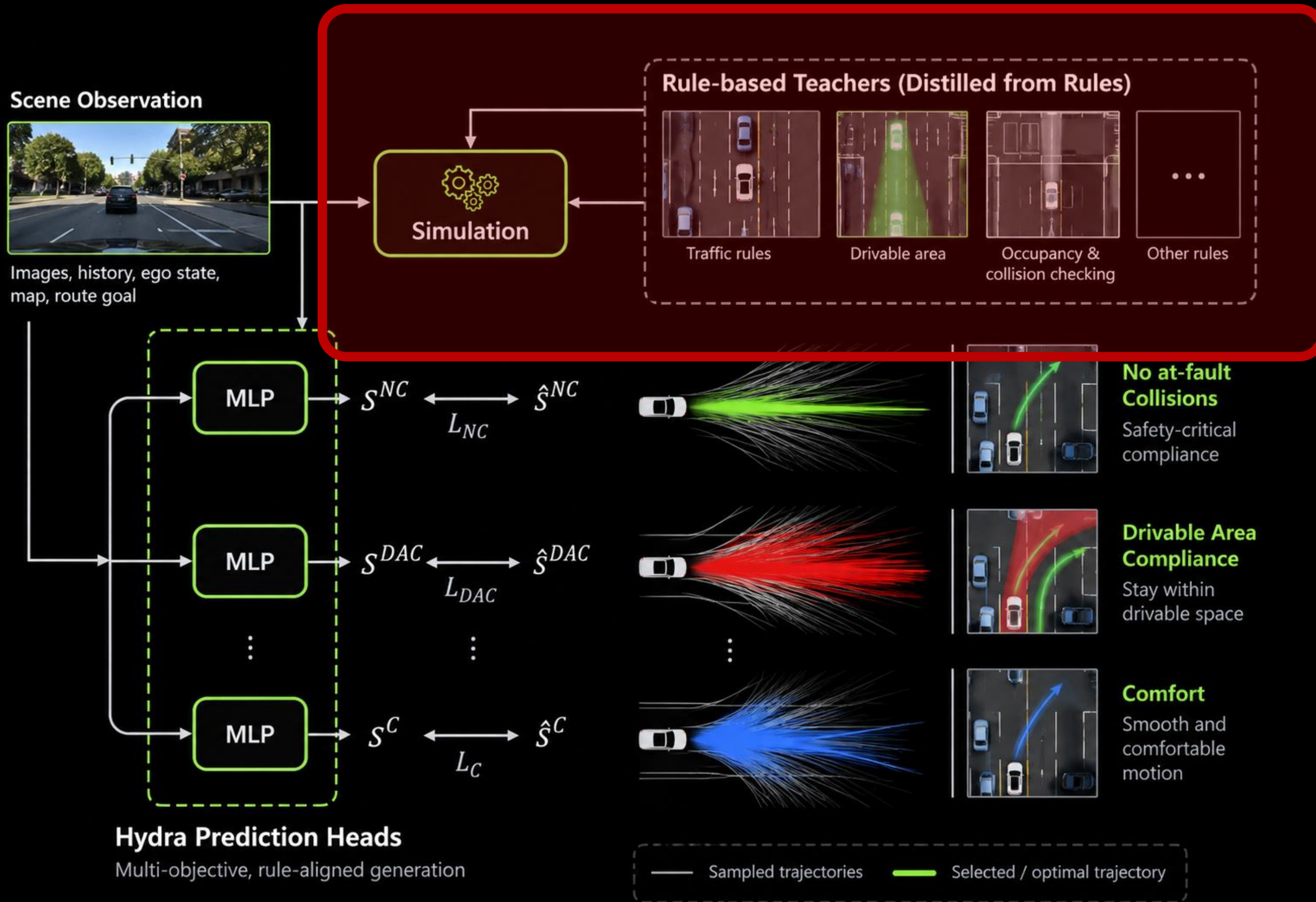
Learn from real-world experience and update the model

Prediction is not enough when the world reacts

Deliberation with Predicted Futures

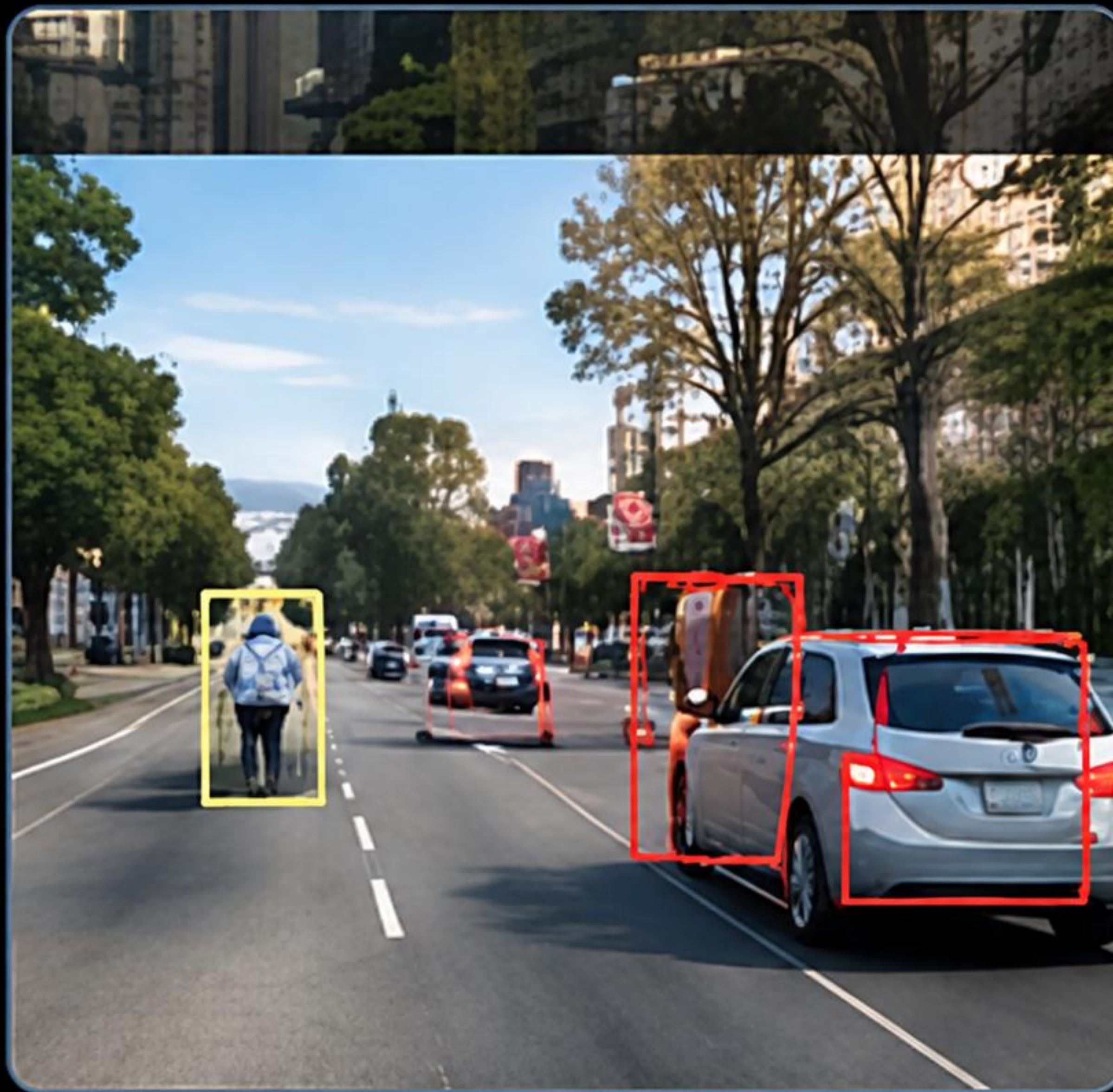


Deliberation with Predicted Futures

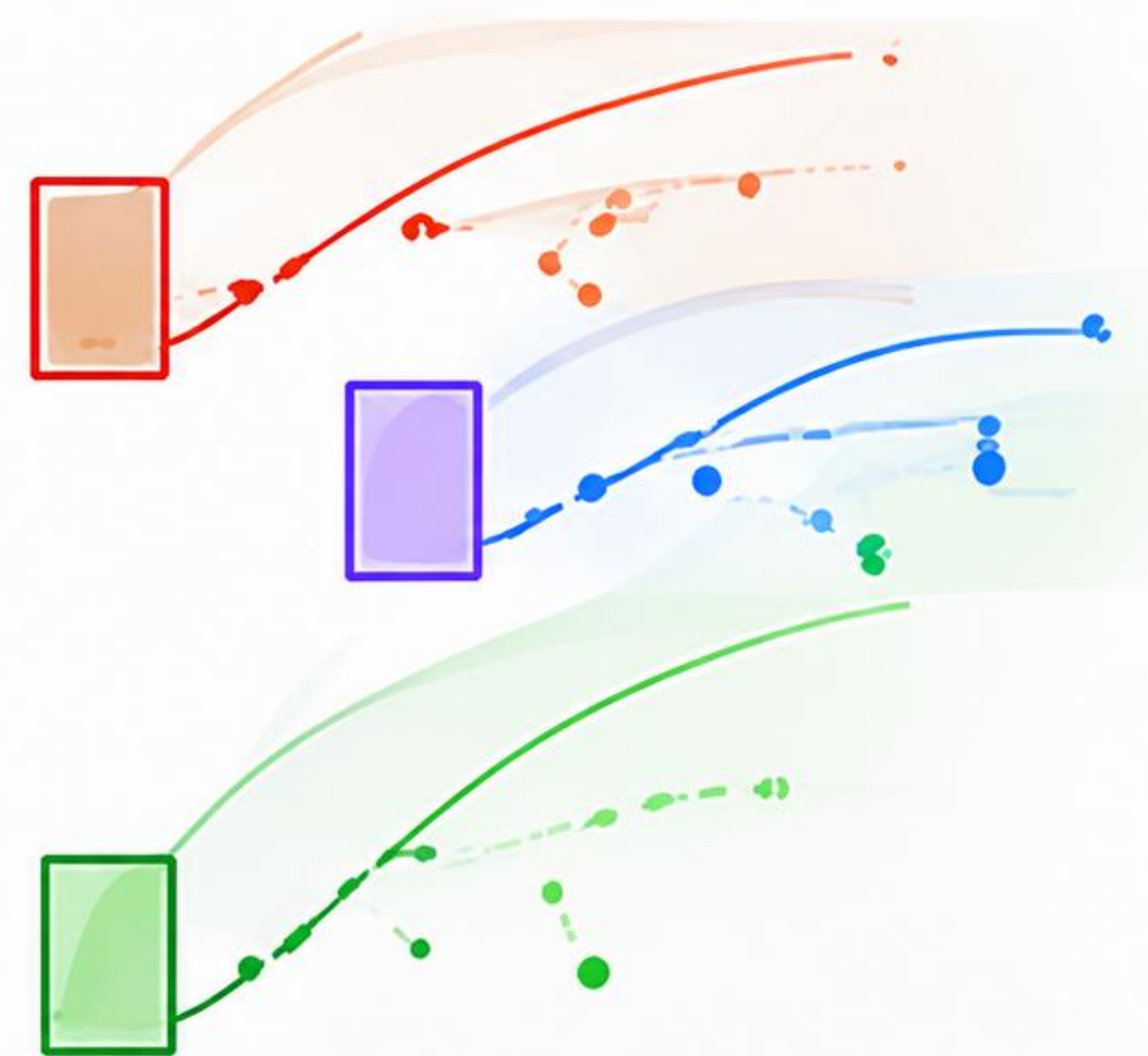


Deliberation with Predicted Futures

Hydra uses a simple model to predict where the other objects will be at inference time



Simple future prediction model
(at inference time)

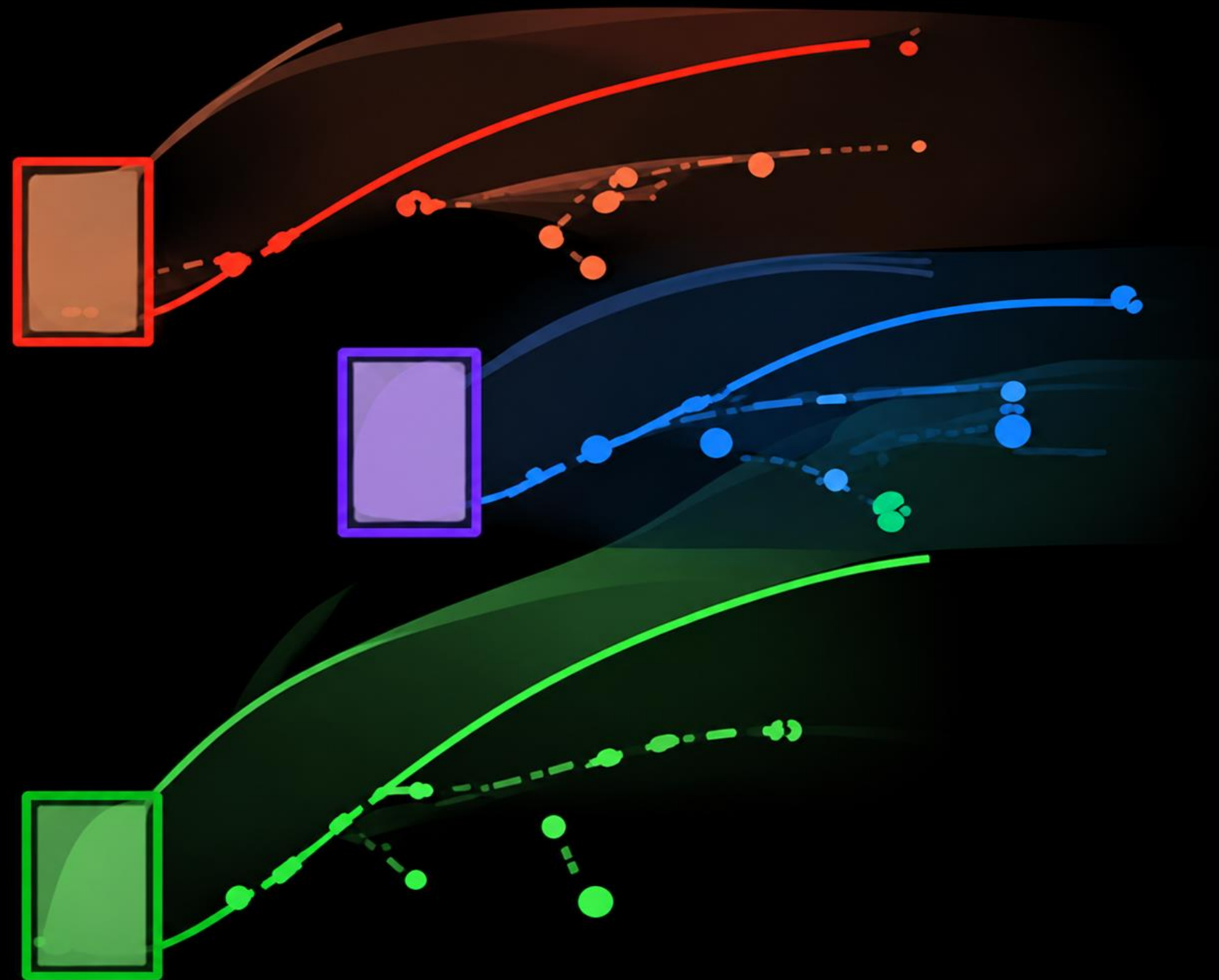


A simple model to dream about the future

World-model-based deliberation

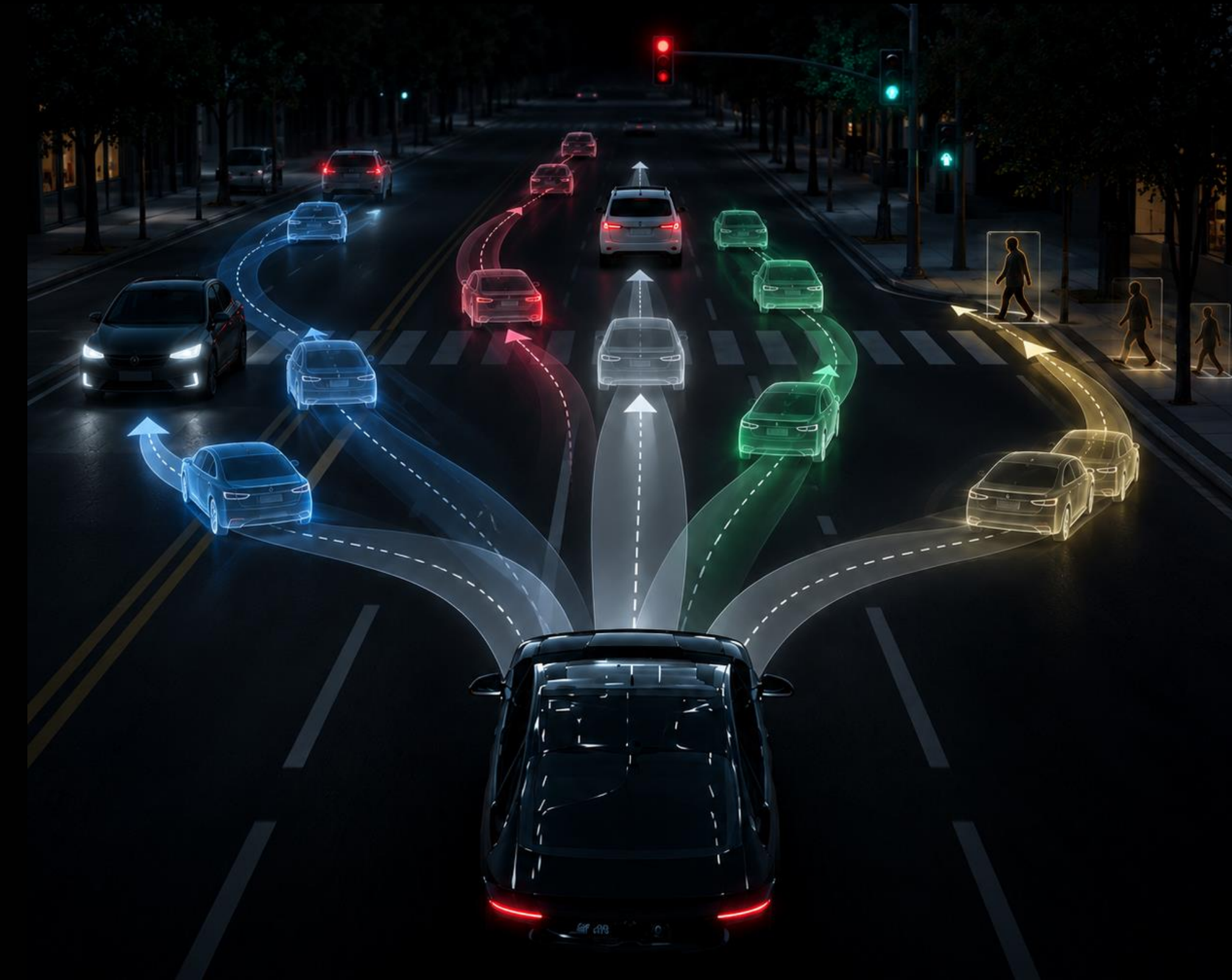
Richer, action-conditioned world models for interactive, multi-agent futures.

Simple
Predict object motion



“what could I do?”

Counterfactual Imagination
Simulate rich interactive worlds



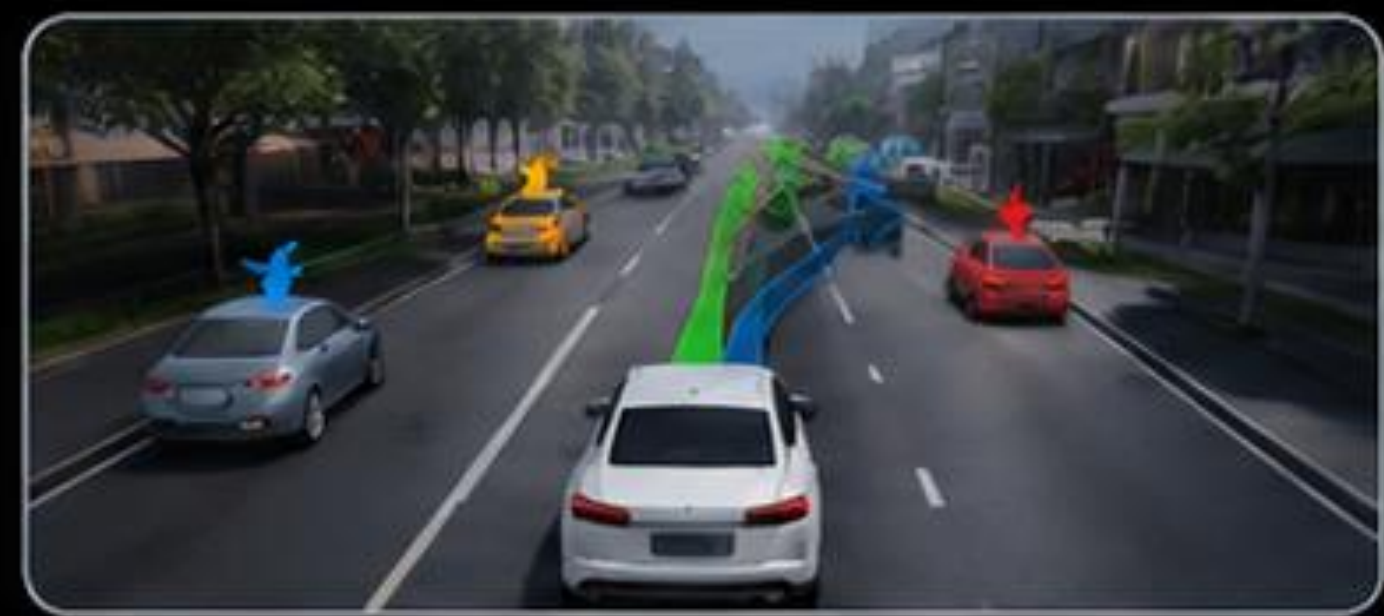
“what would happen if I did this?”

World-model-based deliberation

Think before act in Autonomous Driving

Offline (Training)

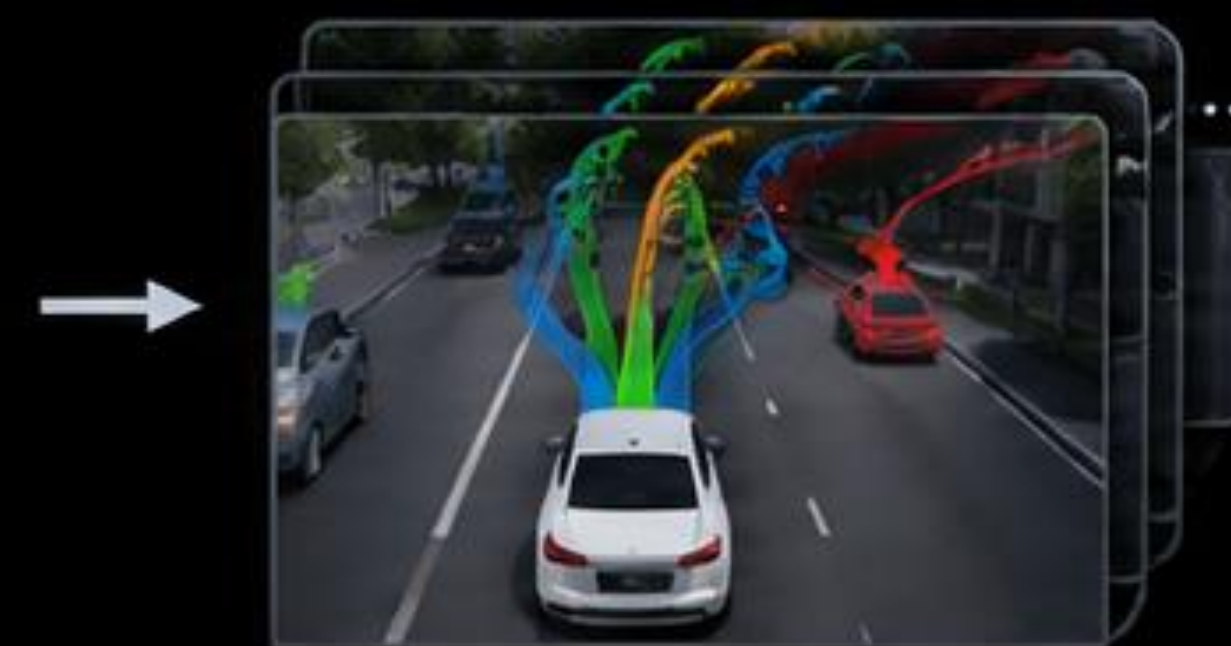
Build reward cache to train the world model



Diverse scenarios with reactive agents
(logged or simulated)



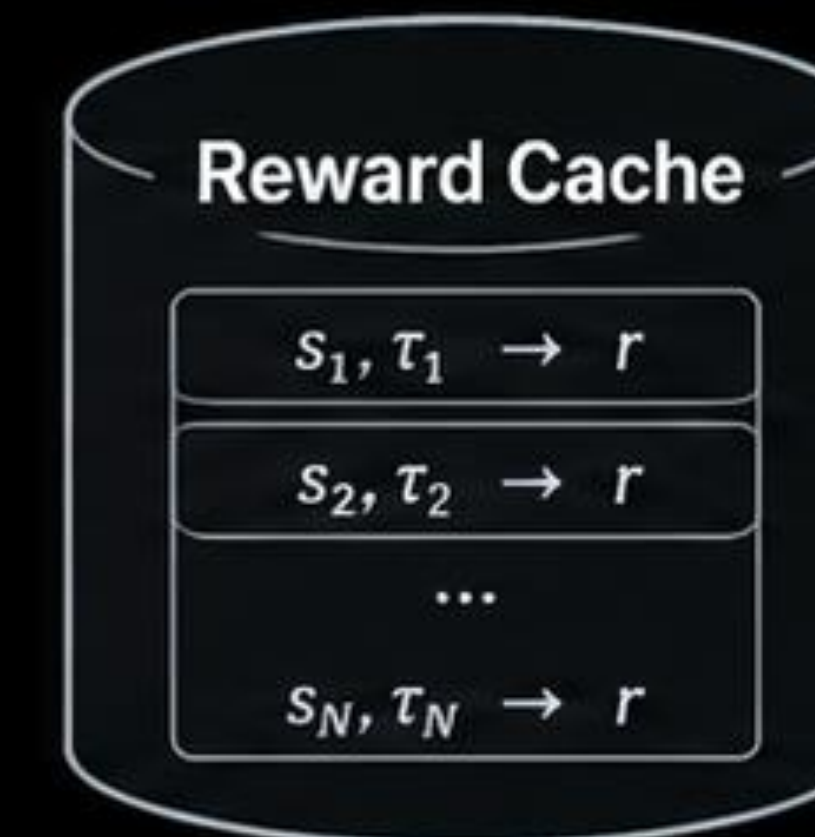
Closed-loop simulation
with reactive agents



Roll out candidate trajectories
and agent reactions



Compute decoupled
reward metrics



Pre-computed rewards
for efficient training



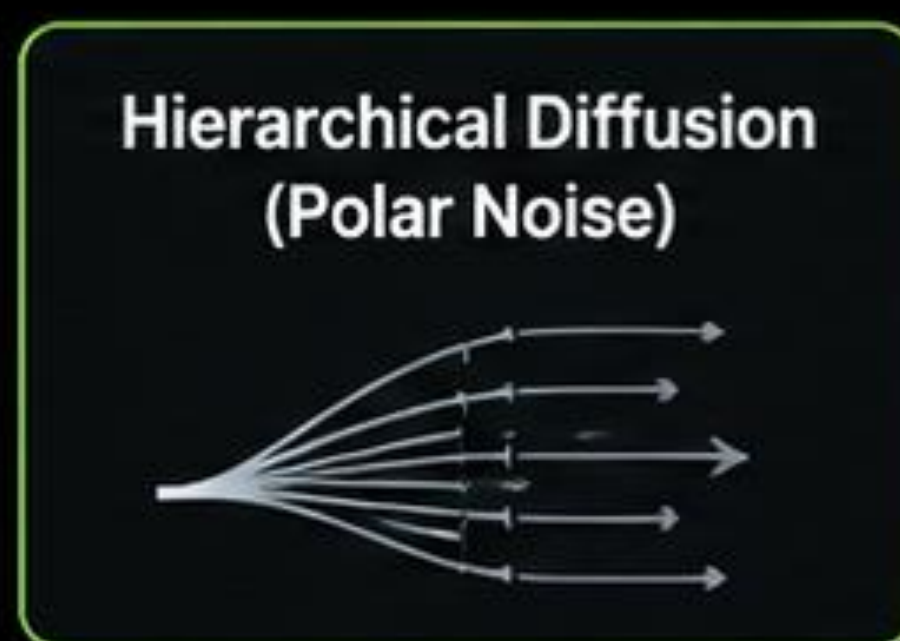
Train a world model
to predict outcomes and rewards
(with reactive agents)

Online (Inference – HAD)

Use world model and reward cache for planning



Current observation



Candidate trajectories



Predict future outcomes
with reactive agents



Score and select
best trajectory



Selected trajectory
for execution

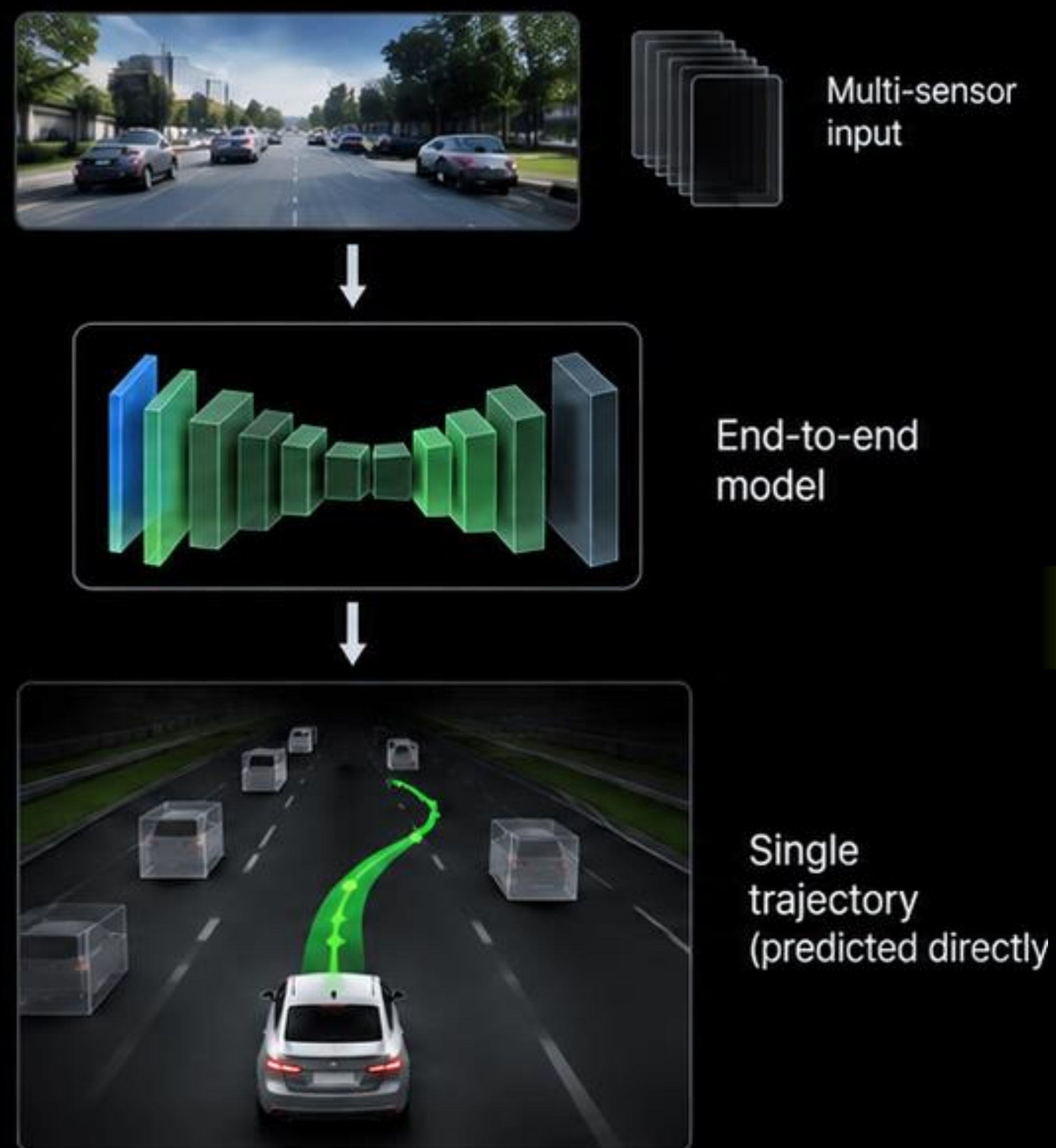
Use during inference

Recap

Recap

Single-shot prediction → Explicit alternatives → Test-time computation

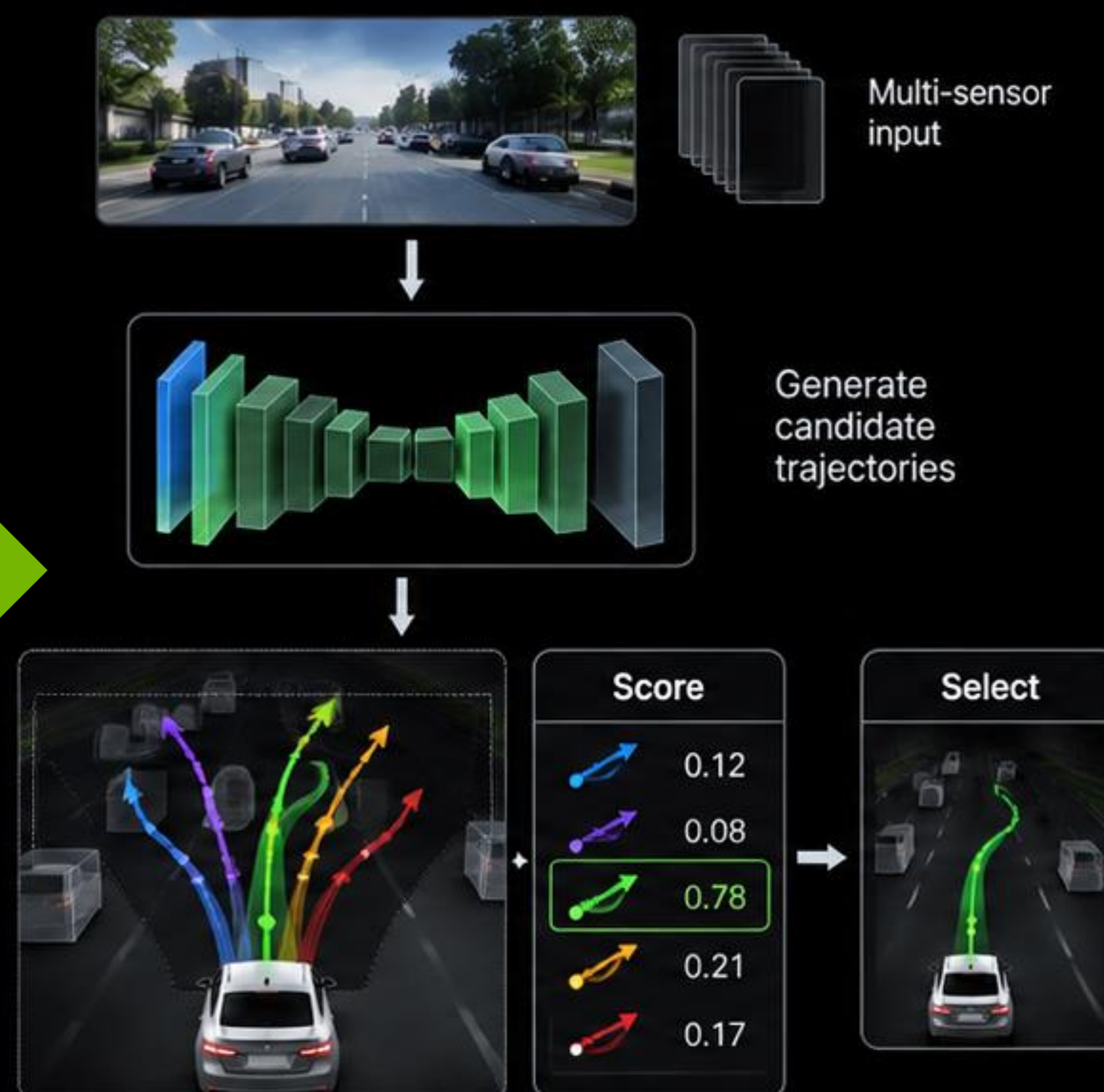
Direct regression



Predict a single trajectory from observation

1st Generation

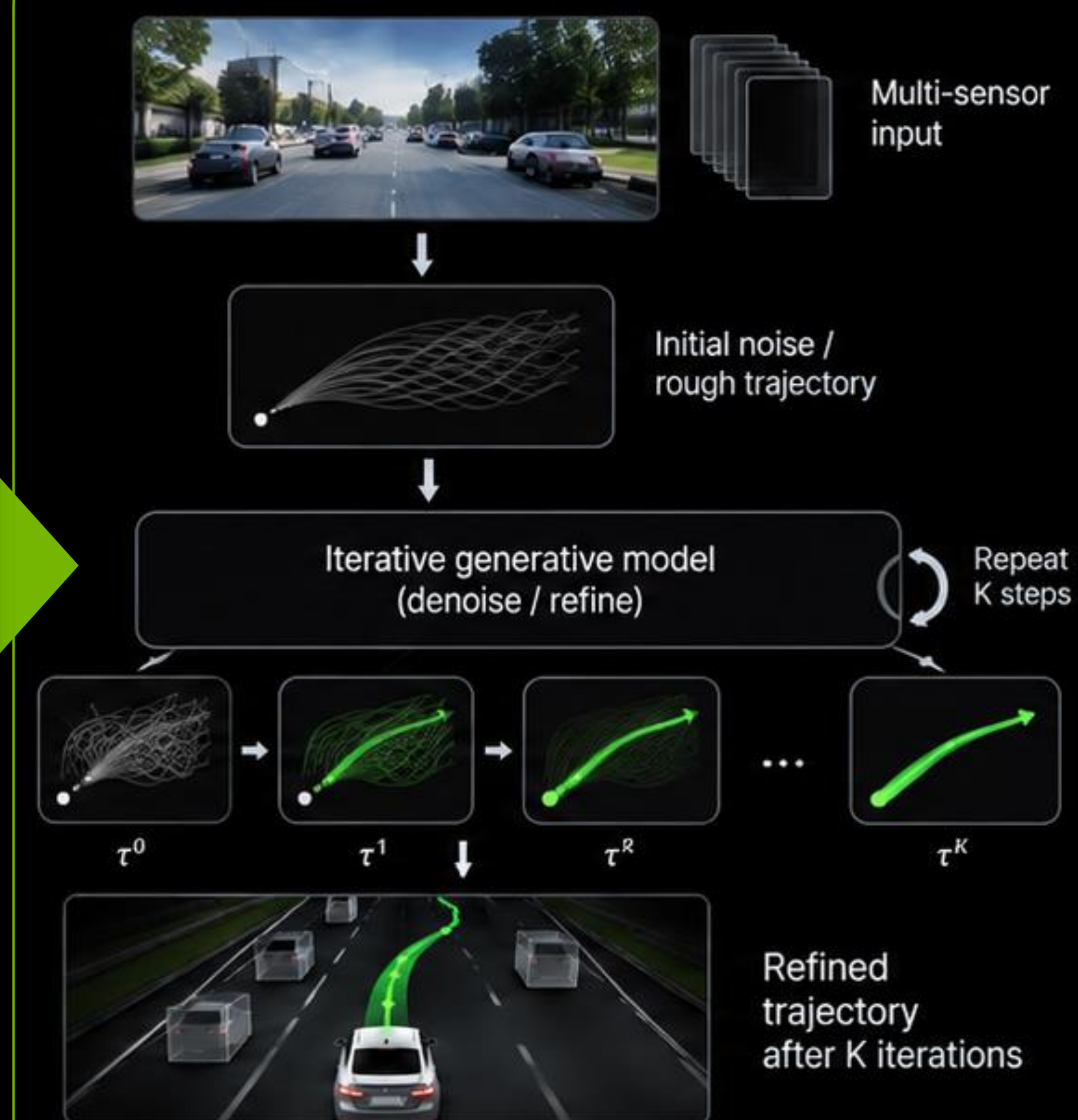
Generate / score / select



Generate multiple plausible trajectories, score them and select one

2nd Generation

Iterative Generative Planning



Iterative search and refine trajectories at test-time

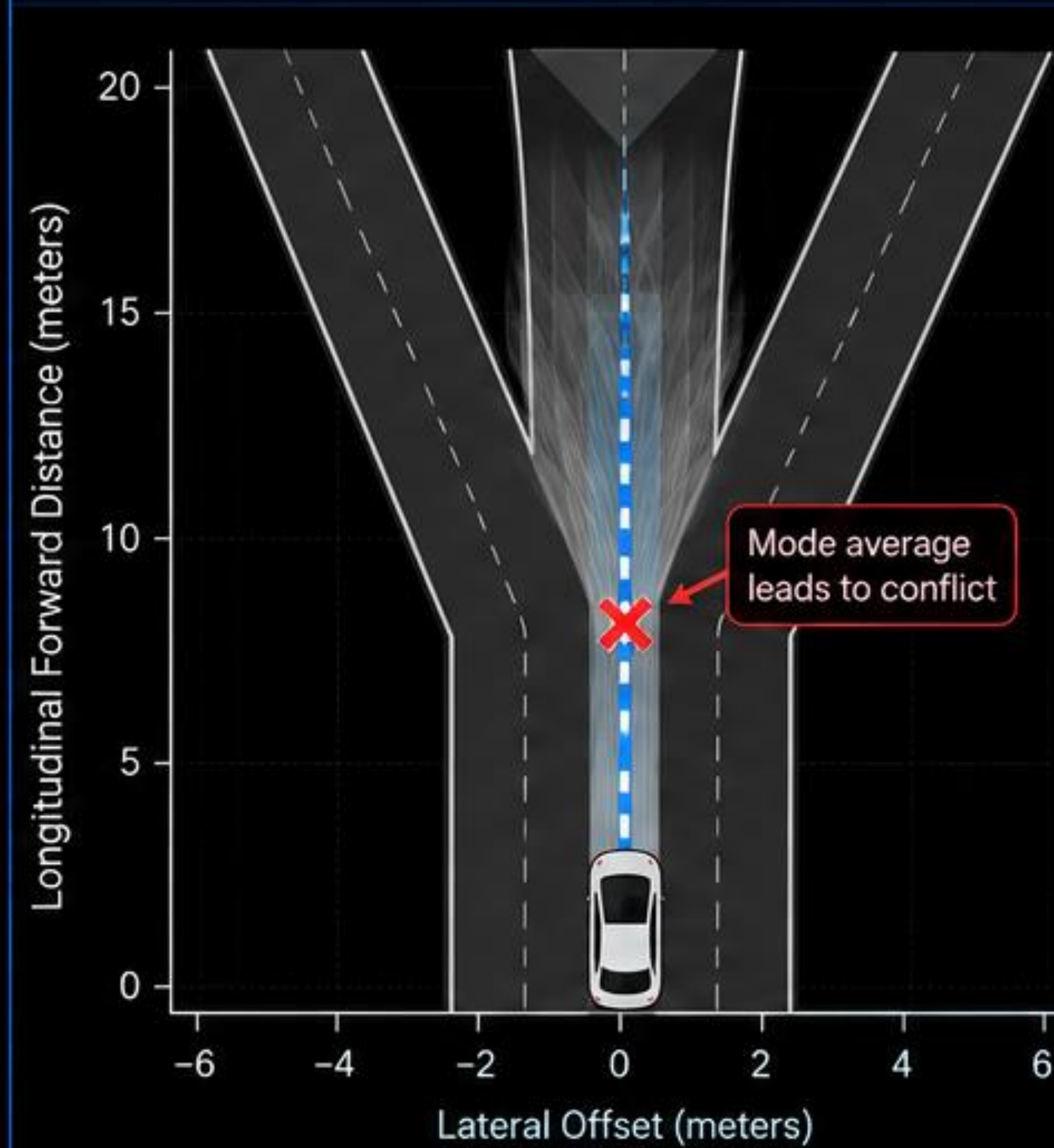
3rd Generation

Trajectory Path Solutions for the 3 Generations

Y-Fork Intersection Problem

Generation 1: Mode Average (Crash)

A single predicted trajectory (imitation learning).



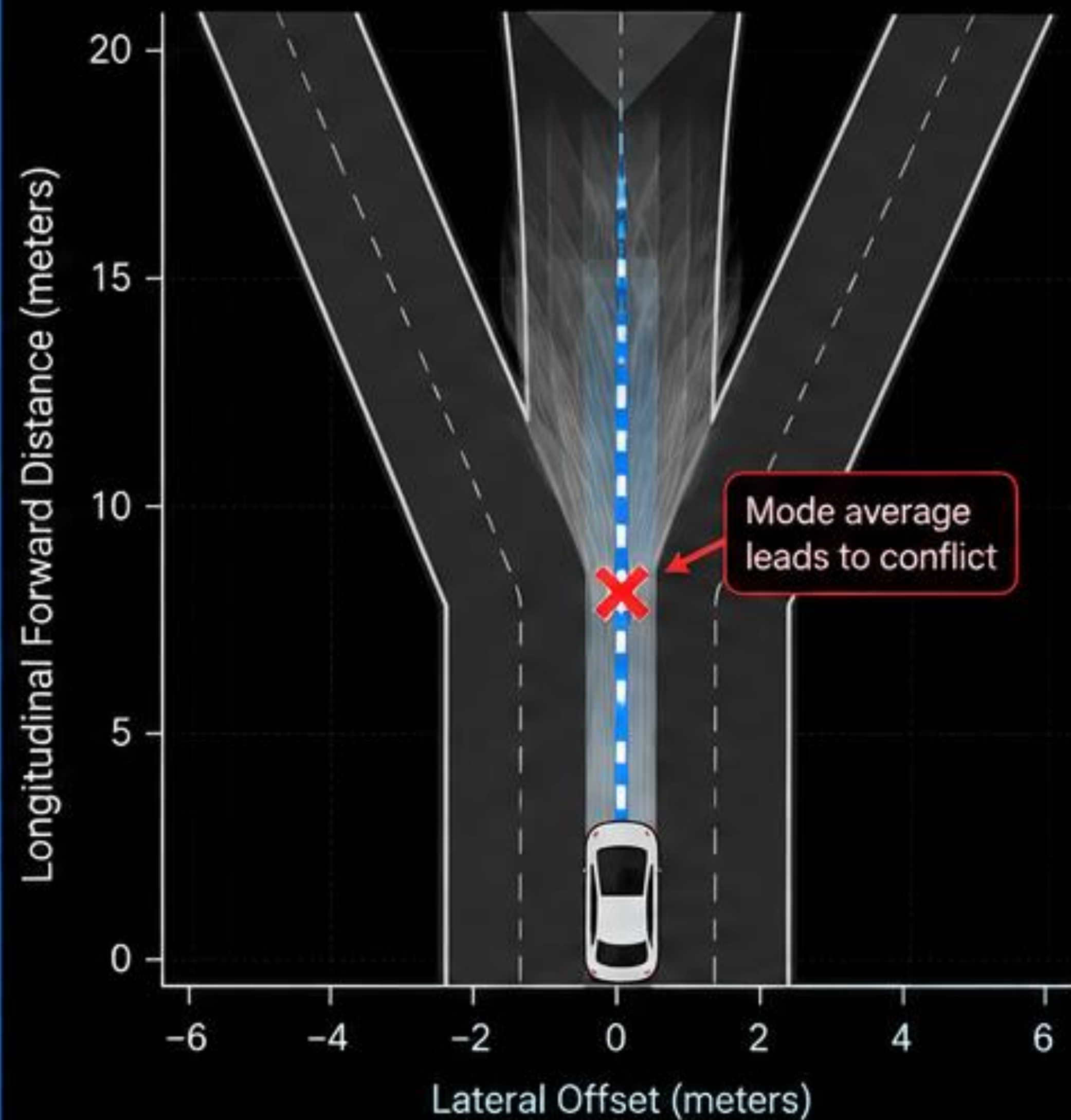
- Single deterministic prediction
- No explicit multi-modality
- Limited closed-loop optimization

Trajectory Path Solutions for the 3 Generations

Y-Fork Intersection Problem

Generation 1: Mode Average (Crash)

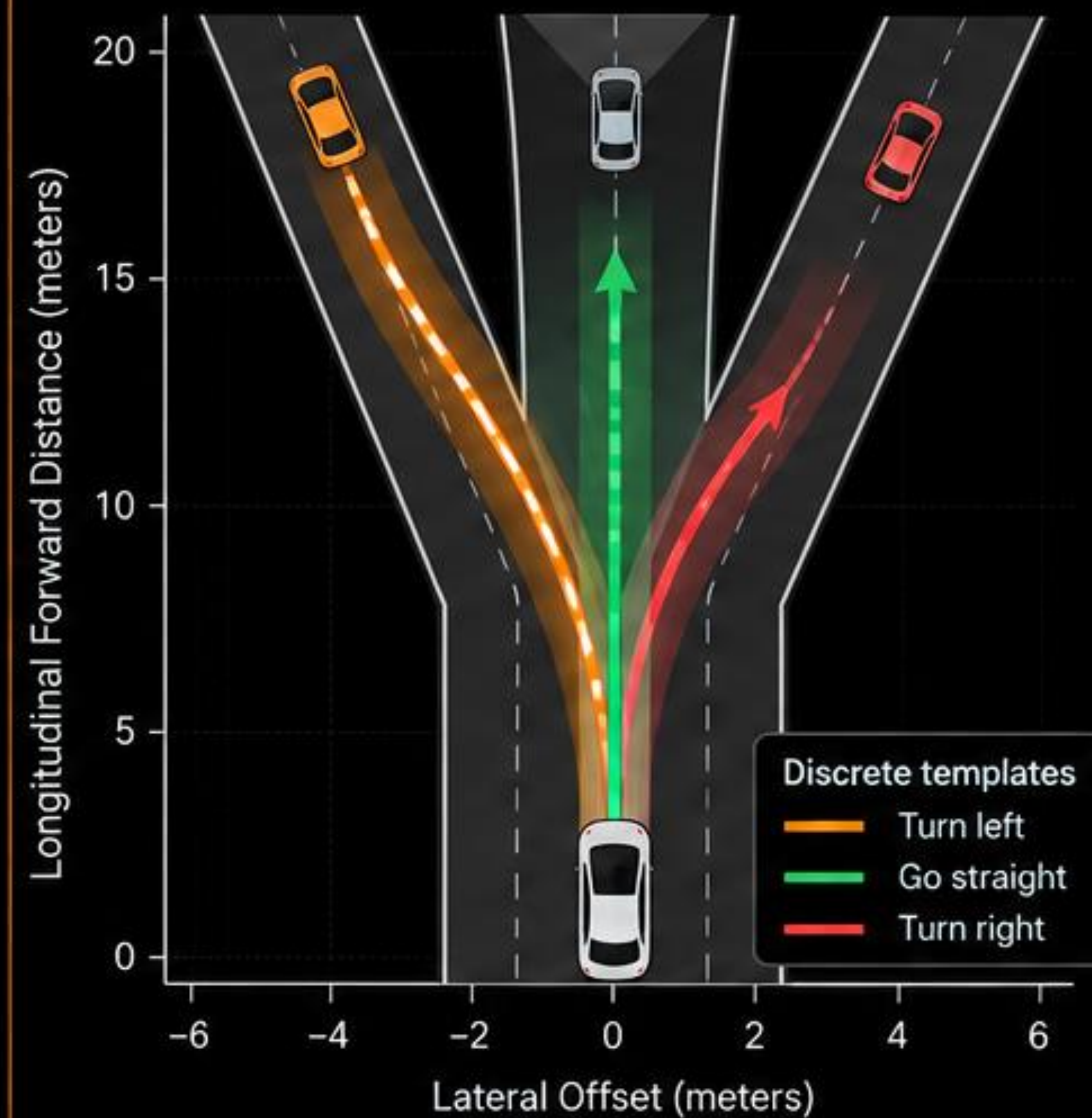
A single predicted trajectory (imitation learning).



- Single deterministic prediction
- No explicit multi-modality
- Limited closed-loop optimization

Generation 2: Static Codebook Template

A discrete set of maneuver templates (e.g., turn left).



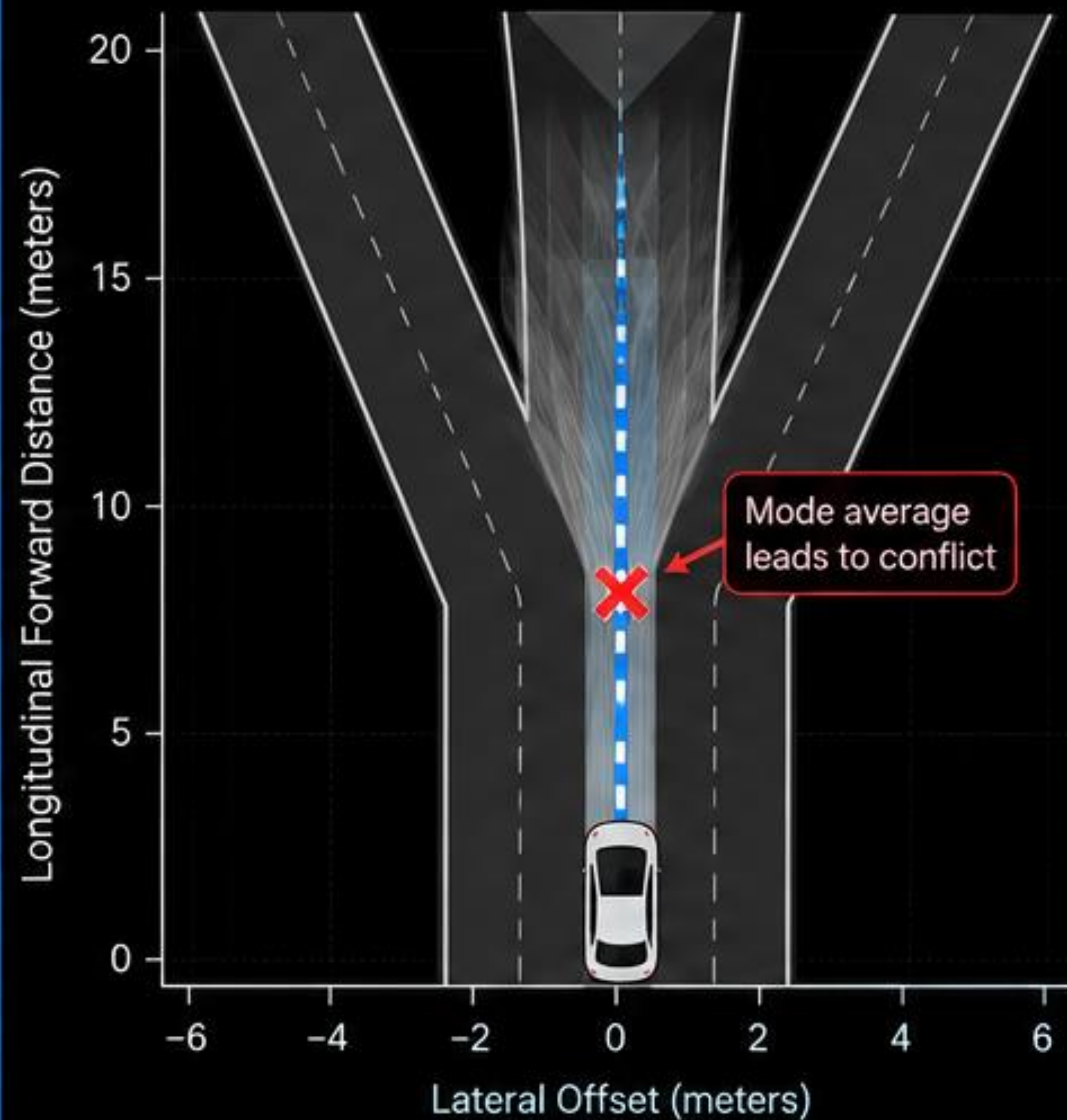
- Structured action space (discrete maneuvers)
- Learned evaluation / value
- Improved closed-loop optimization
- Chooses the best among a few templates

Trajectory Path Solutions for the 3 Generations

Y-Fork Intersection Problem

Generation 1: Mode Average (Crash)

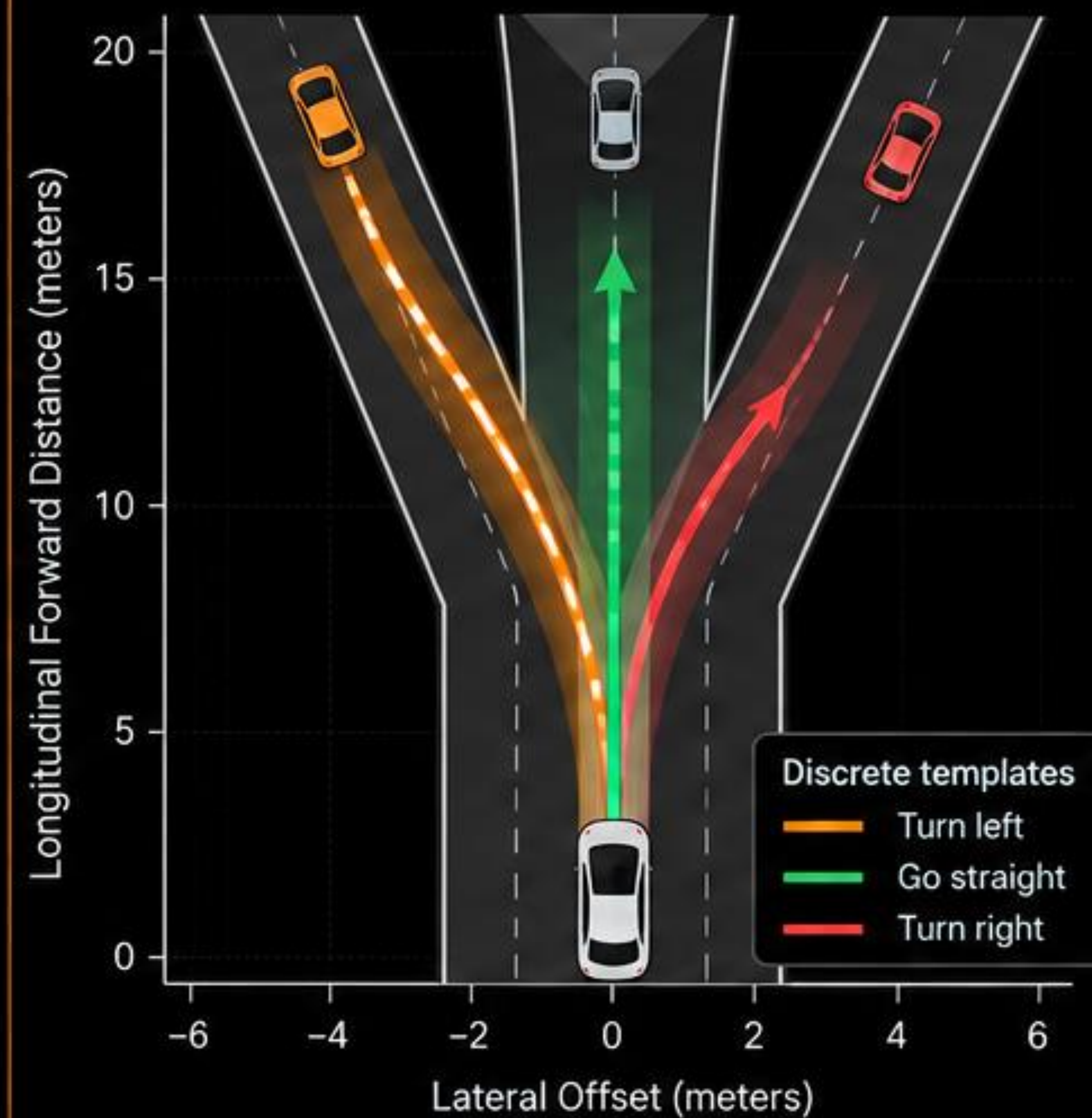
A single predicted trajectory (imitation learning).



- Single deterministic prediction
- No explicit multi-modality
- Limited closed-loop optimization

Generation 2: Static Codebook Template

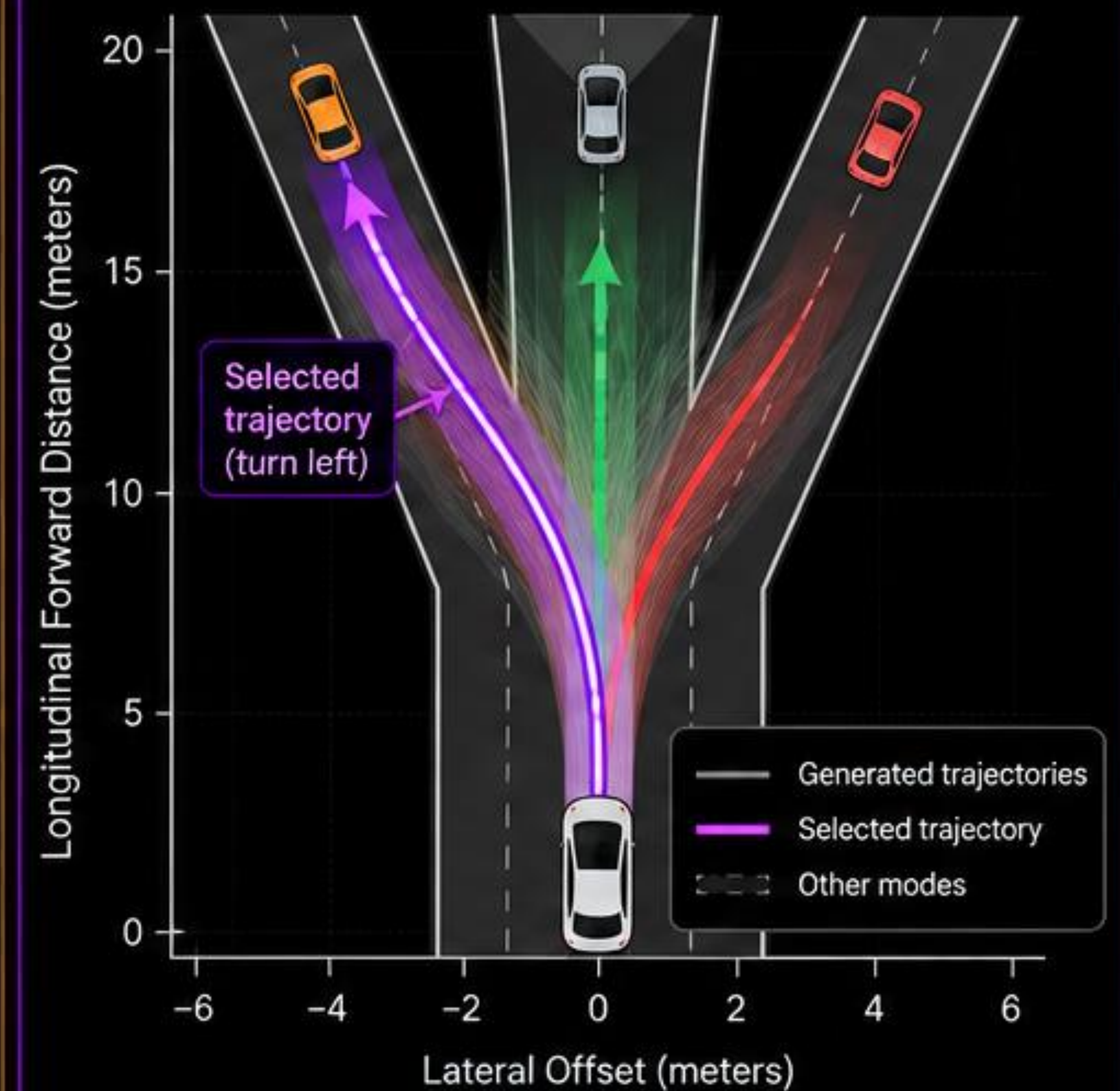
A discrete set of maneuver templates (e.g., turn left).



- Structured action space (discrete maneuvers)
- Learned evaluation / value
- Improved closed-loop optimization
- Chooses the best among a few templates

Generation 3: Hierarchical Hybrid Diffusion (HAD)

Generate, evaluate, and select among diverse plausible futures.



- Generative trajectory distribution (diverse, multi-modal)
- Evaluates with learned and rule-based objectives
- Selects the best action at inference time
- Highest decision capability and safety

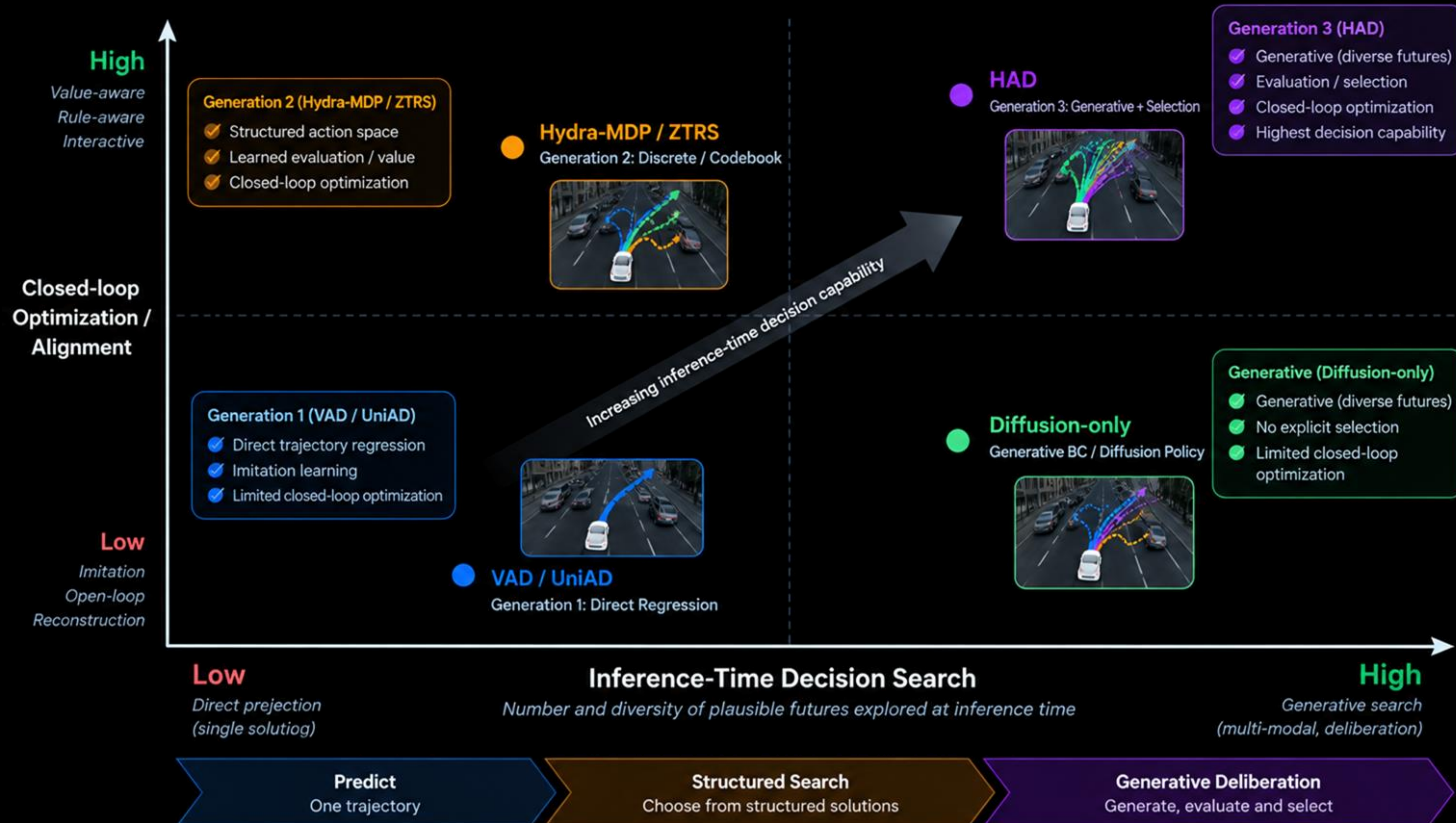
From predicting a trajectory to **think ahead: Reasoning** about the consequences of our actions

References

Part	Core question	Reference	Conceptual step
Direct prediction	“Can we directly predict how to drive?”	Background	Observation → trajectory
Multimodal planning	“One future is not enough—what alternatives should we consider?”	Hydra-MDP, CVPR 2024 (e2e challenge)	Trajectory vocabulary + multi-target learning
Closing the control loop	“Can an open-loop-trained model actually react?”	Hydra-NeXt, ICCV 2025	Trajectory + control + refinement
Generate & evaluate	“Instead of predicting the answer, can we search over answers?”	GTRS, CVPR 2025 (e2e challenge)	Generate → score → select
Selection as the bottleneck	“Can we reliably distinguish nearly identical alternatives?”	DriveSuprim, AAAI 2026	Coarse-to-fine selection
Beyond imitation	“Who says the human trajectory is the optimal target?”	ZTRS, ECCV 2026	Reward-driven planning / RL
Deliberation	“Can generation, evaluation and optimization become an inference-time loop?”	HAD, ECCV 2026	Hierarchical search + structured RL
Edge deliberation	“What should the next generation look like?”	Synthesis	Search → evaluate → optimize → execute → repeat

Evolution of End-to-End Driving Decoders

From prediction to structured decision-making to generative deliberation



*Training builds experience. Test-
time deliberation turns experience
into judgment.*

Thank you!





End-to-End Models for Autonomous Driving:

From Direct Prediction to
Counterfactual Deliberation on the Edge

Jose M. Alvarez,
Director AI Research, NVIDIA
ELLIS Summer School on Autonomous Driving, Sept. 2026

